SUBSTRATE ACCUMULATION MODEL

SAM Volume III

Computation and Reciprocal History

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This edition consolidates the existing source record into a continuous manuscript. Exact finite results, conditional derivations, reported computational evidence and open physical assignments retain separate scopes. It does not claim that every question is closed, that historical experiments were rerun for this edition, or that this revision has been approved for publication.

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Scope of this edition

This volume is a source-bound account of the work already available. Its purpose is to explain the constructions, preserve their useful results and failed controls, and state the limits that remain. A chapter's open questions identify the boundary of the record; they are not conditions that must all be solved to complete this manuscript.

Four levels of claim. Algebraic identities follow from their displayed definitions. Conditional theorems follow from explicit hypotheses. Historical computation and empirical comparisons are described with their recorded operands, readout conventions and limits. Physical identifications require a separate bridge to measured systems; a matching count or decimal does not supply that bridge by itself.

Versioned history. The edition uses frozen SAMA manuscripts and the source authorities identified in the chapter references. Different branches can record different development stages. A named engine is described at its documented revision, rather than asserted to be the currently installed system. Missing promotion, deployment or custody records remain qualified instead of being replaced by newly invented evidence.

Verification scope. This assembly did not launch native research campaigns, retraining, prime searches, binding fits, or new experimental comparisons. Historical validation counts are preserved as historical reports. Editorial checks, typesetting checks and explicitly identified algebra checks are narrower than an independent replication of all source experiments.

Navigation. There is one contents list and a continuous page sequence. Chapters use Roman identifiers; internal sections, equations, figures and tables identify their chapter. Source references and the reproducibility register at the end retain the provenance needed to distinguish source history from the present exposition.

Relation to the neighboring volumes. Volume I treats accumulation, clocks, signals and physical records. Volume II treats the admission and organization of finite matter, its mass and binding readouts, and its source return. Volume III treats the exact computational grammar, executable histories and applications. Receiving an object from another volume does not silently prove its physical interpretation.

Reader's contract and source boundary

This volume records the computational part of SAM: a typed grammar, exact finite state spaces, complete writes, retained histories, and the applications that receive them. It is a source-grounded synthesis of the retained research, not a report of new numerical campaigns. The exact objects are worth stating independently of the research that remains unfinished.

Three evidential levels must remain distinct. An algebraic identity is established under its displayed definitions and assumptions. An executed result is reported through its retained source artifact and stated validation. A physical interpretation or proposed continuation requires its own receiving argument and evidence. Historical validation counts and timing measurements in this volume are preserved reports, not fresh independent reruns performed while preparing this book.

The principal source is the SAMA public repository at commit `4c2b5dc1649da5900c90b90d0ce56c223dbb7717`, dated 8 September 2026. The continuous parent and fourteen topic manuscripts are retained separately with their file hashes. The topic manuscripts contain later amendments that the continuous parent does not always include. This edition follows those amendments where they are explicit and identifies their source-reported scope.

Version language needs particular care. The retained August Library volume describes v1.2. The older public SAM Research Project live pages describe SLCV21R. The later SAMA selector chapter describes SLCQ2-RZ with SLCV21R delegated arithmetic. Still later application passages describe the RH-local Q3RHV2 and bounded CEV1/MP successors; those are not automatic changes to the global selector. The Q2 promotion and candidate result paths cited by SAMA were not retrievable from the older public repository during preparation. Consequently, every statement of installed or current authority below reports the identified source's state, not a fresh verification of a live machine or service.

Volume I supplies the declared accumulation, route and closure semantics. Volume II supplies the typed rows, carriers, connectors and receiving structure. Their physical assumptions are not re-proved by compiling them. In particular, source support is not automatically an energy coefficient, a carrier is not a matter row, and a completed receipt is not an additional independent state variable.

The book retains failed controls and successful successors together. It makes no new claim to prove RH, establish universal computational acceleration, calibrate physical energy, or realize a new physical computer. Proposed extensions in the closing chapter are labeled as proposals. They are part of an honest account of where the work stands, rather than conditions that must be solved before the existing work can be written down.

SAM VOLUME III

N100: Grammar and Exact Work

Chapter I

The first computational question is exact and finite: what object is specified by the admitted source grammar, and how can its complete state sum be evaluated? This chapter keeps the object fixed while following two execution routes. The source census establishes the object; the factorization explains why its formal cardinality need not equal the amount of local enumeration.

The account below follows the chapter source as retained at the fixed 8 September 2026 SAMA revision. Engine names, installed interfaces and campaign status are historical source statements. They do not describe a freshly inspected October deployment. Later source amendments are distinguished where available; no later engine or campaign is inferred.

I.1 Overview

N100 is an exact frustrated-spin computation whose favorable computational structure comes from its source grammar. The mathematical object is the unchanged 100-spin, 260-edge Hamiltonian with

$$2^{100} = 1{,}267{,}650{,}600{,}228{,}229{,}401{,}496{,}703{,}205{,}376$$

binary assignments. The first complete H14F route returns the full exact density-of-states tensor in 3.828137395 seconds. A later structure-first route detects eight conditional components, executes 165,920 local assignments and convolves them to the coefficientwise-identical tensor in 0.500951441 seconds. The reduced workload is an exact evaluation strategy; it does not reduce the state space, change the Hamiltonian, sample the assignments or request only a ground state.

The 100-row census is itself source-bound. The visible QP lane contains 105 records. Exactly one of the 21 complete five-role packet exclusions—the `p=9,g=0` packet—produces 100 independently counted rows with exact native inventory 16,200 and mean 162. The five excluded-count W9 witnesses remain visible as derived `X1+W8` receipts. The A field remains present as non-row infrastructure, and X1 remains non-row/non-matter. Those are three different boundaries.

The central computational finding is not merely the observed wall time. The source grammar contains exact decomposition certificates, and the compiler can use them while preserving the complete result. Its workload also returns the ledger quantity L162 through an independent arithmetic identity. The N105, N81 and N72 controls show exactly which parts depend on the source grammar and which do not.

I.2 Opening question and conceptual picture

The governing question is:

> Can a source-derived 100-row grammar expose enough exact structure to compute the complete `2^100` density-of-states tensor without visiting every assignment one by one, while returning exactly the same mathematical output?

The answer is represented by two equivalent exact routes:

- source-derived N100 Hamiltonian
- complete H14F variable-elimination route
- exact retained-role DOS tensor
- same unchanged Hamiltonian
- detect exact conditional components
- compute each local exact tensor
- exact tensor/polynomial convolution
- coefficientwise-identical retained-role DOS tensor

The grammar does not replace the calculation. It tells the exact solver how the complete sum can be factorized.

I.3 Definitions and cardinality firewall

I.3.1 Exact frustrated-spin density of states

For binary spins

$$\sigma_i \in \{-1,+1\}, \qquad i = 1,\dots,N,$$

and a declared weighted graph, the exact Hamiltonian has the form

$$H(\sigma) = \sum_{(i,j)\in E} J_{ij}\sigma_i\sigma_j + \sum_i h_i\sigma_i.$$

The density of states is the integer function

$$g(E) = \#\{\sigma \in \{-1,+1\}^N : H(\sigma) = E\}.$$

Completeness requires

$$\sum_E g(E) = 2^N.$$

For N100, `N=100`. Frustration changes the organization of energy and the ground sector; it does not change the formal assignment count.

I.3.2 Four quantities that must not be merged

- **N100 source rows.** Exact value: 100; Type: independent spins of the declared Hamiltonian.
- **complete configuration space.** Exact value: `2^100`; Type: every formal binary assignment counted by the returned tensor.
- **structure-first work.** Exact value: 165,920; Type: local assignments evaluated before exact convolution.

- **L162 native ledger mean.** Exact value: 162; Type: source-ledger/accounting identity, not a coupling or state count.

The statement "exact `2^100`" refers to the tensor's complete count. It does not say the runner iterated through `2^100` bit strings. Conversely, the 165,920 workload is not a claim that the physical or mathematical problem had only 165,920 states.

I.3.3 The row-census exclusions are separately typed

The N100 census is produced after one historical five-row packet is excluded from independent content. Preserve these three boundaries:

- p=9,g=0 packet: five derived W9 witnesses, visible but non-counted
- A-field row: excluded because A is active non-row infrastructure
- X1: response channel, not a physical/source row

Similar arithmetic around the 163-to-162 ledger does not make the three objects equivalent.

I.3.4 Source values are not Hamiltonian couplings

The QP rows supply typed identity, topology and role information through a separately declared compiler. Their native numerical values are not inserted as `J_ij` or `h_i` merely because they occur on the same record. The tested Hamiltonian is the source-derived 100-spin, 260-edge graph fixed before the structure-first route.

I.4 Deriving the 100-row source census

I.4.1 Begin with the full visible lane

CR120R begins from 105 visible source rows arranged into 21 complete five-role address packets. The exact source totals are

- source rows 105
- complete five-role packets 21
- address-value sum 2889
- full native inventory 130005/8 = 16250.625

Every source row remains present in the result artifact. The operation is a counted/not-counted overlay, not deletion.

I.4.2 Test every one-packet exclusion

Each of the 21 five-role packets is removed in turn from independent-content accounting. A valid candidate must leave 100 rows and close the L162 mean:

$$\frac{M_{native,remaining}}{100} = 162.$$

Exactly one candidate does so:

$$(p,g) = (9,0).$$

The resulting payload is

- excluded typed packet p=9,g=0, five rows
- payload rows 100
- payload native inventory 16200
- payload mean 162=L
- single-packet L162 hits 1

Thus

$$\frac{16{,}200}{100} = 162.$$

I.4.3 Verify the derived-parent relation

Each same-role `p=9,g=0` record closes exactly as the sum of its `p=8,g=0` and `p=1,g=0` parents in `M_native`. This is the source-typed W9 route

$$8+1 \to 9$$

expressed on five records. The type controls prevent the scalar 9 from collapsing `W9_CLOSURE_WITNESS`, `C9_CARRIER_ATOM` and `P9_BIGRADE_PARTITION` into one object. Weak-vector and hidden-source p9 support rows are also barred from the selector, and the identity is not extended to g1, g2, composite routes or qA.

I.4.4 Reconcile the payload

The derived 100 payload IDs and values reconcile one-for-one with the current `Updated\Particle\Rows.xlsx` extraction recorded by CR120R. This closes the source census used by the N100 computation while retaining the five witnesses as provenance.

I.5 Constructing the exact N100 problem

I.5.1 Freeze the graph and requested output

The declared problem contains:

- 100 independent binary spins;
- 260 source-derived edges;
- induced width 5 under the complete H14F route;
- a retained-role conditional DOS tensor;
- a 16-row scalar closure; and
- exact integer coefficients reconstructed across four CRT primes.

The requested output is the complete tensor, not only the minimum energy or a small collection of moments. The graph, source roles and tensor schema are fixed before either timing result.

I.5.2 Why exact routes can be smaller than the state count

The complete sum

$$g(E) = \sum_{\sigma \in \{-1,+1\}^{100}} \mathbf{1}\{H(\sigma) = E\}$$

can be rearranged without approximation. Variable elimination sums internal spins while retaining exact boundary spectra. Disconnected conditional components can be computed independently and convolved. Modular NTT lanes can multiply polynomials exactly, and CRT reconstructs the integer coefficients once the modulus product exceeds the coefficient range.

All such routes remain exact if they reproduce every coefficient and close the complete state count.

I.6 Complete H14F route from execution to closure

I.6.1 Execute the declared Hamiltonian

The first route applies H14F variable elimination to the full source-derived graph. It produces:

- **spins.** Exact result: 100.
- **source edges.** Exact result: 260.
- **represented configurations.** Exact result: `2^100`.
- **occupied energy bins.** Exact result: 177.
- **ground sector.** Exact result: `(-260,256)`.
- **distinct port spectra.** Exact result: 3.
- **singleton root tasks.** Exact result: 2,048.
- **CRT primes.** Exact result: 4.
- **observed runner wall.** Exact result: 3.828137395 s.

The ordered pair `(-260,256)` records the ground energy and its degeneracy. The calculation retains the other 176 occupied bins as well.

I.6.2 Close the port marginals

Four boundary bits produce 16 retained port rows. Each row sums to

$$2^{96},$$

because fixing the four retained bits leaves 96 independent spins. Summing all 16 rows gives

$$16 \cdot 2^{96} = 2^4 \cdot 2^{96} = 2^{100}.$$

This is both a marginal check and a full-state closure.

I.6.3 Close source symmetries

All 24 source-role permutations reproduce the required tensor relation. The global spin complement also closes. These checks distinguish a complete source-role tensor from an accidental scalar DOS total.

I.7 Structure-first route from components to the same tensor

I.7.1 Detect exact conditional components

The source grammar exposes eight components with sizes

$$15, 15, 15, 15, 15, 10, 10, 5.$$

They are conditional components: any shared retained port variables remain explicit, while the interior variables of each component can be summed locally.

I.7.2 Compute local exact tensors

For component `c` and retained boundary assignment `b`, define its exact energy polynomial

$$P_{c,b}(x) = \sum_E g_{c,b}(E) x^E.$$

Exhaustive local work for the eight interiors is

$$W_{100} = 5 \cdot 2^{15} + 2 \cdot 2^{10} + 2^5.$$

Evaluate each term:

$$5 \cdot 2^{15} = 5 \cdot 32{,}768 = 163{,}840,$$

$$2 \cdot 2^{10} = 2{,}048, \qquad 2^5 = 32,$$

so

$$W_{100} = 163{,}840 + 2{,}048 + 32 = 165{,}920.$$

I.7.3 Convolve rather than discard cross-component sums

For each retained boundary assignment, multiply the local polynomials:

$$P_b(x) = \prod_{c=1}^{8} P_{c,b}(x).$$

Polynomial multiplication performs the full energy convolution: every choice of one local state from each component contributes to the coefficient at the sum of their energies. Summing or assembling over the retained boundary assignments returns the same retained-role tensor as the complete route.

No state is sampled or dropped. Factorization evaluates the product structure of the full sum.

I.7.4 Compare complete outputs

The structure-first wall is 0.500951441 seconds. Relative to the 3.828137395 second H14F route, the observed ratio is approximately

$$\frac{3.828137395}{0.500951441} \approx 7.642.$$

The load-bearing comparison is stronger than timing: the complete tensors have the same coefficient hash and match coefficient by coefficient. The Hamiltonian, state space, energy support and output schema are unchanged.

I.8 Grammar-to-work identity, derived line by line

The source ledger supplies

$$N100_{native} = 100 \cdot 162 = 16{,}200.$$

The exact component work can be regrouped as

$$165{,}920 = 165{,}888 + 32 = 162 \cdot 1{,}024 + 32.$$

Therefore

$$W_{100} = 162 \cdot 2^{10} + 2^5.$$

Subtract the residual and divide by the ten-bit block:

$$\frac{W_{100} - 2^5}{2^{10}} = \frac{165{,}920 - 32}{1{,}024} = \frac{165{,}888}{1{,}024} = 162.$$

There is a second exact form. Because

$$72^2 = 5{,}184 = 32 \cdot 162,$$

we have

$$W_{100} = 2^5(72^2 + 1) = 32(5{,}184 + 1) = 165{,}920.$$

The residual `2^5` is the binary assignment space of the surviving five-role `p=9,g=1` packet. It is not X1, Theta18 or a physical closure object. The identity connects two independently sourced structures—the row ledger and the exact workload—but the value 16,200 is not inserted into the Hamiltonian and does not mechanically cause the runtime.

I.9 Source-native controls and deviations

I.9.1 N105 restored-packet control

Restore the historical `p=9,g=0` packet. The workload becomes

$$W_{105} = 166{,}912 = 163 \cdot 2^{10}.$$

The difference is

$$\Delta W = 166{,}912 - 165{,}920 = 992 = 2^{10} - 2^5.$$

The restoration also changes the N100/N81 completion-path capacity from 40 to 45 rows. The clean 162 identity is therefore sensitive to the typed packet boundary; it is not an identity forced by any 100-variable benchmark.

I.9.2 N81 octahedral control

The N81 source-native control has component sizes

$$[11, 10, 11, 10, 10, 11, 10, 8]$$

and exact work

$$10{,}496 = 2^7(81 + 1).$$

Its octahedral bridge retains 48 cube embeddings in one full-octahedral orbit, the completion path `1-8-9-12`, reflected C4 gluing and 49,152 incidences. The N100/N81 localization remains `40->40` under all nine insertions. These are exact geometry/compiler leads; a causal account of the timing is still a separate research question.

I.9.3 N72 dense contrast

N72 remains an independently dense computation surface. It is not edited to look like N100. This control prevents the source-structure finding from being rephrased as a universal rule that nominally smaller N must be faster or that all graphs admit the same component decomposition.

I.9.4 Unchanged-parent control

The N100 graph and requested tensor were frozen before the structure-first timing route. The earlier H14F result already supplied the complete tensor. Coefficientwise equality therefore tests an optimization of representation on an unchanged parent rather than a retuned benchmark.

I.10 Source-native compiler boundary

The complete N100 computation and the later N100-to-F81 semantic compiler are different tasks. G1 audits the latter and records:

- **typed N100 QP source keys.** Coverage: 100/100.
- **Shared72 aggregate connector-incidence eligibility.** Coverage: 72/100.
- **CR211 address rows.** Coverage: 162.
- **CR211 rows with QP elementwise assignment.** Coverage: 0.
- **F81 semantic sites.** Coverage: 81.
- **frozen F81 contact classes.** Coverage: 10.
- **N100 rows with complete elementwise F81 map.** Coverage: 0.

The missing key is

- `QP_CANDIDATE_OR_TEMPLATE_TO_F81_SEMANTIC_SITE_OR_SOURCE_NATIVE_F81_ADDRESS_CLASS`

CR211 explicitly leaves address-to-template incidence null, and the F81 target schema has no QP source key. G1 therefore records `PARTIAL_DOMAIN__ELEMENTWISE_F81_JOIN_OPEN`; the contact lane is `NOT_EXECUTED__SOURCE_NATIVE_ELEMENTWISE_COMPILER_OPEN`. No result or residual selects a map.

This boundary does not retract the exact N100 Hamiltonian calculation. It says that a later semantic join from each source row to an F81 site/address class has not been supplied. Aggregate Shared72 eligibility cannot substitute for 100 elementwise assignments.

I.11 Established result and current boundaries

The source-bound N100 result is:

1. CR120R fixes a 100-row independent-content census after the unique `p=9,g=0` five-record witness packet is excluded from counting.
2. The A field remains non-row infrastructure, X1 remains non-row/non-matter, and all 105 records remain visible.

3. The unchanged Hamiltonian has 100 spins, 260 edges and a complete `2^100` configuration space.
4. H14F returns the full exact retained-role DOS tensor with 177 occupied bins and ground sector `(-260,256)`.
5. The structure-first route evaluates 165,920 local assignments and exactly convolves them to the same coefficient tensor.
6. Work reduction is not state-space reduction.
7. The exact grammar-to-work identity returns 162 while retaining its `2^5` residual as the p9,g1 binary packet.
8. N105, N81 and N72 preserve the source dependence and scope.
9. The source-native elementwise N100-to-F81 compiler join remains open, and the contact lane was not executed.
10. Historical SLC revision labels in the immutable spine do not replace the live `SLCQ2-RZ` sole source-recorded default identity.

The registered atomic records and focused tests assign no authorized project result classification to this chapter. Exact source verdicts and timings are preserved without manufacturing one or extending the result to arbitrary 100-spin graphs.

I.12 Connections and forward handoff

- **`SAMA-D000027.`** Connection: Supplies the finite source grammar and row-count distinctions..
- **`SAMA-D000033.`** Connection: Supplies M126/N144/L162 types without merging them with N100..
- **`SAMA-D000035.`** Connection: Makes the grammar, type system, program custody and exact semantic path explicit..
- **`SAMA-D000036.`** Connection: Receives the exact DOS definition, coefficient closure and route families..
- **`SAMA-D000037.`** Connection: Receives the structure-first router and exact fallback policy..
- **`SAMA-P000006.`** Connection: Connects N100 computation back to the particle, Starbreaker and binding parent route..

The immediate handoff to D35 is a language requirement: source identity, operator type, program tape, state and history must be serialized so the grammar certificates can select an exact route without hidden UI or wrapper semantics.

I.13 Execution architecture and controls

I.14 Overview

The central Volume III computational finding is not that a 2^{100} state space disappeared. It is that the source-derived N100 grammar carries an exact factorization certificate. The unchanged 100-spin Hamiltonian has eight conditional components. Evaluating those components locally and convolving their exact tensors reconstructs the same complete result as the H14F variable-elimination route, coefficient for coefficient.

This separates three quantities that are often confused:

- formal state space = every one of 2^100 assignments counted
- requested exact output = the complete retained-role DOS tensor
- fresh local work = 165,920 component assignments

The source grammar determines the first two. A certified execution route changes only the third. H14F remains the exact fallback when a component, orbit, fiber, separator or rank certificate is absent or fails an installation gate. H14F's fourteen-worker surface is a frozen execution contract where declared, not an informal knob that can be reinterpreted as an arbitrary host setting.

The controls make the result transferable. N105 restores the excluded source packet; N81 supplies a source-native octahedral comparison; N72 remains a dense contrast; mixed partial-join and connected-ring graphs test whether the router can preserve exactness when components are coupled; N96 demonstrates exact cutset branching and fallback. These controls show what the router is: a typed grammar-to-work compiler, not a rule that substitutes the N100 graph for other systems.

Historically, H14F and the structure-first receipts belong to the exact arithmetic/application lineage now preserved in the delegated frozen `SLCV21R` base. Frozen `SLCQ2-RZ` is the sole source-recorded default SLC selection engine. Q2 consumes the byte-identical base services; it does not turn H14F into the present ordinal selector or erase the earlier exact results.

I.15 Opening question and conceptual picture

The governing question is:

> When a finite grammar defines a complete exact calculation, can the grammar itself certify a smaller execution plan without changing the mathematical state space, Hamiltonian or requested output?

The answer is represented by

- typed source grammar
- declared Hamiltonian and requested tensor
- structural certificate discovery
- admitted exact route
- component convolution
- orbit-sector counting
- separator branching
- rank-aware readout
- or H14F exact fallback
- independent reconstruction of the same tensor

The router never asks which route would make a preferred outcome appear. It asks which exact decomposition is already present in the declared object. If the certificate is missing, the object is not modified to create one.

I.16 Typed definitions and execution contract

I.16.1 Mathematical object versus execution profile

For a fixed Hamiltonian H, define the retained-role tensor

$$G(e,r) = \#\{\sigma : H(\sigma) = e,\ R(\sigma) = r\},$$

where r is one of the declared port-role states. Its scalar DOS is

$$g(e) = \sum_r G(e,r).$$

The mathematical object consists of H, the assignment domain, the role map R, and every requested coefficient $G(e,r)$. H14C and H14F specify how that object is evaluated. They may change elimination order, batching, residue storage, worker persistence and cache policy. They may not change H, delete assignments, reinterpret the roles or replace G with a smaller target.

I.16.2 H14C and H14F

H14C is the completed fourteen-core profile with complete-prime calibration and the accepted retained-capacity contract. H14F adds mixed-width compilation, persistent workers, reduced residue storage and an ephemeral scale-specific projection cache. Both are exact execution profiles.

Where an H14F campaign declares fourteen pinned physical workers, that width is part of the frozen architecture. It cannot silently become "whatever number of host workers is available," because batching, custody and timing receipts are indexed to the declared worker surface. A new worker mapping would be a successor execution profile, not a reinterpretation of the old one.

I.16.3 Structural certificate

A structural certificate is exact evidence that the requested object can be reconstructed from smaller typed objects. Examples include:

- disconnected components after an explicit conditioning boundary;
- a permutation orbit on which energy depends only on marked population;
- a cutset whose fixed assignments reduce admitted elimination width;
- a fee-exempt fiber whose multiplicity can be lifted exactly; and
- a constrained readout with a known independent rank.

The certificate must contain the reconstruction rule. A low-width-looking plot or favorable runtime is not enough.

I.16.4 Router and fallback

The routing policy is

- if conditional components close:
- evaluate each component exactly and convolve
- elif uniform orbit sectors close:
- count exact marked-population sectors
- elif a declared cutset lowers admitted width:
- branch exactly on every cutset assignment
- elif the output is a constrained fiber/readout:
- evaluate independent rank and reconstruct the ambient view
- else:
- execute the exact H14F variable-elimination fallback

Every branch returns the complete requested tensor. Failure to certify one branch is a routing outcome, not permission to approximate.

I.17 The complete H14F N100 result

I.17.1 Source object

The source-derived N100 object has

- **independent spins.** Declared value: 100.
- **source edges.** Declared value: 260.
- **induced width.** Declared value: 5.
- **retained source packet.** Declared value: $p = 1, g = 0$.
- **role order.** Declared value: plus, minus, anti-plus, anti-minus.
- **complete configurations.** Declared value: 2^{100}.

The exact configuration count is

$$2^{100} = 1{,}267{,}650{,}600{,}228{,}229{,}401{,}496{,}703{,}205{,}376.$$

H14F treats the complete 100-spin, 260-edge Hamiltonian as one retained-port variable-elimination object. It executes 2,048 singleton root tasks through four CRT primes.

I.17.2 Retained-role closure

Four binary retained roles give $2^4 = 16$ port rows. Each row sums to 2^{96}, because fixing the four retained role bits leaves 96 independent binary coordinates. Summing all rows gives

$$\sum_{r=0}^{15} \sum_e G(e,r) = 16 \cdot 2^{96} = 2^{100}.$$

The scalar result has 177 occupied energy bins and ground sector ((-260,256)): energy -260 with degeneracy 256. The conditional port rows organize into three spectra according to plus-spin count:

- **0.** Representative state: 0; Ground energy: -260; Ground count: 128; Bins: 174.
- **1.** Representative state: 1; Ground energy: -250; Ground count: 128; Bins: 174.
- **2.** Representative state: 3; Ground energy: -244; Ground count: 256; Bins: 171.
- **3.** Representative state: 7; Ground energy: -250; Ground count: 128; Bins: 174.
- **4.** Representative state: 15; Ground energy: -260; Ground count: 128; Bins: 174.

Thus counts 0/4 share one spectrum, 1/3 share a second and 2 supplies the third. All 24 source-role permutations and the global spin complement close exactly. This is a computationally typed conditional tensor, not an elementwise F81 physical-port assignment.

I.17.3 Execution receipt

The four prime-kernel walls were approximately

$$0.890328, 0.852483, 0.797375, 0.799622 \text{ s},$$

and the total runner wall was 3.828137395 seconds. The scalar DOS semantic hash is `6e404623175ba60629110fb17f75ca0f0398e690db9de7b7a97766d32d0c2d30` and the retained tensor semantic hash is `33ec2e3f5cd3a1e66eb892730181e89bb1aff71d93b97fbbba4959bb238bd78d`. These hashes become the equality target for the structure-first route.

I.18 Deriving the N100 structure-first work

I.18.1 Component census

The unchanged conditional grammar separates into eight exact components of sizes

$$15, 15, 15, 15, 15, 10, 10, 5.$$

The sizes sum to the original independent rank:

$$5 \cdot 15 + 2 \cdot 10 + 5 = 75 + 20 + 5 = 100.$$

For a component of n_a binary variables, direct local enumeration requires 2^{n_a} assignments. Therefore the total fresh component work is

$$W_{100} = 5 \cdot 2^{15} + 2 \cdot 2^{10} + 2^5.$$

Evaluating each term,

$$5 \cdot 32{,}768 = 163{,}840,$$

$$2 \cdot 1{,}024 = 2{,}048,$$

and

$$W_{100} = 163{,}840 + 2{,}048 + 32 = 165{,}920.$$

I.18.2 Why 165,920 does not replace 2^{100}

Let component a return its exact conditional polynomial tensor $P_a(z, r_a)$. The full tensor is reconstructed by summing compatible role assignments and convolving energies:

$$P_{100}(z, r) = \sum_{r_1 \oplus \cdots \oplus r_8 = r} \prod_{a=1}^{8} P_a(z, r_a).$$

At $z = 1$, each local polynomial counts all local assignments, so

$$P_{100}(1) = \prod_{a=1}^{8} 2^{n_a} = 2^{\sum_a n_a} = 2^{100}.$$

The 165,920 number counts fresh local assignments evaluated by the algorithm. The 2^{100} number counts formal global assignments represented by the reconstructed tensor. They answer different questions.

I.18.3 Same-output equality

The structure-first route completed in 0.500951441 seconds. Relative to the 3.828137395-second H14F route, the observed ratio is

$$\frac{3.828137395}{0.500951441} \approx 7.642.$$

This comparison is meaningful because the retained-role tensors are coefficientwise identical and have the same semantic hash. The route did not change the Hamiltonian, sample states, truncate bins or request only the ground sector.

I.18.4 Grammar-to-work identities

The work also admits exact source-grammar decompositions:

$$W_{100} = 162 \cdot 2^{10} + 2^5 = 2^5(72^2 + 1).$$

Indeed,

$$162 \cdot 1{,}024 + 32 = 165{,}888 + 32 = 165{,}920,$$

and $72^2 = 5{,}184 = 32 \cdot 162$. Therefore

$$\frac{W_{100} - 2^5}{2^{10}} = 162, \qquad 2^5 \cdot 162 = 72^2.$$

The residual 2^5 is the formal assignment space of the surviving five-role $p = 9, g = 1$ packet. It is not X1 and not a physical closure object. The identity interprets source grammar and workload; it does not claim that a ledger total was inserted as an energy coefficient.

I.19 Source-native controls and deviations

I.19.1 N105 restoration control

N100 excludes the historical $p = 9, g = 0$ witness packet. Restoring it produces N105 and changes the exact work to

$$W_{105} = 166{,}912 = 163 \cdot 2^{10}.$$

The difference is

$$\Delta W = W_{105} - W_{100} = 166{,}912 - 165{,}920 = 992 = 2^{10} - 2^5.$$

This is a source-native perturbation: it restores an identified packet rather than adding an arbitrary edge until timing changes. It also changes the N100/N81 completion-path capacity from 40 to 45 rows.

I.19.2 N81 octahedral control

The N81 control has component sizes

$$[11, 10, 11, 10, 10, 11, 10, 8]$$

and local work

$$W_{81} = 3 \cdot 2^{11} + 4 \cdot 2^{10} + 2^8$$
$$= 6{,}144 + 4{,}096 + 256$$
$$= 10{,}496 = 2^7(81+1).$$

Its octahedral bridge retains 48 cube embeddings in one full-octahedral orbit, the completion path (1-8-9-12), reflected C4 gluing and 49,152 incidences. This is a geometry/compiler control. Its possible causal relation to runtime remains a research question rather than an installed explanation.

I.19.3 N72 dense contrast

N72 remains dense. It is not edited to reproduce N100's component structure. This is the correct contrast because it asks whether the router recognizes the actual source object rather than imposing one preferred decomposition on every input.

I.19.4 Mixed-connectivity controls

The installation tournament adds two controlled deviations:

- a one-edge partial join, width 5, ground sector ((-261,128)); and
- a connected ring, width 5, ground sector ((-268,2)).

Both complete sixteen-row tensors equal an independent original-component bridge contraction coefficient for coefficient. The partial join retains 128 component-orientation choices; the connected cyclic skeleton leaves the two global orientations. These tests show that a changed coupling topology can be handled exactly when its reconstruction rule is explicit.

I.19.5 Dense N96 fallback

The full N96 graph remains width 24. A frozen five-site cutset creates

$$2^5 = 32$$

exact branches, each admitted to H14F at width 22. This is neither a component convolution nor an approximate shortcut. Every cutset assignment is retained, and H14F performs the exact branch work. The regression ran 4,096 fresh tasks and retained the frozen N96 numerical authority.

I.20 Failure, correction and routing discipline

The principal wrong route would be to treat N100's favorable graph as the universal computational model. That route is rejected in stages:

- wrong control: substitute the N100 component graph for another grammar
- source identity is lost
- correction: inspect the new object's own component/orbit/cutset/fiber/rank certificates
- choose only an admitted exact route
- retest: reconstruct the complete tensor independently
- compare coefficientwise and replay dense regressions
- fallback: if no certificate closes, retain H14F

The installation gates execute this logic rather than merely state it. They record 12/12 route checks, 18/18 mixed arithmetic checks, 9/9 N18/N54 checks, 16/16 N72 checks and 20/20 N96 checks: 75/75 combined executed checks. Their admission was for a separate owner-authorized promotion decision; the gate campaign itself did not silently rewrite a current pointer.

A second deviation is the source-native N100-to-F81 compiler gap. N100 has 100/100 typed source keys and Shared72 has aggregate connector incidence for 72/100 rows, while CR211 exposes 162 address rows but zero elementwise QP assignments. The missing elementwise join cannot be replaced by aggregate counts or external labels. `G:G1@SAM-RESEARCH` preserves that partial-domain boundary in `SAMA-D000034`; it is not a direct H14F result key for this chapter.

I.21 Recorded authority and delegated-base status

Historical H14F, structure-first, Exact Write, W9P/Theta, support/Gram, tensor-orbit and bounded application artifacts remain frozen within the `SLCV21R` arithmetic/application lineage. Recorded authority is

- SLCQ2-RZ
- source-recorded default hybrid ordinal selection engine
- 70 exact source-visible features
- delegates byte-identical arithmetic/application services
- frozen SLCV21R base
- exact H14F fallback and structure-first routing lineage

`SLCV21R` is noncurrent as a revision, but its declared services remain the delegated base. Q1 is the immutable fallback. Q2 does not install physical calibration, Binding, ATOM3D or RH merely by consuming base arithmetic. H14F therefore remains load-bearing without becoming the current selection engine.

I.22 Established result, boundary and forward handoff

The complete N100 result counts all 2^{100} assignments of the unchanged 100-spin, 260-edge Hamiltonian and returns the full exact retained-role tensor. The structure-first route evaluates 165,920 local assignments and reconstructs that same tensor coefficientwise. The execution-work reduction is established for this source grammar; it is not asserted for every arbitrary 100-spin graph.

The router's general result is procedural and exact: detect the source certificate, preserve the requested object, reconstruct independently, and retain H14F when no reduced route closes. N105, N81, N72, mixed-graph and N96 controls define where each branch applies.

The forward handoff goes to `SAMA-D000041`, where Starbreaker's own grammar is routed without substituting the N100 graph. It also supports the current-Q2 ordinal chapter `SAMA-D000055`, where exact base computation and source-visible selection must remain different types.

I.23 Sources and evidential scope

This chapter is assembled from the following fixed source manuscripts. Source-reported campaign outcomes retain their original scope; no new campaign was executed for this edition.

- N100 computational grammar.
- H14f grammar to work factorization.

SAM VOLUME III

SAM Language and Exact Program Semantics

Chapter II

A favorable graph decomposition is useful only if the program and its inputs retain their meanings. The formal language places source identity, operator type, state and history into one execution contract. Its real finite kernel is a substantial result in its own right; its explicit exclusions are part of the specification.

The account below follows the chapter source as retained at the fixed 8 September 2026 SAMA revision. Engine names, installed interfaces and campaign status are historical source statements. They do not describe a freshly inspected October deployment. Later source amendments are distinguished where available; no later engine or campaign is inferred.

II.1 Overview

SAM Language is the semantic boundary between a mathematical grammar and an execution. It owns the parser, resolver, checker, type system, operator definitions, program registry and evaluator. A UI, wrapper or orchestration layer may transport a program, but it cannot acquire authority to recognize a phrase and run hidden scientific logic. Exact program meaning belongs to one declared language path.

That path carries more than a final state. It carries immutable source and program custody, the typed current state, a separately typed accumulated history, deterministic receipts and predecessor hashes. Two programs may end at the same exact state while retaining different route histories. A rejected operation must be atomic, and a tampered receipt must fail custody rather than be silently reinterpreted.

The evidence has three time layers that must remain distinct. SAM Language v0.6 is the preserved formal-simulator campaign whose G01–G16 records test the semantic and custody surfaces in detail. SAM Language v0.7 is the later typed 12-site reflective-tape candidate. Neither is the SLC version or the current SLC release. The live repository authority is `SLCQ2-RZ`; historical language and SLC labels remain source lineage, not current-pointer aliases.

II.2 Opening question and conceptual picture

The governing question is:

> How does a source grammar become one exact, reproducible computation without allowing interfaces, profiles, mutable tapes or hidden dispatch paths to change its scientific meaning?

The required architecture is

- source text or immutable instruction tape
- parser
- resolver
- checker and type system
- operator/program registry
- evaluator
- typed current state + typed history + custody receipt

The forbidden alternative is semantic branching outside the language:

- UI or wrapper recognizes a phrase
- runs an unrelated calculation
- returns a value as if the language produced it

The first architecture makes source, meaning and result traceable. The second creates two semantic authorities and breaks provenance even if the returned number happens to match.

II.3 Version and authority firewall

II.3.1 v0.6: preserved formal-simulator evidence

The v0.6 campaign supplies the focused evidence in this document. It tests one semantic path, formal-profile isolation, the complete 12-site basis, frozen B and X1 operators, ordered-pair schedules, native decompositions, state/history separation, deterministic interfaces, atomic rejection, packaging and 23 wrong controls.

Its source statuses are `PASS` across G01–G16. Those statuses belong to the sealed v0.6 test surface. They are not an authorized SAMA result classification and do not make v0.6 the current language or SLC revision.

II.3.2 v0.7: typed reflective-tape candidate

The Volume III source spine records v0.7 as a later typed, source-linked reflective SLC C1 tape candidate over twelve sites and 4,096 basis addresses. Its first-class types include

- SLCInstructionTape
- SLCTape12
- SLCTapeInspection

Six declared operations create an immutable zero-origin tape, append B/request/response records, execute it and inspect custody. The executor validates the zero-origin hash chain, dispatches through inherited exact state transitions and returns the same `SLCState12` dossier as the corresponding direct program.

The candidate remains bounded to a straight-line 12-site tape. It has no self-hosted parser, recursive source generation or general SAM-Language-to-Dense-Exact adapter. Tape custody and executed state remain different types.

II.3.3 Source-recorded SLC: `SLCQ2-RZ`

The immutable Volume III spine contains historical SLC-current wording from its creation time. The live authority now

controls: `SLCQ2-RZ` is the sole frozen/default SLC, with its delegated arithmetic/application base separately typed. SAM Language, the SAM UI and SLC each have their own version lineage.

Accordingly:

- SAM Language v0.6 = preserved formal-simulator evidence
- SAM Language v0.7 = later typed reflective-tape candidate
- SLCQ2-RZ = sole source-recorded default SLC authority

No row in this document promotes one identity into another.

II.4 Definitions and type boundaries

II.4.1 Source, tape, state and history

- **source text.** Mutability: fixed for a declared execution; Semantic role: human-readable program origin.
- **instruction tape.** Mutability: immutable append/custody object; Semantic role: ordered typed program records and predecessor chain.
- **current state.** Mutability: transformed by accepted operators; Semantic role: exact endpoint dossier.
- **accumulated history.** Mutability: append-only typed receipt; Semantic role: ordered route, including distinctions erased at the endpoint.
- **inspection.** Mutability: read-only; Semantic role: custody and semantic trace readback.

Endpoint equality means

$$s_{final}^{(1)} = s_{final}^{(2)}.$$

It does not imply

$$h^{(1)} = h^{(2)}.$$

That distinction later supports Complete Exact Write and reciprocal history.

II.4.2 The twelve-site address domain

Let a basis state be a twelve-bit vector

$$b = (b_0, \ldots, b_{11}), \qquad b_i \in \{0,1\}.$$

Its canonical integer address is

$$a(b) = \sum_{i=0}^{11} b_i 2^i,$$

so

$$0 \le a < 2^{12} = 4{,}096.$$

The finite address surface permits exhaustive basis, arity, range and site-ID checks.

II.4.3 Operators and profiles

An operator is resolved by identifier through the registry, checked against a typed signature, and executed only after the profile admits it. The formal evidence profile is `FORMAL_CANDIDATE`; normal and research profiles are rejected in G03. This prevents an application mode from injecting semantics into a formal campaign.

II.4.4 Exact receipt

A receipt binds at least the declared program, predecessor lineage, exact state, history and operator trace. The relevant invariants are byte-stable serialization, deterministic ordering and exact coefficients. Recomputing a hash for a mutated object does not authorize it; predecessor custody is part of the chain.

II.5 From source custody to one semantic path

II.5.1 G01 — seal the parent and candidate delta

Before executing semantics, G01 inventories:

- parent files 92
- frozen-source files 150
- candidate executable files 97

Every declared hash matches, no mismatch is reported and the parent remains unchanged. The candidate delta is therefore distinguishable from inherited source. Filename equality or an unsealed working tree would not provide the same custody.

II.5.2 G02 — traverse the only authority path

G02 checks that parser, checker, evaluator, type system and registry exports resolve to one authority. A representative mixed program emits the trace

- RESOLVE
- `QP_ORDERED_PAIR`
- `SLC_ZERO_REGISTER`
- `SLC_PREPARE_REQUEST`

Operator digits resolve through the registry. No legacy sidecar or source-text dispatch is present. The trace matters because a matching endpoint produced through a hidden path would not be the same semantic result.

II.5.3 G03 — isolate the formal profile

The same path is then invoked under three profiles. `FORMAL_CANDIDATE` is accepted; normal and research profiles are rejected. Thus the formal evidence does not borrow undeclared application or exploratory behavior.

II.6 Build and exhaust the formal state surface

II.6.1 G04 — complete basis construction

G04 enumerates all `2^12=4,096` addresses. It verifies the exact dimension, site-ID order and binary range, then rejects

wrong arity, out-of-range values, wrong register size and unknown/reordered site identifiers.

This establishes a finite formal address space. It does not identify the twelve sites with physical hardware or assign coupling magnitudes.

II.6.2 G05 — exact canonical-state invariant

The evaluator returns an exact sparse state. G05 checks:

- canonical sparse-key order;
- integer coefficients;
- the engine contract hash; and
- exact norm.

Reordering terms before hashing, converting coefficients to floats or using an undeclared norm would change the contract even if a display looked similar.

II.6.3 G06 — freeze B request and X1 response

G06 installs B prepare/request and X1 response as differently typed operators. For every declared basis column it verifies frozen hashes, matrix shapes, prepare columns, response mapping and involution. B contains a binary flip but is not identical to the flip; request and response cannot be exchanged.

This is the formal premise used by D39. It defines executable operators inside the candidate without assigning a physical B-to-X1 causal weld.

II.7 Complete reciprocal and decomposition schedules

II.7.1 G07 — ordered-pair lift

Twelve sites produce

$$12 \cdot 11 = 132$$

ordered non-self placements. Each placement acts on all 4,096 basis states in both directions:

$$132 \cdot 4{,}096 = 540{,}672$$

forward cases and the same number in reverse. Therefore

$$2 \cdot 132 \cdot 4{,}096 = 1{,}081{,}344.$$

G07 closes all 1,081,344 classicalized schedule cases. Using unordered pairs, one direction or a sample would be a different schedule.

II.7.2 G08 — request sign and gauge diagnostics

The four frozen real request candidates `REQ02`, `REQ08`, `REQ09` and `REQ15` are each tested on all 132 placements:

$$4 \cdot 132 = 528.$$

All 528 placement cases pass, leaving four surviving gauges under the installed positive-first-row convention. The diagnostic therefore preserves ambiguity among the four real gauges instead of selecting one from a desired downstream result.

II.7.3 G09 — every declared native decomposition

The grammar generates five factor-shape families with exact assignment counts:

- **`1`×`12`.** Assignments: 1.
- **`3`×`4`.** Assignments: 5,775.
- **`4`×`3`.** Assignments: 15,400.
- **`6`×`2`.** Assignments: 10,395.
- **`12`×`1`.** Assignments: 1.

Their sum is

$$1 + 5{,}775 + 15{,}400 + 10{,}395 + 1 = 31{,}572.$$

All 66 rooted-forest parent cases also pass, and the two induction modes agree. The decomposition is native to the declared grammar rather than an external factorization imported after execution.

II.7.4 G10 — peel and restore the extra edge

G10 executes 204 exact controls:

- 1×12 : 132 cases
- 3×4 : 36 cases
- 4×3 : 24 cases
- 6×2 : 12 cases
- total: 204 cases

For each case, the declared extra edge is peeled, the exposed native decomposition is evaluated, the same edge is restored, and the original terminal state returns exactly. Restoring a different edge or accepting graph isomorphism without exact state equality would weaken the control.

II.7.5 G11 — frozen classical-limit boundary

G11 checks all eight forward/reverse two-bit rows in the frozen classical limit. Every row matches. The result is deliberately bounded: a complete classical table is not evidence that complex-phase or general quantum semantics have been installed.

II.8 State, history and interface custody

II.8.1 G12 — same endpoint, different route

Chain and star programs produce the same exact coefficients and the same terminal-state hash, but their history hashes differ. In symbols,

$$s_{chain} = s_{star}, \qquad \mathrm{hash}(h_{chain}) \neq \mathrm{hash}(h_{star}).$$

The component partition is not used to erase the route. This is the formal reason an endpoint-returning computation can retain a nontrivial receipt.

II.8.2 G13 — deterministic API and CLI receipts

The same declared program is executed repeatedly through API and CLI. Both interfaces are byte-deterministic, their receipts match each other, the CLI exits successfully and no traceback occurs. Interface adapters therefore do not introduce independent semantics or serialization defaults.

II.8.3 G14 — atomic rejection and tamper detection

G14 mutates coefficient, exponent, operator argument, predecessor hash and receipt content. Every mutation is detected. Invalid operations reject atomically, leaving no accepted prefix or partially changed state.

This is a semantic property as well as a custody property: malformed input does not acquire meaning merely because its first tokens were valid.

II.8.4 G15 — inherited and new replay on a clean wheel

The candidate suite contains

$$154 \text{ inherited} + 29 \text{ new} = 183 \text{ tests.}$$

All 183 pass. The clean wheel contains 37 members, and 13 CLI cases pass with API/CLI parity, no sidecar or UI dependency, no repository-path dependency and no traceback. This retest establishes that the candidate package preserves the parent while carrying the new semantics outside the source checkout.

II.8.5 G16 — keep the wrong controls and open boundaries

All 23 declared wrong controls produce their diagnostic outcomes. The release also retains five open domains:

1. complex-phase semantics;
2. measurement and publication semantics;
3. physical connectivity;
4. coupling magnitude and cost; and
5. hardware realization.

A passing formal candidate remains scoped because these boundaries remain in the artifact rather than being erased after the regression run.

II.9 The v0.7 reflective-tape semantics

The v0.7 candidate makes the instruction sequence itself typed data. A typical execution can be understood as:

1. create a zero-origin `SLCTape12` and record its origin hash;
2. append a typed B/request record whose predecessor is the current tape tip;
3. append a typed X1/response record under the same custody law;
4. seal the straight-line tape;
5. validate the complete predecessor chain;
6. dispatch each record through the inherited exact transition;
7. return `SLCState12` and a separate `SLCTapeInspection`.

If the direct program and tape program contain the same declared operations, their current-state dossiers match. Their custody still arrives through different source objects, so the tape is not replaced by the endpoint.

This reflective layer prepares exact program composition and history without making the language self-hosting. Recursive source generation, a general parser inside the tape and a complete typed-IR-to-Dense-Exact adapter remain open.

II.10 Source-native N100-to-F81 compiler boundary

G1 asks a different question from G01–G16: whether the exact language and N100 source rows already carry a complete elementwise semantic map to F81. The audit finds:

- 100/100 typed QP source keys
- 72/100 Shared72 aggregate connector-incidence rows
- 162 CR211 address rows
- 0 CR211 QP elementwise assignments
- 81 F81 semantic sites
- 10 F81 contact classes
- 0 N100 rows with a complete elementwise F81 map

The missing relation is

- QP candidate/template
- F81 semantic site or source-native F81 address class.

No outcome or residual selects a map. The contact lane is not executed and the status remains `PARTIAL_DOMAIN__ELEMENTWISE_F81_JOIN_OPEN`.

The boundary shows why formal determinism and semantic completeness are different. The language can parse, type, execute and receipt a program while a particular source-to-target scientific join remains absent.

II.11 Controls, deviations and promotion discipline

The full chain prevents several distinct deviations:

- **parent source changes under the candidate.** Detecting evidence: G01 hashes and inventories; Preserved correction or boundary: parent remains sealed; candidate delta stays explicit.
- **hidden semantic dispatch.** Detecting evidence: G02 path and trace; Preserved correction or boundary: one parser/checker/evaluator/registry path.
- **profile leakage.** Detecting evidence: G03; Preserved correction or boundary: formal candidate accepts only its declared profile.
- **malformed address gains meaning.** Detecting evidence: G04; Preserved correction or boundary: range, arity, size and ID rejection.
- **request and response collapse.** Detecting evidence: G06; Preserved correction or boundary: frozen hashes, shapes and typed mappings.

- **sampled schedule called complete.** Detecting evidence: G07; Preserved correction or boundary: all 1,081,344 cases executed.
- **preferred gauge selected after outcome.** Detecting evidence: G08; Preserved correction or boundary: all four survivors preserved.
- **endpoint equality erases history.** Detecting evidence: G12; Preserved correction or boundary: equal state hash, unequal history hash.
- **interface changes semantics.** Detecting evidence: G13; Preserved correction or boundary: byte-identical API/CLI receipts.
- **partial mutation survives rejection.** Detecting evidence: G14; Preserved correction or boundary: atomic rollback and tamper detection.
- **source-tree dependency hidden by tests.** Detecting evidence: G15; Preserved correction or boundary: clean-wheel replay.
- **open physical domains disappear after PASS.** Detecting evidence: G16; Preserved correction or boundary: 23 wrong controls and five boundaries retained.
- **aggregate incidence becomes elementwise mapping.** Detecting evidence: G1; Preserved correction or boundary: zero assignments and unexecuted contact lane preserved.

Passing validation is not promotion. v0.6 stays a preserved formal-simulator surface, v0.7 stays a reflective-tape candidate, and SLC Recorded authority is resolved through the live `SLCQ2-RZ` pointer rather than a historical filename.

II.12 Established result and current boundaries

The source-bound language result is:

1. one declared parser/resolver/checker/type/registry/evaluator path owns exact program semantics;
2. twelve sites give a complete 4,096-address finite basis;
3. B request and X1 response are frozen, differently typed operators;
4. all 1,081,344 ordered-pair schedule cases and all 31,572 declared native decompositions close under their respective tests;
5. current state, immutable source tape and accumulated history remain separate;
6. API and CLI produce deterministic identical receipts;
7. malformed operations reject atomically and declared tampering is detected;
8. 154 inherited plus 29 new tests pass in the clean candidate wheel;
9. 23 wrong controls pass while complex phase, measurement/publication, physical connectivity, coupling/cost and hardware realization stay open;
10. v0.7 supplies a later typed reflective tape but not general Dense-Exact integration; and
11. the N100-to-F81 elementwise semantic join remains open and unexecuted.

The focused G artifacts carry source status `PASS` or the explicit partial G1 status. The registered atomic records assign no authorized project result classification to D35, so this revision does not create one.

II.13 Connections and forward handoff

- **`SAMA-D000034.`** Connection: Supplies the N100 source grammar and exact structure-first certificates that language must serialize..
- **`SAMA-D000038.`** Connection: Receives immutable programs, signed `Z4^9` operations, separate state/history and exact readback for Complete Exact Write..
- **`SAMA-D000039.`** Connection: Uses the typed B request, X1 response and W9 resolution route..
- **`SAMA-D000042.`** Connection: Receives the separate history receipt for directional event-reel computation..
- **`SAMA-P000006.`** Connection: Uses the language as the computation interface for particle grammar, Starbreaker and binding..

The forward handoff is to Complete Exact Write: compile a declared signed program to its exact group action, apply it without mutating the tape, and retain a reconstructible history even when the endpoint returns.

II.14 Sources and evidential scope

This chapter is assembled from the following fixed source manuscripts. Source-reported campaign outcomes retain their original scope; no new campaign was executed for this edition.

- Sam language and exact program semantics.

SAM VOLUME III

Exact State Spaces and Thermodynamics

Chapter III

This chapter moves from one graph to the family of exact methods that may evaluate a declared state-space object. It then follows the resulting density of states into dimensionless thermodynamics. The distinction between an exact numerical object and a physically calibrated energy is maintained throughout.

The account below follows the chapter source as retained at the fixed 8 September 2026 SAMA revision. Engine names, installed interfaces and campaign status are historical source statements. They do not describe a freshly inspected October deployment. Later source amendments are distinguished where available; no later engine or campaign is inferred.

III.1 Overview

An exact state-space computation answers a stronger question than an optimizer: not merely which assignment has the lowest energy, but how many assignments occupy every possible energy. For a declared binary Hamiltonian this complete answer is the density of states, or DOS. Its coefficients are integers, their sum is the complete formal configuration count, and their arrangement retains the energy organization created by the graph, couplings and fields.

The computation can be exact without visiting every assignment one by one. Gray enumeration, variable elimination, component convolution, modular polynomial arithmetic, orbit sectors and rank-aware readout are different routes to the same requested object. Their right to be called exact comes from reconstruction and equality obligations: coefficientwise agreement, state- count closure, moments, symmetries, port marginals, hashes and independent readers. Runtime is useful evidence about an execution route; it is not the definition of exactness.

This distinction has a second consequence. Exact mathematics and executable custody are separate obligations. SLCX023 retained exact exhaustive and NTT/variable-elimination authorities but failed its frozen same-run custody gates. That failure remains part of the record. SLCX023A corrected the custody mechanism in a separate successor, reproduced the unchanged coefficient authorities and left the failed parent immutable. The chain demonstrates the Courtroom method at its best: retain what failed, identify the type of failure, change only the defective layer, replay the original question and preserve both receipts.

The chapter ends by deriving exact thermodynamic readers from the frozen N96 DOS. Those readers are exact and dimensionless while the physical energy scale remains symbolic. A polynomial can determine partition functions and moments without thereby choosing joules, kelvin, a contact orientation or a physical binding map.

III.2 Opening question and conceptual picture

The governing question is:

> How can a finite computational system account for every formal assignment, preserve the exact integer answer under different execution routes, and carry that answer into thermodynamics without importing an undeclared physical scale?

The full chain is

- declared typed grammar and Hamiltonian
- exact integer DOS g(E)
- independent route equality and state-count closure
- custody-qualified frozen artifact
- exact polynomial P(x)
- exact partition, moment and response readers
- symbolic-scale physical boundary

Every arrow changes the representation or the question, not the source object. A component factorization may reduce work but may not reduce the state space being counted. CRT may store residues but must reconstruct the original integers. A thermodynamic reader may turn coefficients into rational functions or numerical display values but may not retroactively change the DOS. A physical unit conversion may occur only after a source-authorized scale exists.

III.3 Typed definitions and domains

III.3.1 Source grammar, assignment and Hamiltonian

Let $V = \{1, \ldots, N\}$ be the declared independent binary sites, let $E_G \subseteq V \times V$ be the declared interaction edges, and let

$$\sigma = (\sigma_1, \ldots, \sigma_N) \in \{-1, +1\}^N.$$

For integer edge weights J_{ij} and integer local fields h_i, the Ising-type Hamiltonian used here is

$$H(\sigma) = \sum_{(i,j)\in E_G} J_{ij}\sigma_i\sigma_j + \sum_{i\in V} h_i\sigma_i.$$

The types are load-bearing:

- N counts independent binary coordinates;
- E_G is topology, not an energy histogram;
- J_{ij} and h_i are declared Hamiltonian coefficients;
- a native grammar-row value does not become J_{ij} merely because it is numerical; and
- a constrained occurrence ledger is not automatically an independent N-spin state space.

This is why the N100 compiler keeps source-row selection separate from Hamiltonian coefficients and why later M126/N144/L162 readouts retain their rank and sidecar types.

III.3.2 Density of states

For each attainable integer energy e, define

$$g(e) = \#\{\sigma \in \{-1,+1\}^N : H(\sigma) = e\}.$$

The function g is an exact integer ledger. Summing over all energies partitions the complete configuration space:

$$\sum_e g(e) = \sum_e \sum_\sigma \mathbf{1}[H(\sigma) = e] = \sum_\sigma 1 = 2^N.$$

This derivation is elementary but decisive. Every assignment has exactly one energy, so it contributes once. Frustration changes which energies occur and how assignments are distributed among them. It does not change the cardinality 2^N of the formal assignment domain.

A ground-state solver keeps only

$$e_0 = \min_\sigma H(\sigma), \qquad g(e_0),$$

whereas a DOS retains every pair ((e,g(e))). Interior bins, odd moments, response sectors and finite-temperature behavior would be lost if the output were reduced to an optimum.

III.3.3 Exactness, execution and custody

Three types must not be collapsed:

- **Mathematical exactness.** Question answered: Are the requested coefficients the exact integers?; Required evidence: equality, closure, moments, symmetries, reconstruction.
- **Execution policy.** Question answered: How was the exact object evaluated?; Required evidence: route declaration, worker/batch receipts, performance data.
- **Artifact custody.** Question answered: Did the sealed code, inputs, receipts and outputs retain their declared identities?; Required evidence: hashes, domains, manifests, independent verification.

An exact tensor with a failed receipt is not a successful custody packet. A fast route with no equality proof is not established as exact. A hash-correct artifact does not become the right mathematical object unless the declared calculation also closes.

III.4 Deriving the exact route family

III.4.1 Gray exhaustive authority

The direct definition suggests enumerating all 2^N assignments. A Gray code orders them so adjacent assignments differ in one spin. If spin k changes sign, only terms incident to k change, giving the incremental update

$$\Delta H_k = -2\sigma_k \left(h_k + \sum_{j:(k,j)\in E_G} J_{kj}\sigma_j \right).$$

Thus one full energy need not be recomputed from scratch at every state. The route is still exponential in the number of visited assignments, but for small N it is transparent and supplies an independent authority against more structured routes. In SLCX023, `GRAY_EXHAUSTIVE` covers the complete address space through $N = 24$.

III.4.2 Variable elimination

Write a local factor for every edge and field contribution. Eliminating a spin means summing its two values while retaining an exact table over the still-live neighboring boundary. If B_k is that boundary and z marks energy, a schematic elimination step is

$$F_{k+1}(\sigma_{B_k}; z) = \sum_{\sigma_k=\pm 1} z^{\Delta H_k(\sigma_k,\sigma_{B_k})} F_k(\sigma_k, \sigma_{B_k}; z).$$

The coefficients remain integers. Cost follows the largest induced boundary width w, roughly through tables of size 2^w, rather than nominal arity alone. Different elimination orders therefore change work while leaving the Hamiltonian and the final polynomial fixed.

SLCX023 deliberately used two distinct exact versions: a reference NTT/VE route with a frozen min-degree order and a typed NTT/VE route with an independently frozen cell-aware min-fill order. Their boundary tables were retained so equality could be checked below the final scalar output.

III.4.3 Exact component convolution

Suppose the declared graph splits into independent components $G_1, \ldots, G_m$ after conditioning on an explicit boundary. If component a has DOS polynomial

$$P_a(z) = \sum_e g_a(e) z^e,$$

then additivity of energy gives

$$P_G(z) = \prod_{a=1}^{m} P_a(z),$$

and therefore

$$g_G(e) = \sum_{e_1+\cdots+e_m=e} \prod_{a=1}^{m} g_a(e_a).$$

This is not a sample or truncation. Every combination of component assignments appears once in the convolution. At $z = 1$, closure is inherited:

$$P_G(1) = \prod_a P_a(1) = \prod_a 2^{N_a} = 2^{\sum_a N_a}.$$

The N100 structure-first calculation in `SAMA-D000037` uses this exact identity. The same logic also explains why a reduced work count can still return a complete 2^N census.

III.4.4 NTT lanes and CRT reconstruction

Large integer polynomial products can be evaluated modulo primes. For a suitable prime p_ell, an NTT transforms coefficients, multiplies pointwise and applies the inverse transform, all in exact modular arithmetic. One lane returns

$$g(e) \bmod p_\ell.$$

With pairwise-coprime primes $p_1, \ldots, p_L$, CRT returns the unique integer coefficient modulo

$$M = \prod_{\ell=1}^{L} p_\ell.$$

If the declared coefficient bound is less than M (or less than $M/2$ for a signed reconstruction), the original integer is unique. The modular lanes are therefore an exact representation of the coefficient only when the prime product, reconstruction convention and bound all close. A displayed floating-point polynomial is downstream and cannot replace these obligations.

III.4.5 Orbit sectors and rank-aware readout

When a dense graph is uniform under a declared permutation action, assignments can be grouped by marked population. A sector with k marked sites has multiplicity

$$\binom{N}{k},$$

provided the Hamiltonian is constant on that orbit or the remaining typed coordinates are retained. This replaces assignment traversal with an exact orbit census.

Likewise, an ambient readout with m slots may have only $r < m$ independent coordinates. The exact route evaluates the 2^r independent domain and applies the declared readout map. It must not report the ambient 2^m ceiling as independent state count. This distinction becomes critical for N144 independent state and L162 constrained occurrence views.

III.4.6 Route-acceptance checklist

Two routes may differ internally and still compute the same object. Their equivalence is established through a layered checklist:

1. identical occupied-energy support;
2. identical integer coefficient in every bin;
3. $sum_e g(e) = 2^N$;
4. matching exact raw moments $\sum_e e^k g(e)$;
5. declared parity, complement and role-permutation symmetries;
6. matching retained-port marginals and conditional tensors;
7. stable semantic hashes; and
8. an independent reconstruction or reader.

A runtime comparison is interpreted only after these checks establish same-object, same-output work.

III.5 Deviation chain: SLCX023 to SLCX023A

III.5.1 The frozen question

SLCX023 asked whether a frozen exact-DOS development ladder could close across six instances of sizes (12,24,36) using independent Gray, reference NTT/VE and typed NTT/VE authorities while also preserving execution and receipt custody. The equality standard was exact: same occupied energies and same integer coefficient in every bin.

The campaign also contained eight deliberately wrong controls and eighteen metamorphic checks. Wrong controls were not noise to be deleted. They were expected-invalid transformations used to demonstrate that the validation surface could distinguish the declared computation from plausible but wrong alternatives.

III.5.2 Preserved failure

The run completed all six instances, detected all eight wrong controls and passed all eighteen metamorphic checks. Yet its frozen verdict remained

- SLCX023_FAIL__EXACT_DOS_MISMATCH_OR_CUSTODY

because the first coefficient, custody, invariant or execution failure was a stop condition and same-run repair was forbidden. The historical G02 and G11 custody gates failed. This typed outcome matters: the available exact lane structure did not authorize rewriting a failed executable receipt as a pass.

The parent preserved its precommit, executable and execution-ledger hashes and recorded `same_run_repair\=\false`. The failed packet therefore remained reconstructible.

III.5.3 Correction hypothesis

The appeal did not change the scientific question, Hamiltonians, route definitions or coefficient authorities. It isolated two implementation and custody mechanisms:

- execute the official retest from an isolated retained capsule whose runner, adapter and engine bytes are checked before and after work; and
- distinguish the external lane domain of legacy receipts from the boundary-table schema recorded by corrected receipts.

This is a narrow correction. It predicts that the unchanged scientific outputs will reproduce while the custody gates now close.

III.5.4 Separate retest and cumulative record

SLCX023A executed as a successor rather than modifying SLCX023. It reproduced all 16 of 16 lane-instance DOS hashes, passed 154 of 154 exactness checks, 64 of 64 moment checks and 18 of 18 metamorphic controls. Twelve of those

metamorphic controls reran transformed solvers and six diagnosed cell repartitioning. All 432 corrected receipts verified with their domain and boundary-table schema separated and with signed links back to the legacy hashes.

The complete chain is therefore

- SLCX023 frozen design
- exact lanes execute
- custody G02/G11 do not close
- parent remains FAIL and immutable
- SLCX023A precommits two custody corrections
- unchanged 16/16 scientific hashes reproduce
- exactness, moment, metamorphic and 432 receipt checks pass
- corrected successor supplements the parent

No hidden $N = 48$ or $N = 60$ instance was opened. The appeal did not install a SAM Language operator or create a physical Ising/gravity claim.

III.6 Frozen N96 worked example

III.6.1 Complete configuration count

For $N = 96$,

$$2^{96} = 79{,}228{,}162{,}514{,}264{,}337{,}593{,}543{,}950{,}336.$$

The frozen H14C computation completed four CRT primes and 1,024 root-batch checkpoints. It retained sixteen port rows, each summing to 2^{92}, hence

$$16 \cdot 2^{92} = 2^4 \cdot 2^{92} = 2^{96}.$$

The scalar DOS occupies 348 energy bins from -346 through 348, with ground-state degeneracy 12. State-count, support, parity and raw-moment checks one through four passed. Both active 512×512 fiber factors remained installed and all four scalar marginals equal the exact N96 result.

The execution completed in 40,307.938817 seconds with the frozen DOS semantic hash `d74fb93facbebf18fa3cfe3ab2d33a0ec013f27b7ab6f326414bd5cdce4a11cb`. The later H14F profile reproduced that DOS hash under a different execution policy. That comparison concerns work routing; it does not replace the N96 Hamiltonian or create a second physical model.

III.6.2 Shifted exact polynomial

The energy spacing is two, so write

$$E_k = -346 + 2k, \qquad 0 \le k \le 347,$$

and define

$$P_{96}(x) = \sum_{k=0}^{347} g(E_k) x^k.$$

Then

$$P_{96}(1) = \sum_k g(E_k) = 2^{96},$$

with $a_0 = 12$ and $a_{347} = 7$. The degree records the complete occupied energy span after shifting; its coefficients remain the exact degeneracies.

III.7 From DOS to exact thermodynamics

III.7.1 Partition function derivation

Let the dimensionless inverse scale be $b = \beta\varepsilon_E$, where $epsilon_E$ remains symbolic, and set

$$x = e^{-2b}.$$

The partition function is

$$Z(b) = \sum_k g(E_k) e^{-bE_k} = \sum_k g(E_k) e^{-b(-346+2k)}.$$

Since $e^{346b} = x^{-173}$ and $e^{-2bk} = x^k$,

$$Z_{96}(x) = x^{-173} P_{96}(x).$$

This factorization separates the exact coefficient ledger from the chosen thermodynamic reader.

III.7.2 Euler derivatives and moments

Let

$$\Theta = x\frac{d}{dx}.$$

Then

$$\Theta P = \sum_k k g(E_k) x^k, \qquad \Theta^2 P = \sum_k k^2 g(E_k) x^k.$$

The mean shifted index is $\langle k \rangle = \Theta P / P$. Because $E = -346 + 2k$,

$$\langle E \rangle = -346 + 2\frac{\Theta P}{P},$$

and

$$\mathrm{Var}(E) = 4\left(\frac{\Theta^2 P}{P} - \left(\frac{\Theta P}{P}\right)^2\right).$$

Higher raw moments follow by expanding $(-346+2k)^m$ and replacing k^j with $\Theta^j P / P$. At $x = 1$, T1 records

$$\langle E \rangle = 0, \quad \langle E^2 \rangle = 1286, \quad \langle E^3 \rangle = 534, \quad \langle E^4 \rangle = 4{,}952{,}792.$$

The nonzero third moment retains an exact asymmetry hidden by the zero mean. The microcanonical mode is $E = 0$ with

$$g(0) = 1{,}761{,}659{,}846{,}545{,}153{,}896{,}312{,}195{,}809.$$

At the positive low-temperature end, the reader approaches $E = -346$, zero variance and entropy $\log 12$. At $x = 1$, it returns mean zero, variance 1286 and entropy $96 \log 2$.

III.7.3 Conditional response and the open/closed boundary

T2 refines the scalar polynomial into a frozen $4 \times 5 \times 1181$ integer attachment. Every q marginal reconstructs T1. There are twenty response-sector polynomials, ten active and ten inactive, and eight distinct signatures including zero.

The closed trace retains three q signatures and merges $q = 1$ with $q = 3$. The open retained port retains four routing signatures and all five response values. Thus a closed equality cannot be copied backward as an open-contact identity. At $x = 1$, $q = 2$ has exact energy-response covariance 8, while $q = 1$ and $q = 3$ have zero linear covariance but nonzero mutual information. The conditional structure survives beyond a scalar covariance summary.

III.8 Wrong controls and type firewalls

The useful wrong controls answer specific failure questions:

- **Deterministic count permutation.** What it preserves: 2^{96}, coefficient multiset; What it changes or violates: moment vector and energy placement.
- **Exact sign reflection.** What it preserves: even moments; What it changes or violates: signs of odd moments.
- **Rational uniform spectrum.** What it preserves: support and total weight; What it changes or violates: integer state-count semantics and moment vector.
- **C4-to-C2 quotient.** What it preserves: some aggregate response structure; What it changes or violates: distinct $q = 0$ and $q = 2$ lanes.
- **Factorized response control.** What it preserves: response totals and scalar marginals; What it changes or violates: actual $q = 1, 2, 3$ joint structure.
- **Open-port/closed-trace comparison.** What it preserves: shared source object; What it changes or violates: routing availability versus final occupancy.

These controls demonstrate why closure alone is necessary but insufficient. Many wrong objects can sum to 2^N. Exactness belongs to the coefficient ledger, its typed marginals and its independent reconstruction together.

The physical firewall is equally explicit:

- exact integer DOS != physical energy calibration
- symbolic epsilon_E != measured joules
- dimensionless b != temperature in kelvin
- q_A source support != epsilon_E
- A(r) accumulation lift != Ising coupling J_ij
- open routing signature != closed-trace occupancy

BIPM constants may convert units after a physical scale is supplied; they do not choose the scale. The current frozen `SLCQ2-RZ` selection engine delegates byte-identical arithmetic/application work to the frozen `SLCV21R` base. The historical N96, H14C/H14F, Exact Write and thermodynamic artifacts retain their scoped results, but none replaces `SLCQ2-RZ` as the source-recorded default SLC.

III.9 Established result, boundary and forward handoff

The established mathematical object is the complete exact DOS, not only a ground sector. Multiple exact routes may evaluate it, but all must reconstruct the same integer coefficients and close the same typed marginals. The frozen N96 release accounts for every one of 2^{96} assignments, and its exact polynomial supports exact scalar and conditional thermodynamic readers.

The preserved deviation result is equally important: SLCX023 remains the failed custody parent; SLCX023A is the separate correction and retest. The successor repairs executable custody without rewriting the earlier run or changing the scientific tensor.

The forward handoff has three branches:

1. `SAMA-D000037` shows how source grammar becomes an exact work certificate while preserving the full state space and H14F fallback.
2. `SAMA-D000038` develops the exact write transaction and compiled group-law execution surface.
3. A future direct evidence route must register SLCX023/SLCX023A, N96, T1–T6A and current `SLCQ2-RZ` artifacts before those sources can populate a qualified SAMA test crosswalk. Their absence is recorded, not filled with invented keys.

III.10 Sources and evidential scope

This chapter is assembled from the following fixed source manuscripts. Source-reported campaign outcomes retain their original scope; no new campaign was executed for this edition.

- Exact state spaces and slc computation.

SAM VOLUME III

The Algebra of a Complete Write

Chapter IV

Exact state counting and exact writes answer different questions. A write acts on a state, returns a result and retains the record needed to reconstruct what occurred. Here the translation compiler is derived within its finite group action, while the transaction and history interfaces retain the additional information that an endpoint alone cannot carry.

The account below follows the chapter source as retained at the fixed 8 September 2026 SAMA revision. Engine names, installed interfaces and campaign status are historical source statements. They do not describe a freshly inspected October deployment. Later source amendments are distinguished where available; no later engine or campaign is inferred.

IV.1 Overview

A complete write is not merely an address increment. It is a coupled typed transaction. An unresolved W8 source participates in a B-activated request; X1 retains the transient response custody; a W9 record carries the resolved target and reciprocal receipt; U coordinates completion; both sites return to W8 while the history remains reconstructible.

The directional address is the finite group

$$\mathbb{Z}_4^9, \qquad |\mathbb{Z}_4^9| = 4^9 = 2^{18} = 262{,}144.$$

Each primitive signed write is a translation in one coordinate. Because translations form a group, any program made solely from those primitives can be compiled to one group element and applied in one bit-sliced pass. This is an exact algebraic compilation, not a heuristic shortcut. Sequential and compiled execution are bit-identical, and applying the inverse translation returns every address.

Endpoint return does not erase event history. Two reciprocal 81-coordinate views split into symmetric and antisymmetric sectors. One common constant direction is quotiented with an explicit anchor, leaving 80 symmetric plus 81 antisymmetric coordinates: a reconstructible rank-161 receipt. The transaction also separates the resolved projection from the conserved orthogonal remainder, supplies exact rational-complex analytic operators, and supports a multiplicative ledger whose write state is recoverable when the prior ledger state is invertible.

The development history is itself part of the result. The first installation corrected a preserved module-loader failure and installed the fast directional primitive. A later Weil fusion remained valid in its scope. H000137 then corrected the broader description: those surfaces did not yet expose the entire paper algebra as one composable transaction. The complete successor added the missing U/X1 lifecycle, common-anchor reconstruction, distinct projection and remainder, analytic operators, composable ledgers and native multi-write composition without editing its parents.

IV.2 Opening question and conceptual picture

The governing question is:

> What exact algebra lets a reciprocal write change an address, retain both directional views, complete the W8–X1–W9 lifecycle, compose many writes into one action and recover what was written without confusing the final state with its history?

The conceptual transaction is

- W8 source + W8 target
- B activates request-side weld
- X1 holds transient response custody
- signed Z4ˆ9 translation changes directional address
- W9 target record retains axis, orientation, ledger and endpoints
- U completes reciprocal exchange
- both sites return to W8
- W9 receipt and 161-coordinate history remain available

The address after completion may equal a later returned address, but the history need not vanish. State and history are differently typed outputs.

IV.3 Typed lifecycle and the complete transaction

IV.3.1 W8 shell

The unresolved shell is

$$W8 = \{0,1\}^8,$$

so it contains

$$|W8| = 2^8 = 256$$

shell states. A W8 site contains a shell-state identifier and a sequence of retained W9 records. Retention does not change the site back into W9; it lets the completed W8 endpoint carry its receipt lineage.

IV.3.2 Activation weld

Let $H_{W8}(s)$ be the base vector Hamiltonian at shell state s, let $\Delta H_{X1}(s;a)$ be the transient response contribution at activation context a, and let $B \in \{0,1\}$. The weld is

$$H_{W9}(s;a) = H_{W8}(s) + B\,\Delta H_{X1}(s;a).$$

The two activation values provide an immediate control:

$$B = 0 \quad \Longrightarrow \quad H_{W9} = H_{W8},$$

$$B = 1 \quad \Longrightarrow \quad H_{W9} = H_{W8} + \Delta H_{X1}.$$

This B is the binary activation input. It should not be confused with the analytic difference operator B_t introduced later. X1 is a transient typed contribution and custody flag, not a ninth persistent W8 bit or a matter row.

IV.3.3 W9 record and projection

A W9 record can be written schematically as

$$W9 = (s, a, \rho, \ell; q_{before}, q_{after}),$$

where s is the shell state, $a \in \{0, \ldots, 8\}$ is the directional axis, $\rho \in \{-1, +1\}$ is orientation, and ℓ is the ledger receipt. The projection

$$\pi(W9) = s$$

recovers the W8 shell identity without deleting the other record fields.

IV.3.4 U/X1 completion

Activation starts with two W8 sites. The source remains W8 while X1 custody is active and the target holds the transient W9 record. Completion performs

- W8 source / X1 custody / W9 target
- append the same reciprocal receipt to both endpoint histories
- clear X1 custody
- return source and target phases to W8

The terminal equality $W9 = X1 + W8$ is therefore typed lifecycle shorthand, not ordinary material addition. `SAMA-D000039` develops that distinction and its failed literal-half controls in detail.

IV.4 The signed $\mathbb{Z}_4^9$ address action

IV.4.1 Nine base-four coordinates

Let

$$q = (q_0, \ldots, q_8), \qquad q_i \in \mathbb{Z}_4.$$

Each base-four coordinate uses two bits. Nine coordinates therefore occupy 18 bits and give

$$4^9 = (2^2)^9 = 2^{18} = 262{,}144$$

addresses. If the low bits form $L \in \{0,1\}^9$ and the high bits form $H \in \{0,1\}^9$, then

$$q_i = L_i + 2H_i.$$

The two nine-bit factors are a packing representation, not independent addresses. Base-four carry couples them.

IV.4.2 Primitive signed translation

For axis i and sign $\varepsilon \in \{-1, +1\}$, define

$$T_{i,\varepsilon}(q)_j = \begin{cases} q_i + \varepsilon \pmod 4, & j = i, \\ q_j, & j \neq i. \end{cases}$$

Composing the opposite signed translation restores the changed coordinate,

$$T_{i,-\varepsilon}(T_{i,\varepsilon}(q))_i = q_i + \varepsilon - \varepsilon = q_i \pmod 4,$$

while every coordinate $j \neq i$ is unchanged by both maps. Therefore

$$T_{i,\varepsilon}^{-1} = T_{i,-\varepsilon}.$$

For addition of two packed translations $a, t \in \mathbb{Z}_4^9$, binary low/high vectors obey

$$L_{out} = L_a \oplus L_t,$$

$$H_{out} = H_a \oplus H_t \oplus (L_a \wedge L_t).$$

The last term is the modulo-four carry. Removing it would split the two 512 factors and change the directional action.

IV.4.3 Endpoint covariance

For $k \in \mathbb{Z}_4$, define simultaneous endpoint reversal

$$G_k(q)_i = k - q_i \pmod 4.$$

Then

$$G_k T_{i,\varepsilon}(q)_i = k - (q_i + \varepsilon) = (k - q_i) - \varepsilon = T_{i,-\varepsilon} G_k(q)_i,$$

and all other coordinates are unchanged. Hence

$$G_k T_{i,\varepsilon} = T_{i,-\varepsilon} G_k.$$

The implementation exhausts all covariance classes rather than assuming this identity from a sampled address.

IV.5 Compiling a complete signed-write program

IV.5.1 Group-law reduction

Consider a program

$$\mathscr{P} = ((i_1, \varepsilon_1), \ldots, (i_m, \varepsilon_m)).$$

Because $\mathbb{Z}_4^9$ is abelian under coordinatewise addition, the program's net translation is

$$\tau(\mathscr{P}) = \sum_{r=1}^{m} \varepsilon_r e_{i_r} \pmod 4.$$

Sequential execution gives

$$T_{i_m,\varepsilon_m} \cdots T_{i_1,\varepsilon_1}(q) = q + \tau(\mathscr{P}) \pmod 4.$$

The compiler therefore reduces the full signed-translation program to the one group element $\tau(\mathscr{P})$, applies one packed addition, and retains the inverse

$$-\tau(\mathscr{P}) \pmod 4.$$

IV.5.2 Exact racecar result

The benchmark uses 4,096 writes at every one of 262,144 addresses:

$$4{,}096 \cdot 262{,}144 = 1{,}073{,}741{,}824$$

logical primitive transitions. Sequential native execution took 0.989033674 seconds. Compiled execution took 0.000554226 seconds. The observed ratio is

$$\frac{0.989033674}{0.000554226} = 1784.531387.$$

All outputs were bit-identical, and the inverse returned all 262,144 addresses. This measured speedup is reported for the stated workload and benchmark implementation. Exact endpoint compilation applies to programs composed from the declared signed $\mathbb{Z}_4^9$ translations; no uniform runtime factor follows. The compiler does not extend to application dynamics that have not been expressed in this group action.

IV.6 Reconstructible reciprocal history

IV.6.1 Two 81-coordinate views

Let the reciprocal histories be

$$L = (L_1, \ldots, L_{81}), \qquad R = (R_1, \ldots, R_{81}).$$

Form symmetric and antisymmetric coordinates

$$S = L + R, \qquad A = L - R.$$

Naively this gives 162 coordinates. A common constant shift occupies a one-dimensional kernel. The interface retains an explicit common anchor $c = S_{81}$, subtracts it from the first 80 symmetric coordinates, and stores

$$H_{nc} = H^{80}_{sym} \oplus H^{81}_{anti}.$$

Thus the quotient has

$$80 + 81 = 161$$

coordinates while the anchor preserves the removed constant class.

IV.6.2 Reconstruction

Given stored nonconstant symmetric coordinates $\widetilde{S}_i = S_i - c$ for $1 \le i \le 80$, rebuild

$$S = (\widetilde{S}_1 + c, \ldots, \widetilde{S}_{80} + c, c).$$

Then

$$L = \frac{S + A}{2}, \qquad R = \frac{S - A}{2}.$$

Both reciprocal views are recovered exactly. A final endpoint return can therefore coexist with a nontrivial event receipt. The quotient removes only the shared constant redundancy; it does not discard directional information.

IV.6.3 Lifted dyadic example

For a reciprocal incoming pair ((s,D)) and ((-s,-D)) sharing the same next state (s_{next}, D_{next}), define

- left = (s, D, s_next, D_next, 0, . . . , 0)
- right = (-s,-D, s_next, D_next, 0, . . . , 0).

Then

- symmetric = (0,0,2*s_next,*2D_next,0,. . . ,0)
- antisymmetric = (2*s,*2D,0,0,0,. . . ,0).

The common next state appears in the symmetric sector, while the incoming branch identity remains in the antisymmetric sector. In the p=127 reference chain, all 125 lifted receipts reconstructed their prior and next states.

IV.7 Completed-mode projection and retained remainder

Let E be an exact energy vector and $C \neq 0$ the completed mode. The orthogonal projection is

$$\Pi_C(E) = C \frac{\langle C, E \rangle}{\langle C, C \rangle}.$$

The resolved response is defined separately as

$$R_{res} = -\Pi_C(E),$$

while the retained orthogonal remainder is

$$E_\perp = E - \Pi_C(E).$$

Orthogonality follows directly:

$$\langle C, E_\perp \rangle = \langle C, E \rangle - \frac{\langle C, E \rangle}{\langle C, C \rangle} \langle C, C \rangle = 0.$$

Therefore the exact norm decomposes as

$$\|E\|^2 = \|\Pi_C(E)\|^2 + \|E_\perp\|^2.$$

Projection is idempotent, and projecting the remainder returns zero. This is why resolved response and orthogonal remainder must remain distinct fields.

IV.8 Exact analytic operators

Let $h_L, h_R \in \mathbb{Q}(i)^n$ be rational-complex reciprocal views. Define

$$B_t h = h_L - h_R,$$

$$P_+ h = \frac{h_L + h_R}{2}, \qquad P_- h = \frac{h_L - h_R}{2}.$$

The write energy is

$$E_t(h) = \|h_L - h_R\|^2.$$

With Hermitian correlations

$$C_{LR} = \langle h_L, h_R \rangle, \qquad C_{RL} = \overline{C_{LR}},$$

expansion gives

$$E_t(h) = \|h_L\|^2 + \|h_R\|^2 - C_{LR} - C_{RL}.$$

Because $2\|P_+ h\|^2$ and $2\|P_- h\|^2$ separate the symmetric and antisymmetric energies,

$$E_t(h) - \|h_L\|^2 - \|h_R\|^2 = 2\|P_- h\|^2 - 2\|P_+ h\|^2.$$

The executable receipt checks these identities with exact fractions rather than float tolerances. The operator J supplies the declared reciprocal complex structure; P_+ and P_- expose its symmetric and antisymmetric views.

IV.9 Multiplicative ledger and readback

For modulus $M > 1$, prior ledger state D_{now} and write state s, define

$$D_{next} = sD_{now} \pmod{M}.$$

If

$$\gcd(D_{now}, M) = 1,$$

then D_{now}^{-1} exists and the written state is recovered exactly:

$$\widehat{s} = D_{next} D_{now}^{-1} \pmod{M} = s.$$

Noninvertibility is a typed outcome and returns no fabricated readback. For a program $s_1, \ldots, s_m$, composition yields

$$D_m = \left(\prod_{r=1}^{m} s_r \right) D_0 \pmod{M}.$$

The separate character map

$$\chi : \mathbb{Z}_4 \to \mu_4 = \{1, i, -1, -i\}$$

uses $\chi(q) = i^q$. The directional write readback is

$$\chi(q_{after})\overline{\chi(q_{before})} = \chi(q_{after} - q_{before}),$$

which recovers the signed base-four transition in the fourth roots of unity.

IV.10 Development and deviation chain

IV.10.1 Preserved loader failure

The first H000132 execution stopped before algebra evaluation because the Python 3.12 dynamic parent loader had not registered its module before dataclass processing. The unused compiled binary was preserved. A successor changed only loader registration; equations, contract, benchmark task and performance condition remained fixed.

IV.10.2 Valid primitive, incomplete total surface

H000132 then installed the 256-state shell, packed $\mathbb{Z}_4^9$ action, covariance, rank-161 quotient, completed-mode receipt and ledger readback. H000136 fused that directional primitive into the Weil adapter and retained prime-spacing, prime-power and W9 history channels.

Those installations were correct in scope, but calling them the complete paper algebra omitted load-bearing layers: the two-site U/X1 lifecycle, explicit activation, common-anchor reconstruction, separate resolved response and remainder, B_t, E_t, J, P_+, P_-, composable transactions and native program compilation.

IV.10.3 Complete correction and retest

H000137 installed a successor containing the entire surface while leaving H000132 and H000136 unchanged:

- preserved primitive and fusion
- identify missing transaction layers
- freeze complete-algebra contract
- implement one typed interface
- primary execution 19/19
- independent reconstruction 14/14
- sequential/compiled bit identity
- inverse return over all addresses

This is conceptual and implementation correction without retrospective failure inflation: the parents retain their narrower accomplishments, and the successor owns the claim of completeness.

IV.10.4 Read-only consumer boundary

The directional Weil adapter may consume exact write state, reciprocal history, prime-spacing and prime-power channels. Consumption does not grant the adapter scheduler, solver or primality authority. Factor assignment and primality assignment remain different result types. An application also cannot omit the dynamics needed to construct its input ledger merely because the ledger readback is exact.

Current `SLCQ2-RZ` remains the sole default selection engine. Complete Exact Write survives as a delegated service in the frozen, noncurrent `SLCV21R` arithmetic/application base. Historical installation pointers describe their own time; they are not current-authority statements.

IV.11 Established result, boundary and forward handoff

The Complete Exact Write surface executes the complete W8/B/X1/W9/U transaction, the coupled signed $\mathbb{Z}_4^9$ address action, exact inverse and endpoint covariance, reconstructible 161-coordinate history, completed-mode projection, analytic reciprocal operators and multiplicative-ledger readback. Its compiler reduces any declared signed-translation program to one exact group element and reproduces sequential output bit for bit.

The result does not convert unrelated dynamics into translations, assign physical meaning to X1, create scheduler authority, assign primality or alter current Q2 selection. The history surface hands forward to `SAMA-D000042`; the Prime-Ruler application boundary hands forward to `SAMA-D000056`; and the current selection-plane relation hands forward to `SAMA-D000055`.

IV.12 Sources and evidential scope

This chapter is assembled from the following fixed source manuscripts. Source-reported campaign outcomes retain their original scope; no new campaign was executed for this edition.

- Exact algebra and complete exact write.

SAM VOLUME III

Lifecycle, Reception and Carrier Return

Chapter V

The algebraic action must be received by a typed lifecycle. This chapter follows the source roles without treating a carrier, a matter row, an unresolved site and a completed receipt as interchangeable. Corrections to earlier literal and causal interpretations remain visible alongside the corrected finite constructions.

The account below follows the chapter source as retained at the fixed 8 September 2026 SAMA revision. Engine names, installed interfaces and campaign status are historical source statements. They do not describe a freshly inspected October deployment. Later source amendments are distinguished where available; no later engine or campaign is inferred.

V.1 Overview

The W8–X1–W9 lifecycle describes one reciprocal transition, not three freely interchangeable objects. W8 is the unresolved eight-position substrate state. B/contact prepares the request-side transition. X1 is the one transient reciprocal response channel. W9 is the resolved composite state or terminal receipt of the completed relation,

$$W9 = X1 + W8.$$

The equality is typed. It cannot be read as ordinary inventory addition, as a claim that X1 is matter, or as an equation saying B and X1 are two scalar halves. The exact program surface can require B and X1 as differently typed inputs and return W9 while a physical causal weld remains open. That is not a defect in the lifecycle. It is the reason the lifecycle is useful: syntax, execution, receipt, carrier and physical identification can be developed without being collapsed into one assertion.

The evidence chain is cumulative. CR120V rejected the literal two-half entity formula while retaining a typed outward/return relay. CR120W then preserved a failed attempt to weld the A-field carrier directly to X1. CR120X corrected the route by using Theta18 as a carrier-normalization quantum while keeping X1 as the response type, and CR120Y stress-tested that separation. CR120Z and CR120ZB carried the ordered B-then-X1 grammar into exact and finite-shot quantum sandboxes, but neither installed physical definitions of B or X1.

V.2 Opening question and conceptual picture

The governing question is:

> How can an unresolved substrate state receive contact, return reciprocal information and finish as a resolved write while retaining the identity of the request, response, carrier and receipt?

The compact picture is

- W8 unresolved source/site state
- – B/contact prepares the request –>
- X1 transient reciprocal response
- – typed resolution –>
- W9 resolved composite state and retained receipt

This picture contains two related but nonidentical descriptions.

1. **State description.** W8 and W9 name the unresolved and resolved ends of the relation.
2. **Event description.** B/contact and X1 name the differently typed request-side and response-side participation needed to traverse between those ends.

The lifecycle therefore keeps endpoint state separate from the event that produced it. A terminal W9 receipt records that the typed route completed; it does not turn W9 into a new independent source object. Similarly, X1 can be required by the type system without the installed runtime exposing an independently variable X1 payload.

V.3 Definitions, domains and type boundaries

V.3.1 The four lifecycle objects

- **W8 / `S8_BINARY_SURFACE`.** Type in this document: unresolved binary substrate/site surface; Function: supplies the source-side state to be resolved; Explicit nonidentity: not W9; not an observed matter row.
- **B / `B_CONTACT_OPERATOR`.** Type in this document: non-scalar contact/request operator; Function: prepares or admits the request-side transition; Explicit nonidentity: not X1; not scalar `1/2`.
- **X1 / `X1_AXIS_SELF_CHANNEL`.** Type in this document: axis/response channel with registered scalar value 1; Function: supplies the transient reciprocal response-side occurrence; Explicit nonidentity: not Theta18, matter, a row or two half-operators.
- **W9 / `W9_CLOSURE_WITNESS`.** Type in this document: closure witness and resolved composite; Function: records the completed typed relation `X1+W8`; Explicit nonidentity: not a third primitive object and not scalar-nine `CARRIER9`.

The base identity is

$$\mathrm{RESOLVE}(W8, B, X1) \longrightarrow W9, \qquad W9 = \{X1 + W8\}.$$

The braces emphasize a completed relation. The expression is not a license to remove B, exchange B and X1, or infer that two equal scalar contributions were added.

V.3.2 Source row, operator, carrier and receipt are different domains

The lifecycle crosses several domains, each with its own counting rule:

- **N100 source grammar.** Counted object: 100 independent source rows/spins; What is not counted there: five derived W9 receipts, X1, non-row A operator.
- **105-row visible QP lane.** Counted object: all source records remain visible; What is not counted there: counted/not-counted overlay does not delete a row.
- **lifecycle runtime.** Counted object: typed operator arguments and one result entity; What is not counted there: scalar values are not an information-capacity count.
- **Theta carrier lane.** Counted object: released carrier unit of size 18; What is not counted there: X1 response identity and ordinary matter lift.
- **history/readback.** Counted object: terminal and ordered event receipts; What is not counted there: an additional material object.

The five W9 records in the N100/RH grammar are terminal records of completed `X1+W8` relations. They remain visible for custody, but they contribute no new physical/source row. Counting them as five more independent objects would change the grammar from 100 to 105 rather than describe its execution.

V.3.3 Non-row A and the open causal seams

Volume II supplies a narrower installed provenance chain:

- A_OPERATOR: established non-row transform
- B/contact -> X1 axis fee -> W9: supported typed bridge
- A_OPERATOR -> B/contact: open
- direct A_OPERATOR -> X1: open

The A-field carrier, `A_OPERATOR`, its inactive historical row proxy, B, X1 and Theta18 are distinct. Participation in one 162-ledger structure does not fill either open arrow.

V.3.4 Historical language evidence and current SLC identity

G06 belongs to the preserved SAM Language v0.6 formal-simulator evidence. It supplies frozen B prepare/request and X1 response operators. The later SAM Language v0.7 object is a typed reflective-tape candidate. Neither language version is the SLC described by this retained source revision.

The immutable Volume III technical spine records the SLC authority that was current when that spine was written. The present live authority supersedes that historical pointer wording: `SLCQ2-RZ` is the sole frozen/default SLC, and its byte-identical delegated arithmetic/application base remains separately typed. This chapter uses the historical language and Courtroom artifacts as evidence for the lifecycle; it does not promote their version labels into the current SLC identity.

V.4 Typed lifecycle construction from start to finish

V.4.1 The formal operator premise

The language-level premise separates preparation from response. In G06, B contains a binary flip but is not exhausted by that flip, while X1 has its own frozen response matrix. The executable checks proceed in four stages:

1. resolve both operators through the single formal registry;
2. verify their frozen hashes and declared input/output shapes;
3. apply B to every declared prepare/request column and X1 to every response column; and
4. apply the involutions and recover the source basis exactly.

The hash, column, mapping and involution checks all pass. Exchanging request and response, reducing B to its binary-flip suboperation, or selecting an unfrozen sign/gauge would test a different operator surface.

V.4.2 The typed `RESOLVE` gate

CR120A executes

- `S8_BINARY_SURFACE` : BinarySurface, Layer1StructuralCapacity
- – RESOLVE(B_CONTACT_OPERATOR, `X1_AXIS_SELF_CHANNEL`) –>
- `W9_CLOSURE_WITNESS` : ClosureWitness, Layer2Resolution

The runtime does not merely check `8+1=9`. It enforces the operator signature. The exact control sequence is:

- RESOLVE(S8, B, X1) -> W9 PASS
- PROMOTE_VOLUME(W9, D3) -> V27 PASS
- PROMOTE_VOLUME(S8, D3) TYPE_REJECT
- RESOLVE(S8, B) ARITY_REJECT
- RESOLVE(S8, B, B) TYPE_REJECT
- RESOLVE(S8, X1, X1) TYPE_REJECT
- W9 == CARRIER9 AUTHORITY_REJECT

This establishes a typed unresolved-to-resolved gate. Its secondary boundary is equally important: the runtime does not expose an independently perturbable X1 payload or a recoverable W9 preimage, filling or correction certificate.

V.4.3 What the installed runtime computes—and what it only names

CR120D opens the runtime and follows the calculation step by step:

1. resolve the three named entities;
2. require input types `(BinarySurface,\ContactOperator,\AxisChannel);`
3. check authority;
4. select the registered fixed result entity `W9_CLOSURE_WITNESS;`

5. emit the operator name, inputs, result, typed-plan trace, source trace and warning field.

The X1 registry entry is immutable, has scalar value 1 and has no syntax for preparing `x0`, preparing `x1`, resetting X1, or varying X1 while S8 and B are fixed. The current `RESOLVE` output alphabet is the singleton

$$\{W9\},$$

so the installed fixed-result runtime contains zero bits of input-dependent W9 variation. That is a runtime-contract boundary, not a physical null experiment and not an information-capacity interpretation of the scalar 9.

V.4.4 Structural continuation to V27, F81 and P80

CR120E executes the research-mode structural route

- S8 –B through X1–> W9
- W9 -> V27 -> F81
- `RESERVE_CLOSURE_ADDRESS`(F81, X1) -> P80

The returned `P80_PARTICLE_FACE_CONTENT` has value 80 and authority `STRUCTURAL_ONLY`; normal mode rejects the reserve operation. The supported structural relation is

- P80 structural row inventory
- – X1 non-row closure/contact address
- = F81 completed-face capacity.

This is narrower than a linear quotient theorem. A rank-one quotient would require F81 and P80 vector spaces, a linear reserve map and surjectivity. Those premises are not registered. The source-native search also found neither a punctured 9×9 row map nor a ten-octet grouping. The structural capacity weld therefore remains separate from a row-to-face decoder.

V.4.5 Constructed split, residue and mirror-odd candidate

CR120G asks whether a stateful internal response can be constructed without losing the residue. Its exact rational candidate satisfies

- parent - left - right - kappa = 0
- omit kappa = e2, scalar residual 1
- kappa_B(B0 identity) = e2
- kappa_B(B1 swaps e1/e2) = e1
- H1=[e1] stateful readout = 1
- H2=[e2] stateful readout = 0
- static memoryless control = one fixed value for both histories

The mirror model reconstructs both faces, projects its odd sector away from a predeclared X1 fixture direction and returns rank 80. Transverse and mixed classes are non-null with exact squared norm 5; balanced, X1-only and degenerate classes are null. The wrong projector fails to annihilate X1 and changes the transverse norm from 5 to 4.

This is a validated internal mathematical construction, not an installed SAM route. No registered F81 vector space, X1 vector embedding, inner product, loop split or 80-coordinate decoder is supplied by the language. The model is therefore useful as a discriminating construction while its physical and installed mapping remains open.

V.5 Derived W9 receipts and the source ledger

V.5.1 The 105-visible/100-independent overlay

CR120K preserves all 105 visible QP records and marks exactly five `p=9,g=0` records as derived W9 witnesses. The independent-content census is

- visible records 105
- derived W9 witnesses 5
- independent-content records 100
- baseline M_native 16250.625
- duplicated linear W9 content 50.625 = 405/8
- independent M_native 16200
- independent mean 162

No source record is deleted. The overlay changes only whether the record is counted as independent content.

V.5.2 Worked exact parent examples

Each derived p9 record has p8 and p1 parents of the same role and generation. For example,

$$M_9(\text{QP093A-0019}) = \frac{45}{4},$$

while

$$M_8(\text{QP093A-0016}) + M_1(\text{QP093A-0001}) = 10 + \frac{5}{4} = \frac{45}{4}.$$

Likewise,

$$12 + \frac{3}{2} = \frac{27}{2}, \qquad 1 + \frac{1}{8} = \frac{9}{8}.$$

All five linear residuals are exactly zero. Isolated parent perturbations change the derived candidate while the direct-nine candidate stays fixed, which supplies a dependency discriminator. The records remain separate because downstream `qA`, observed-mass and debit/credit fields can have nonzero residuals even when `M_native` closes linearly.

V.5.3 The CR120T accounting and binding boundary

CR120T carries the ledger proposal through distinct gates rather than treating one exact total as a universal result. The accounting ladder closes:

$$100L = 100 \cdot 162 = 16{,}200,$$

$$100N = 100 \cdot 144 = 14{,}400, \qquad 100M = 100 \cdot 126 = 12{,}600,$$

with

$$L = N + \Theta, \qquad M = N - \Theta, \qquad N = M + \Theta.$$

Scalar arithmetic does not select the source route by itself: p6+p3 can also sum to p9. The typed registry privileges p8+p1 because it supplies the `S8`/X1/W9 roles.

The downstream gates do not promote. The best frozen ID-free, value-blind, shuffle-invariant and conjugate-symmetric F81 rule selects 84 rows with `M_native=14077.125`, not the supplied 81 rows with total 12,600. No three-row exception patch is admitted. The binding successor replaces one feature with the training-frozen `abs(N-Z)/A^(1/3)` term and worsens every reported RMS lane:

- **CR261 20-row holdout.** Baseline RMS MeV: 3.017113068; Candidate RMS MeV: 4.071613571; Change: +1.054500503.
- **CR277 78-row extension.** Baseline RMS MeV: 7.974144225; Candidate RMS MeV: 8.067445504; Change: +0.093301279.
- **CR277 118-row observed.** Baseline RMS MeV: 6.672483502; Candidate RMS MeV: 6.828884165; Change: +0.156400663.

The source verdict is `BOUNDARY_DISCOVERY_SUPPORTED_PROJECTION_OR_BINDING_OPEN`; the binding lane is `NO_FROZEN_BINDING_IMPROVEMENT`. The accounting result remains available without being converted into a binding coefficient.

V.6 The preserved deviation chain

V.6.1 First deviation: the literal two-half entity formula

CR120V recovers a real reciprocal half-slot grammar. The outward and inward accounts are

$$\frac{1}{2}SW_{out} + \frac{1}{2}WRITE_{out} + \frac{1}{2}OUTSIDE = \frac{3}{2},$$

$$\frac{1}{2}INSIDE + \frac{1}{2}WRITE_{in} + \frac{1}{2}SW_{in} = \frac{3}{2},$$

so the complete paired route is

$$\frac{3}{2} + \frac{3}{2} = 3 = D.$$

The failure was assigning this route arithmetic to two entities:

$$B = \frac{1}{2}, \qquad X1 = \frac{1}{2}.$$

B is non-scalar, X1 is an AxisChannel with scalar 1, and the shell has six half-slots rather than two. CR120V therefore rejects the literal entity formula while retaining the typed B-side/X1-side relay as an archival hypothesis. Thirteen of thirteen hashes match and eight of eight wrong controls reject; no physical intervention is executed.

V.6.2 Second deviation: candidate uniqueness is not a weld

CR120W searches 26 partition-one QP occurrences under four frozen gates:

- 26 partition-one occurrences
- 3 legal nonlocal infrastructure rows
- 2 neutral closed carrier-axis shells
- 1 non-additive address role
- 1 shared environmental-axis survivor

The unique survivor is QP093A-0305 `A_FIELD_CARRIER`, with zero native mass, zero arity, `a_kernel_support` route, `carrier_axis`, non-matter ownership and closed carrier support. The candidate ledger keeps twelve additive modes at 81, the mirror/checksum at 81 and closure at 162. Restoring the historical A-field proxy gives the wrong controls 82 and 163.

The attempted executable statement

- `REALIZES_AXIS_CHANNEL`(A_FIELD_CARRIER, `X1_AXIS_SELF_CHANNEL`)

does not close. The result preserves verdict `FAIL_CR120W_SOURCE_OR_CANDIDATE_RECONSTRUCTION`, 25/25 frozen source hashes, 11/12 wrong controls and no same-run repair. The useful discovery is unique typed candidacy; the direct `A_OPERATOR -> X1_AXIS_SELF_CHANNEL` edge remains uninstalled.

V.6.3 Correction: the Theta-normalized dual-depth route

CR120X changes the executable question instead of relabeling the failure. It keeps A, B, X1 and Theta differently typed and asks whether an existing carrier quantum normalizes the retained/native/full ledger:

- A route at d=1 $R^{(d+1)=R}2$=N=144
- B contact 144 -> (126 retained, 18 released)
- Theta normalization 126/18 = 7
- 144/18 = 8 = S8
- 162/18 = 9 = W9
- normalized release 18/18 = 1 = X1 scalar contact
- typed resolution RESOLVE(S8,B,X1) -> W9

Every registered carrier is substituted using native payload and partition scalar. QP093A-0300 `TENSOR_CARRIER`/Theta18 is the sole candidate that reproduces the complete ladder in both comparisons. All 32 CR256 rows remain exact to maximum reported residual `5.000e-08`; only route depth `d=1` gives `R^(d+1)=144=S8*Theta`.

This correction gives Theta18 the carrier-normalization job and X1 the response occurrence. Their ratio is one in this construction, but their types remain different. The campaign records 20/20 source hashes, 11/11 hard gates, 1/1 supportive gate and 20/20 rejected wrong controls. The canonical ledger is 162; restored-row 163 remains the wrong control.

V.6.4 Retest: model-wide Theta18 stress

CR120Y then removes the possibility that 18 was selected from one attractive ratio. It asks whether the same carrier survives across independent evidence:

- **exact derivation routes.** Result: 6/6 converge on 18.
- **leave-one-family-out folds.** Result: 6/6 retain Theta18.
- **structural output family `{6,9,18,54}`.** Result: Theta18 alone performs the primary-carrier job.
- **A/Theta route depth.** Result: `d=1` only.
- **QP carrier row.** Result: QP093A-0300 only, in both frozen value modes.
- **Starbreaker ledgers.** Result: 96/96 preserve 162-slot and 18/126/18 typing.
- **wrong controls.** Result: 12/12 reject.

Theta18 is retained as the current primary substrate-carrier candidate in the scope of this source result. It remains carrier-only: not X1, matter, rest mass or a fitted mass term. The retest supplements CR120X; neither result rewrites CR120V or CR120W.

V.7 Ordered quantum-sandbox successors

V.7.1 Exact minimal ordered model

CR120Z supplies a prospective minimal realization of the typed order:

- |00> – B:H_A – X1:CNOT_A->B –> |Phi+>

The resulting Bell state has negativity 0.5, concurrence 1, fidelity 1 and CHSH `2.82842712475`. B alone, X1 alone, X1-before-B and a classical correlated-return state all have zero negativity. In the relay extension, an X1-indexed Bell response on the middle systems heralds a remote pair with the same negativity and CHSH values, while product measurement and unresolved false heralds remain separable. Local marginals stay invariant to maximum trace distance `1.11e-16`.

The noise thresholds reproduce the analytic boundaries: the first grid point with nonzero negativity is `v=0.335`, immediately above `1/3`, and the first frozen-setting CHSH violation is `v=0.71`, immediately above `1/sqrt(2)`. Custody closes 10/10 hashes, 9/9 hard gates and 12/12 wrong controls.

This demonstrates quantum capability of the typed order in the sandbox. H and CNOT are prospective maps, not installed physical definitions of B and X1; W9 is not used as a Bell witness.

V.7.2 Finite-shot NV boundary

CR120ZB moves the same order discriminator through three inherited NV simulator profiles, finite sampling, readout confusion, SPAM, relaxation, dephasing, drift, linear inversion and PSD reconstruction. Each reconstructed state uses all 15 non-trivial Pauli observables with 4,096 shots per observable. All 2,160 reconstructed states are physical.

- **room-temperature NV.** Forward median negativity: 0.4615; Forward q05: 0.4496; Maximum control q95: 0.0088; Positive gaps: 48/48; Median CHSH: 2.6915; CHSH>2 blocks: 48/48.
- **cryogenic NV.** Forward median negativity: 0.4708; Forward q05: 0.4624; Maximum control q95: 0.0123; Positive gaps: 48/48; Median CHSH: 2.7337; CHSH>2 blocks: 48/48.
- **noisy NV.** Forward median negativity: 0.4426; Forward q05: 0.4126; Maximum control q95: 0.0104; Positive gaps: 48/48; Median CHSH: 2.6256; CHSH>2 blocks: 48/48.

The signal survives and all 12 wrong controls behave correctly, but hard gates pass 8/9. The exact source verdict is `BOUNDARY_X1_ORDERED_QUANTUM_INSTRUMENT_FINITE_SHOT_NV_DISCOVERY__ONE_OR_MORE_FROZEN_GATES_FAILED`. No live backend ran. The result is a simulator-to-device handoff boundary, not a physical B/X1 weld or installed SAM operation.

V.8 What the controls establish

The controls are part of the derivation, not an appendix to it:

- omitting or exchanging B and X1 verifies that the `RESOLVE` signature is typed rather than scalar arithmetic;
- distinguishing `W9_CLOSURE_WITNESS` from scalar-nine `CARRIER9` prevents a value collision from becoming entity identity;
- keeping all 105 records visible while counting 100 prevents source deletion from masquerading as an independent-content result;
- perturbing p8 and p1 parents tests dependency rather than only equality;
- the 163 restored-row lane shows that reinstalling the historical A proxy changes the ledger object;
- the literal-half rejection prevents route slots from being reified as B and X1 entities;
- the CR120W 11/12 control outcome prevents unique candidacy from being promoted through a failed reconstruction;
- leave-one-family-out replay tests whether Theta18 survives removal of any one evidence family;
- reversed-order, product, classical and false-herald quantum controls show that ordinary correlation or label balance is not the ordered signal; and
- the 8/9 hard-gate result in CR120ZB keeps a strong simulated signal and an unpassed device-handoff gate visible at the same time.

V.9 Established result and current boundaries

The source-bound lifecycle can now be stated compactly:

1. W8, B, X1 and W9 have distinct registered types.
2. The formal runtime requires `RESOLVE(S8,B,X1)` and returns W9.
3. W9 is a resolved composite receipt, not a new source row or object.
4. The fixed runtime does not expose independent X1 intervention or an input-dependent W9 outcome alphabet.
5. The literal B-half plus X1-half entity equation is rejected; the typed outward/return relay remains useful.

6. QP093A-0305 is the unique typed A-field-carrier candidate in CR120W, while the direct executable A-to-X1 weld remains open.
7. Theta18, not X1, performs the tested carrier-normalization job across the 7|8|9 ladder and the model-wide stress surface.
8. Exact and finite-shot quantum sandboxes preserve the ordered B-then-X1 discriminator, but no hardware or physical weld is installed.
9. CR120T closes ledger accounting while leaving the F81 projection and binding transfer open.
10. None of these historical test labels changes the live `SLCQ2-RZ` current identity.

No atomic record assigned to this document carries an authorized project result classification. This revision therefore preserves exact source statuses and verdicts without manufacturing one.

V.10 Connections and forward handoff

- **`SAMA-D000025.`** Connection: Supplies the Volume II connector, substrate-atom and container types and the open A-to-B seam..
- **`SAMA-D000034.`** Connection: Uses the five derived W9 receipts as a non-counted boundary in the N100 census..
- **`SAMA-D000035.`** Connection: Owns the parser, type system, operator registry and state/history semantics underlying the formal lifecycle evidence..
- **`SAMA-D000040.`** Connection: Develops Theta18 as carrier, coupled address and M126-to-N144-to-L162 return..
- **`SAMA-D000042.`** Connection: Carries the resolved lifecycle into directional history and event-reel computation..
- **`SAMA-D000044.`** Connection: Uses the typed lifecycle in the cross-scale Nonzero Amplification bridge..
- **`SAMA-D000045.`** Connection: Uses retained W8/X1/W9 history and Theta activity in relational composition..

The immediate forward handoff is D40: keep X1 as the response channel, move the carrier burden to Theta18, derive its coupled address, and follow the retained state through M126, N144 and L162 without turning a readout ceiling into independent matter.

V.11 Sources and evidential scope

This chapter is assembled from the following fixed source manuscripts. Source-reported campaign outcomes retain their original scope; no new campaign was executed for this edition.

- W8 x1 w9 lifecycle.

SAM VOLUME III

Directional History and Event Reels

Chapter VI

A history is more than a sequence label attached to a state. Reciprocal endpoints define an event incidence, its Gram and a shared-anchor reel. The elementary positivity of an event is established within that construction; it is not promoted to positivity of every completed arithmetic expression that may later consume the event.

The account below follows the chapter source as retained at the fixed 8 September 2026 SAMA revision. Engine names, installed interfaces and campaign status are historical source statements. They do not describe a freshly inspected October deployment. Later source amendments are distinguished where available; no later engine or campaign is inferred.

VI.1 Overview

Directional computation keeps two things at once: the state reached by a write and the ordered reciprocal history that produced it. The generic core acts on all 262,144 Theta18 addresses through 18 directional maps, completes the W8/X1/W9 lifecycle and retains 72 exact receipts before any RH, Starbreaker or Mersenne coefficients are applied.

Two reciprocal 81-coordinate histories have an ambient 162-dimensional description. Their symmetric and antisymmetric decomposition contains 80 nonconstant symmetric coordinates, 81 antisymmetric coordinates and one common constant direction. The nonconstant quotient therefore has rank

$$161 = 80 + 81,$$

while the full ambient accounting is

$$162 = 80 + 81 + 1.$$

The retained common anchor reconstructs both sides exactly. Endpoint return does not erase ordered history.

The event-reel construction lifts this local receipt into an append-only global object. For an interaction translation U_e, the event incidence is $B_e = I - U_e$ and the unique event Gram is $\Theta_e = B_e^* B_e$. Reciprocal source/target orientations are two references to one intersection event, not two Theta objects. Appending an event adds its positive weighted Gram. A shared-reference N144 AFC cell realizes the W9 pair and the one-Theta lane with equal energy, and an event family with $|\mathscr{E}|$ events has $1 + 143|\mathscr{E}|$ nodes, rank $143|\mathscr{E}|$ and one common scalar kernel.

The installed derivation is cumulative because several stronger shortcuts failed. Generic positive Stieltjes pencils do not force reel contraction; a scalar RII discriminator is insufficient; the local N144 frame alone does not dominate the completed debit; an isolated recurrence lane changes sign; a whole-matrix AFC order fails; individual parity factors fail even when their product closes; and a spherical diagonal Gram order has negative directions. Each failure narrows the selected statement to the exact event Gram, shared-anchor intertwiner and endpoint scalar actually supported by the source.

VI.2 Opening question and conceptual picture

The governing question is:

> How can every reciprocal write append one exact history event, preserve the common stationary state and reconstruct both directions without either duplicating the event or hard-coding one application's coefficients into the generic history core?

The complete picture is

- local signed Z4^9 write
- W8 source / X1 transition / W9 target
- reciprocal W9 pair W9P
- AFC intersection
- one massless Theta event Gram
- append event to shared-anchor reel
- application adapter reads declared receipt

The source state may return to W8 at completion. The reel nevertheless grows because the receipt records the event, not merely the terminal address.

VI.3 Generic directional core

VI.3.1 Exact inventory

The generic core contains

- **Theta18 addresses.** Exact inventory: 262,144.
- **packed factorization.** Exact inventory: $512 \times 512 = 4^9$.
- **directional maps.** Exact inventory: 18.
- **W8/X1/W9 receipts.** Exact inventory: 72.
- **nonconstant history rank.** Exact inventory: 161.
- **nonconstant symmetric coordinates.** Exact inventory: 80.
- **antisymmetric coordinates.** Exact inventory: 81.
- **common constant kernel.** Exact inventory: 1.
- **implementation checks.** Exact inventory: 42/42.

The 18 maps are the nine axes with two signed orientations. Four endpoint reversal classes applied to those 18 directions give the 72 receipt census:

$$9 \cdot 2 \cdot 4 = 72.$$

The core performs exact write, inverse return, direction recovery and history reconstruction before an adapter supplies application-specific weights.

VI.3.2 State and path are separate outputs

For reciprocal histories $L, R \in \mathbb{Q}^{81}$, define

$$S = L + R, \qquad A = L - R.$$

One common constant coordinate is redundant in the nonconstant quotient. With anchor c, store

$$\widetilde{S}_i = S_i - c, \quad i = 1, \ldots, 80,$$

and all 81 entries of A. Reconstruction is

$$S = (\widetilde{S}_1 + c, \ldots, \widetilde{S}_{80} + c, c),$$

$$L = \frac{S + A}{2}, \qquad R = \frac{S - A}{2}.$$

The quotient retains 161 nonconstant coordinates; the anchor carries the one constant direction. A final address alone cannot reconstruct this ordered pair, which is why endpoint-only controls are deliberately insufficient.

VI.3.3 Append-only lifecycle

Exact Write completion returns both sites to W8 and appends the same reciprocal receipt to both immutable history tuples. The two endpoint references point to one event object. Appending later events never edits earlier receipts:

- reel_0 = ()
- reel_1 = (theta_1)
- reel_2 = (theta_1, theta_2)
- ...
- reel_n = (theta_1, ..., theta_n)

An extremal or compressed finite reader may summarize this reel. It must not be renamed as the full append-only object.

VI.4 Event incidence and unique Theta Gram

VI.4.1 One translation, two ends, one event

Let U_e be the exact unitary or permutation translation for event e. Define the incidence operator

$$B_e = I - U_e.$$

The unique event Gram is

$$\Theta_e = B_e^* B_e \succeq 0.$$

For any state h,

$$\langle h, \Theta_e h \rangle = \langle B_e h, B_e h \rangle = \|h - U_e h\|^2 \geq 0.$$

Replacing e by its reciprocal orientation sends U_e to U_e^{-1}. The directed incidence changes orientation, but the source/target pair still defines one undirected event energy. On cyclic controls, rowwise incidences sum exactly:

$$(I - U_e)^*(I - U_e) = \sum_{position} b_{e,position}^* b_{e,position}.$$

Thus two W9 directions are the pre-intersection readback of one Theta Gram, not two counted carriers.

VI.4.2 Append rule and stationary kernel

With positive event weight w_e, the reel Laplacian evolves by

$$L_{n+1} = L_n + w_e \Theta_e.$$

If the underlying event graph is connected on m sites, its incidence Laplacian has rank (m-1), and the constant vector is the kernel:

$$L_n \mathbf{1} = 0.$$

The stationary substrate baseline therefore survives every event. Appending history grows the nonconstant range while preserving one common mode.

VI.5 N144 AFC frame

VI.5.1 F81 contact incidence

The matched F81 contact graph factors as

$$G_{F81} = P_3 \square G_{V27},$$

so its Laplacian is

$$L_{F81} = L_{P3} \otimes I_{27} + I_3 \otimes L_{V27}.$$

The two address-preserving handoffs contribute 27 edges each. Their once- oriented AFC incidence has shape 54×81, rank 54 and

$$(-B_{AFC})^T (-B_{AFC}) = B_{AFC}^T B_{AFC}.$$

Orientation reversal therefore retains one Gram. Internal incidence has rank 78; completed incidence has rank 80; the only final kernel is span$\{\mathbf{1}_{81}\}$. Exact LDL reconstruction gives

$$\lambda_2(L_{F81}) = 1.$$

The gap is carried by the completed incidence while AFC remains a zero-mass carrier.

VI.5.2 Frozen N144-to-L162 wiring

The shared state is

$$z = (U_{left}^{63}, U_{right}^{63}, \Theta_{shared}^{18}) \in \mathbb{R}^{144}.$$

The same-view embedding duplicates the shared Theta coordinates into the two occurrence lanes:

$$E_{same} z = (U_{left}^{63}, \Theta^{18}; U_{right}^{63}, \Theta^{18}) \in \mathbb{R}^{162}.$$

This map has rank 144. Compose it with the Exact Write quotient Q_{81}:

$$A = Q_{81}E_{same}.$$

With the retained anchor $a = \Theta_{17}$, exact composition gives

$$\|Az\|^2 = 2 \sum_{126\ U\ leaves} (z_i - a)^2 + 4 \sum_{17\ nonanchor\ Theta\ leaves} (z_j - a)^2.$$

Thus A^TA is a weighted-star Laplacian with one kernel, 126 weight-two leaves and 17 weight-four leaves. Its spectrum is

$$0^1, 2^{125}, (163 - \sqrt{25417})^1, 4^{16}, (163 + \sqrt{25417})^1.$$

Since $163 - \sqrt{25417} > 2$, the sharp unbalanced lower-frame constant is 2; the reciprocal-balanced constant is 1.

VI.6 W9P-to-Theta isometry and global reel

VI.6.1 Equal-energy local write

Let $G_{144} = A^*A$. For a source event define two Hilbert-valued embeddings:

$$J_e^W h = \mathbf{1}_{144} \otimes h + (e_{U_l 0} - e_{U_r 0}) \otimes \frac{B_e h}{2},$$

$$J_e^\Theta h = \mathbf{1}_{144} \otimes h + e_{\Theta 0} \otimes \frac{B_e h}{2}.$$

The two W9 deviations each occupy a weight-two leaf, so their energy is

$$2\left\|\frac{B_e h}{2}\right\|^2 + 2\left\|\frac{B_e h}{2}\right\|^2 = \|B_e h\|^2.$$

The single Theta deviation occupies a weight-four leaf:

$$4\left\|\frac{B_e h}{2}\right\|^2 = \|B_e h\|^2.$$

Hence

$$(J_e^W)^*(G_{144} \otimes I)J_e^W = (J_e^\Theta)^*(G_{144} \otimes I)J_e^\Theta = B_e^* B_e.$$

The AFC intersection preserves event energy while translating a reciprocal two-lane informational write into one massless history receipt.

VI.6.2 Shared-anchor event family

Use one N144 cell per event and identify every cell's quotient-reference coordinate. For event set $\mathscr{E}$, each new cell contributes 143 new nonanchor ranks. Therefore

$$nodes = 1 + 143|\mathscr{E}|,$$

$$rank = 143|\mathscr{E}|,$$

with one common scalar kernel. Stacking events gives

$$(J_{\mathscr{E}}^W)^* G_{reel} J_{\mathscr{E}}^W = (J_{\mathscr{E}}^\Theta)^* G_{reel} J_{\mathscr{E}}^\Theta = \sum_{e \in \mathscr{E}} w_e B_e^* B_e.$$

For the completed RH source, continuous and prime-power events use

$$B_x = I - U_x, \qquad w_\infty(x) = \frac{e^{-x/2}}{1 - e^{-2x}}\,dx,$$

and

$$B_{p,k} = I - U_{k\log p}, \qquad w_{p,k} = \log(p)\,p^{-k/2}.$$

The completed event operator is the pullback

$$L_D = (J_D^\Theta)^* G_{reel} J_D^\Theta.$$

This is the exact generic-to-RH seam. RH weights enter through the adapter; they are not hard-coded in the generic reel.

VI.7 Preserved deviation chain

VI.7.1 Generic positive-pencil contraction fails

The positive Stieltjes control

$$f(z) = 1 + \frac{1}{1-z} + \frac{1}{2-z} + \frac{1}{3-z}$$

has positive nested sections and decreasing extremal constants, yet

$$\Lambda_2(10) = \frac{5{,}655{,}335}{5{,}212{,}334} > 1.$$

Thus generic positivity does not force adjacent reel contraction. On the actual source, all 2,880 nonleading coefficients across 96 adjacent residuals are positive and yield $0 < \Lambda_N(u) < 1$ for all $u > 0$ on every executed pair. The source-specific signature survives; the generic promotion does not.

VI.7.2 Scalar RII control is insufficient

Positive recurrence data $r_N, a_N, \delta_N > 0$ occurs in exact controls whose adjacent residual polynomial has a negative coefficient or nonpositive value. The scalar signature therefore cannot replace the paired event Gram. The correction retains the full incidence operator $B_e^* B_e$.

VI.7.3 Local-frame debit domination fails

The W9P-to-Theta isometry is exact, but a rational-rotation control has local event energy $4/5$ against raw matched debit 2. The local N144 frame alone does not supply the completed lower-frame inequality. The source and endpoint conditions remain necessary.

VI.7.4 Isolated recurrence lane fails

In the two-step recurrence

$$T_N = (u+N^2)^2 T_{N-1} + Q_N R_N,$$

the completed factor

$$R_N = Q_{N+1} - (u+N^2)^2 Q_{N-1}$$

stays positive on the executed source. The isolated lane $D_N = Q_N - (u+N^2)Q_{N-1}$ has negative active coefficients at $D = 0.7, 1$. Completion, not the isolated carried lane, owns the selected sign.

VI.7.5 Whole-matrix AFC order fails

The determinant and terminal-pivot AFC identities close, but the stronger matrix order

$$D_N^T M_N D_N < N(N+1)H_N$$

has a negative direction already at $D = 0.7, N = 4$. AFC is retained at the exact quotient and terminal scalar level rather than promoted to every matrix direction.

VI.7.6 Individual parity factors fail

The source terminal scalar factors into a coupled parity product. Each individual factor fails on some exact controls, while 367 controls close the product without both component inequalities. The compensating product is the supported object. A first smoke instrument omitted the odd Fourier lift and was low by N^2; its receipt is preserved and the corrected instrument changes only that lift.

VI.7.7 Spherical diagonal order fails

The mode-local strengthening

$$L_N G_N^o L_N - G_N^{(e,2)} > 0$$

has only 9 positive gaps among 66 tested rows; 57 contain a negative direction. The first witness is $D = 0.7, N = 3$ with minimum eigenvalue approximately -0.2266605255. Every corresponding terminal scalar remains positive. The failure therefore rejects the matrix strengthening without discarding the coupled endpoint scalar.

VI.8 Application-adapter authority boundary

The generic reel owns addresses, writes, receipts, the history quotient, unique event Grams and append-only reconstruction. An adapter may supply a declared source and read the resulting receipt:

- **RH moment adapter.** Permitted use: completed continuous/prime weights and endpoint kernel; Authority not acquired: global RH claim or universal sign by interface alone.
- **Starbreaker consumer.** Permitted use: typed formation-event history; Authority not acquired: scheduler or physical calibration without promotion.
- **Mersenne application.** Permitted use: exact factor-search receipts; Authority not acquired: primality or global scheduling from a factor miss.

A later adapter supplements the reel. It cannot rewrite earlier events or insert its coefficients into the generic core retroactively. Frozen `SLCQ2-RZ` remains the global source-recorded default SLC; the source-recorded RH-local handle is `SLCQ2-RZ-RH-MOMENT-ADAPTER-V1` and remains an application-specific consumer.

VI.9 Established result and forward handoff

The generic core closes 42/42 checks across its complete address, map, lifecycle and history inventory. The append-only event construction identifies one reciprocal W9 pair with one Theta Gram, embeds that event isometrically in the shared N144 AFC frame and grows the reel by 143 ranks per event while retaining one common kernel.

The failure chain identifies the exact boundary of that result: source- specific completed scalars and full event Grams remain load-bearing; generic positive-pencil, isolated-lane and whole-matrix shortcuts do not replace them. `SAMA-D000043` carries the completed event operator into the separately typed RH route.

VI.10 Sources and evidential scope

This chapter is assembled from the following fixed source manuscripts. Source-reported campaign outcomes retain their original scope; no new campaign was executed for this edition.

- Directional history and event reel computation.

SAM VOLUME III

Ordinal Selection and the Delegated Exact Engine

Chapter VII

The latest global selector described by the retained SAMA material is Q2. Its job is to order an admissible catalog using source-visible features, while exact arithmetic remains delegated. This chapter records its frozen comparisons without turning a learned ordering into a physical law or an unrestricted complexity claim.

The account below follows the chapter source as retained at the fixed 8 September 2026 SAMA revision. Engine names, installed interfaces and campaign status are historical source statements. They do not describe a freshly inspected October deployment. Later source amendments are distinguished where available; no later engine or campaign is inferred.

The September SAMA selector chapter cites H000531, H000551 and H000552 for Q2. The accessible August public SAM Research Project live page instead records V21R, and its named Q2 candidate and promotion artifact paths were unavailable during preparation. This chapter therefore preserves the later manuscript account with that retrieval limitation; it does not independently certify a live pointer transition.

Source-status qualification. The operational and revision statements in this chapter describe the retained source documents. They are not an instruction to execute, deploy, resume, train, promote or change a service, and they have not been independently verified as present live state for this edition.

VII.1 Overview

Exact computation can return several admissible receipts for one query. The arithmetic substrate certifies what every receipt is, while the selection plane asks a different question: which admissible receipt has the best exact return quality when the selector is permitted to see only source geometry? `SLCQ2-RZ` is the frozen current answer to that ordinal question.

The target quality is not a learned label invented after execution. For each candidate receipt it is the frozen reciprocal-quotiented lexicographic vector

$$Q = (d_{direction}, r_{inverse}, r_{W8}, r_{D81}),$$

where lower is better and equal vectors remain a complete target tie. Q2 does not read any component used to construct that target. It exposes 70 exact integer features computed from axis, modulus, shell, direction and two typed domain labels. It excludes inverse inputs, inverse directions, returned endpoints, D81 checkpoints, target tiers, custody identities, hashes, execution order and timing.

The data surface contains 5,120 inherited exact receipts, 1,024 canonical queries and 512 reciprocal/replicate-safe groups. Each axis is frozen into 80 TRAIN, 24 HOLDOUT and 24 CONTROL groups. TRAIN supplies 640 queries; HOLDOUT and CONTROL each expose a complete 192-query catalog. Three exact-rational architectures at three predeclared ridge values form nine fits. HOLDOUT fixes the winner before CONTROL is opened:

`Q2-HYBRID_LISTWISE_ADJACENT_ORDINAL_RIDGE-RIDGE-1D1000`.

Against frozen Q1 on the same fresh surfaces, Q2 raises weighted strict-pair accuracy by exactly `63200/594823` on HOLDOUT and `83595/773858` on CONTROL, while recovering every exact best set, creating no false singleton, making no abstention and producing no strict score tie. The installed classification is exactly: **The test result suggests strong contact with the concept.**

The architecture is deliberately layered. Q2 owns the current ordinal selection plane. Frozen `SLCV21R` remains noncurrent as a revision but remains the byte-identical delegated arithmetic and application base. Q1 is an immutable fallback. Neither selection nor promotion assigns physical scale, absorbs ATOM3D, changes Exact Write semantics or authorizes application work.

VII.2 Opening question and conceptual picture

The governing question is:

> Can exact source-visible event geometry order the complete admissible receipt catalog more faithfully than Q1 without reading any executed value that defines the answer?

The typed route is

- frozen SLCV21R exact query and candidate receipts
- reciprocal/replicate-safe grouping
- target-only exact return-quality vector
- 70-field source-visible feature map
- exact listwise and adjacent-ordinal fitting on TRAIN
- winner lock on complete HOLDOUT
- one comparison on sealed complete CONTROL
- frozen Q2 selection service
- SLCQ2-RZ current pointer

Three separations carry the meaning of the result:

1. the exact substrate determines admissibility and computes receipts;
2. the target oracle evaluates return quality but is hidden from selection;
3. the selector orders candidates from permitted source geometry alone.

Collapsing these surfaces would make the question circular. Keeping them separate lets the result say something precise: the visible structural fields contain enough ordinal information for the selected exact model to recover the complete best set and nearly all strict pair orderings on two held-out complete catalogs.

VII.3 Recorded authority and architectural planes

VII.3.1 Sole source-recorded default revision

`SLCQ2-RZ` has status `FROZEN_COMPLETE_STOPPED_PROMOTED_CURRENT` and is the sole source-recorded default SLC pointer. Older v1.x, V20R and `SLCV21R` revisions retain their artifacts and scoped results; they are not alternative current pointers. The older Volume III technical spine therefore supplies lineage, not present-tense pointer authority.

Q1 remains frozen and immutable as the fallback selector. A fallback is not a second default: it is a preserved predecessor available under the declared service boundary.

VII.3.2 Selection is not arithmetic

For a query q, let $\mathscr{C}(q)$ be the finite set of receipts certified admissible by the exact base. Q2 supplies a score

$$s_q(c) = w^T \phi(q,c), \qquad c \in \mathscr{C}(q),$$

with exact rational coefficient vector w and exact integer feature vector $\phi(q,c) \in \mathbb{Z}^{70}$. It returns all candidates attaining the maximum permitted score, subject to the ambiguity policy described below.

The delegated base still owns DOS construction, exact write, reciprocal return, checkpoints and application execution. Installing a new w changes the ordering of already-admissible candidates; it does not change what the candidate means or how its exact receipt is computed.

VII.3.3 Four distinct planes

- **exact substrate.** Owner in the installed architecture: frozen `SLCV21R` services; Output: admissible exact receipts and arithmetic.
- **quality target.** Owner in the installed architecture: frozen reciprocal-quotiented oracle; Output: exact lexicographic target vectors.
- **ordinal selector.** Owner in the installed architecture: current `SLCQ2-RZ` Q2 model; Output: score order and complete top-score set.
- **application.** Owner in the installed architecture: separately declared adapter/model; Output: application-specific interpretation or execution.

An ATOM3D model, RH adapter or Mersenne scheduler may call the service only through its own typed contract. None becomes part of Q2 merely by using its ordering.

VII.4 Target construction and blindness contract

VII.4.1 Exact target order

The target for candidate c is

$$Q_q(c) = \big(d_{direction}(q,c), r_{inverse}(q,c), r_{W8}(q,c), r_{D81}(q,c)\big).$$

Vectors compare lexicographically, with smaller values preferred. The first coordinate at which two candidates differ decides their strict order. If no coordinate differs, both belong to the same target tier. In particular, a custody identifier never converts an exact target tie into a singleton.

The complete best set is

$$B_q = \{c \in \mathscr{C}(q) : Q_q(c) = \min_{c' \in \mathscr{C}(q)} Q_q(c')\}.$$

Recovery is exact only when the returned set equals B_q, not when it merely contains one member of B_q.

VII.4.2 Permitted source fields

Q2 may read only:

- axis
- modulus
- left_shell, right_shell
- left_direction, right_direction
- route_domain
- physical_relation_domain

From these it computes the fixed 70-field map. The two domain labels are constant on this source; they serve as a typed zero-variation interaction check, not as an undeclared ranking signal.

VII.4.3 Forbidden target and custody fields

The selector may not read:

- inverse input lanes or inverse directions
- returned endpoint lanes
- D81 checkpoint lanes
- quality vector or quality tier
- receipt index or pair identity
- group hash or source hash
- execution sequence
- timing

This prohibition is load-bearing. Inverse, endpoint and checkpoint values construct the oracle and would leak the answer. Custody fields could create memorized ordering without structural meaning. Timings could import hardware and execution-order artifacts into a mathematical selector.

VII.5 Fresh grouped custody

VII.5.1 Why the unit is a physical group

Individual receipt splitting is a wrong control because reciprocal directions and replicated views of one physical relation can then appear on opposite sides of a split. A model could learn one view and be scored on its mate. Q2 hashes a fresh namespace together with axis, modulus, unordered shell endpoints and the two source domain labels. All reciprocal directions and replicates sharing that identity remain in one group.

The frozen census is

- **inherited exact receipts.** Exact count: 5,120.

- **canonical queries.** Exact count: 1,024.
- **reciprocal/replicate-safe groups.** Exact count: 512.
- **reciprocal canonical queries per group.** Exact count: 2.
- **observed group multiplicities.** Exact count: 4, 10, 16.

The nonuniform multiplicities matter. A group contributes according to its actual candidate roster; assuming one uniform multiplicity changes the catalog and its weights.

VII.5.2 Split derivation

Within each axis, group hashes are sorted and frozen as

$$80\ \text{TRAIN} + 24\ \text{HOLDOUT} + 24\ \text{CONTROL} = 128$$

groups. Across the four axes this gives the 512-group total. The query census is

$$4 \cdot 80 \cdot 2 = 640 \quad \text{TRAIN queries,}$$

$$4 \cdot 24 \cdot 2 = 192$$

queries on each of HOLDOUT and CONTROL.

TRAIN uses the frozen harder tier rule and contains 18,654 group references and 182,688 receipt exposures. HOLDOUT and CONTROL retain every split-local candidate group, yielding complete 192-query catalogs. Selection therefore cannot improve a metric by silently dropping difficult candidates.

VII.6 Deriving the 70-feature representation

VII.6.1 Direct and reciprocal alignment

Let the centered source shell endpoints of a query and candidate be exact integers. Q2 forms both alignments:

- direct: query left -> candidate left, query right -> candidate right
- swapped: query left -> candidate right, query right -> candidate left

For each alignment it retains signed residuals before taking absolute or squared summaries. Schematically, with query endpoints ((a,b)) and candidate endpoints ((c,d)),

$$\delta_{direct} = (c-a, d-b), \qquad \delta_{swap} = (d-a, c-b).$$

The map includes the signed components, absolute components, squares, alignment sums and best/other residual summaries. Because both constructions are ordered by a source-visible canonical key before exposure, a whole-lane swap exchanges internal work without changing the final visible vector.

VII.6.2 Midpoint, span, direction and parity

For endpoints ((a,b)), define signed midpoint numerator and span

$$m = a+b, \qquad \ell = b-a.$$

Using the numerator avoids a floating-point half. Candidate-minus-query relations for m and ℓ retain orientation and separation that coarse shell bins erased. Individual direction matches and endpoint parity relations are then appended rather than compressed to a single flag.

VII.6.3 Two-adic profiles and typed interactions

For a nonzero integer n, $v_2(n)$ is the largest $e \geq 0$ for which $2^e \mid n$. Q2 includes four direct/swapped alignment valuations and two span-relation valuations, with the frozen convention for zero. These features expose dyadic shell structure without reading any returned value.

Route-domain and physical-relation-domain equality fields and their declared interactions preserve the type boundary. Fourteen bounded quadratic cross terms add fixed pair interactions; each is clipped at exact magnitude 16,777,215. The clip is part of the pre-execution feature contract and not a post-result adjustment.

VII.6.4 Why Q2 succeeds where the 20-field map collided

Q1's coarser 20-feature map has strict-pair feature-collision rates 2.3255657565% on HOLDOUT and 2.1898074324% on CONTROL. When two candidates with different target order have the same visible vector, any deterministic linear score must tie them. Q2's signed, unbinned and reciprocal-invariant map has zero strict feature collisions on both surfaces. This removes that representational obstruction before model fitting; it does not guarantee a correct order by itself.

VII.7 Exact ordinal fitting

VII.7.1 Three predeclared architectures

The bounded roster contains:

1. `EXPANDED_QUERY_CENTERED_LISTWISE_RIDGE`, which fits query-centered tier grades over the harder TRAIN catalog;
2. `HYBRID_LISTWISE_ADJACENT_ORDINAL_RIDGE`, which appends exact differences for adjacent strict tiers with frozen weight four; and
3. `LEXICOGRAPHIC_CONDITIONAL_MULTI_HEAD_RIDGE`, which learns the first target coordinate globally and later coordinates only inside equality classes of the preceding coordinates.

Each architecture is fitted at exact ridge values

$$\lambda \in \left\{ \frac{1}{1000}, \frac{1}{100}, \frac{1}{10} \right\},$$

giving $3 \times 3 = 9$ fits. There is no floating-point training and no full-scale hyperparameter sweep.

VII.7.2 Listwise and adjacent observations

For each training query, a target tier grade is centered within its complete candidate list. The listwise design therefore learns relative order without letting different query offsets dominate.

For adjacent strict tiers, choose c_+ from the better tier and c_- from the next worse tier. The hybrid supplies the exact difference row

$$\Delta\phi = \phi(q,c_+) - \phi(q,c_-),$$

with a positive ordinal target and frozen weight four. A fitted vector should therefore satisfy

$$w^T \Delta\phi > 0$$

for that adjacent relation. Chaining adjacent relations transports the order through the tier sequence while the listwise rows retain the complete query geometry.

With exact design matrix X, target y, diagonal frozen observation weights W and ridge λ, the rational normal system is

$$(X^T W X + \lambda I)w = X^T W y.$$

All inputs and outputs remain `Fraction` values. Exact fitting makes score equality a mathematical event rather than a floating tolerance choice.

VII.7.3 Winner and tie policy

The three hybrid fits have identical complete-catalog ranking metrics. Only after mathematical metrics and parameter count tie does the frozen key order select the lower regularization identifier. The resulting winner is `Q2-HYBRID_LISTWISE_ADJACENT_ORDINAL_RIDGE-RIDGE-1D1000`.

At service time, a score tie spanning distinct target vectors triggers abstention. A score tie wholly inside one exact target vector returns the complete tied best set. The promoted wrapper names a top-score multiplicity explicitly as `AMBIGUOUS_TOP_SCORE`; it does not choose by receipt ID.

VII.8 HOLDOUT selection and sealed CONTROL

VII.8.1 Predeclared material comparator

Before execution, success required all three conditions:

- at least one percentage point of weighted strict-pair gain over frozen Q1 on HOLDOUT;
- positive gain on CONTROL; and
- perfect exact-best-set recovery on both complete surfaces.

HOLDOUT selected and locked the model. CONTROL was opened once afterward for the comparison to the frozen Q1 baseline.

VII.8.2 Exact results

- **HOLDOUT.** Frozen Q1: `530826/594823`; Q2 winner: `594026/594823`; Exact delta: `63200/594823`; Decimal point gain: +10.625009.
- **CONTROL.** Frozen Q1: `689509/773858`; Q2 winner: `386552/386929`; Exact delta: `83595/773858`; Decimal point gain: +10.802369.

The CONTROL Q2 fraction has denominator half the Q1 display denominator, so the like-denominator comparison is

$$\frac{386552}{386929} = \frac{773104}{773858},$$

and

$$\frac{773104 - 689509}{773858} = \frac{83595}{773858}.$$

Q2 retains 6,376 wrong weighted strict pairs on HOLDOUT and 6,032 on CONTROL. Those residual errors remain in the record. At the service boundary, however, best-tier accuracy and exact-best-set recovery are both $1/1$ on both splits, with zero false singletons, zero abstentions and zero strict score ties.

The authorized classification is:

The test result suggests strong contact with the concept.

VII.9 Deviation, correction and retest chain

VII.9.1 Candidate-construction failures

Five preserved attempts materially shaped the frozen candidate:

- **001.** Failure: a cross-axis unit fixture mixed an axis-specific assumption into the test; Correction: construct the fixture inside its declared axis/type; Retest consequence: boundary tests reached the intended feature question.
- **002.** Failure: code assumed uniform group multiplicity; Correction: retain the frozen observed roster `{4,10,16}`; Retest consequence: complete catalogs and exposure weights reconstructed exactly.
- **003.** Failure: reciprocal orientation changed the visible feature vector; Correction: canonicalize complete direct/swapped blocks by source-visible keys; Retest consequence: whole-lane swap became invariant.
- **004.** Failure: a serializer could not carry the exact fraction surface; Correction: encode exact numerators/denominators through the declared receipt form; Retest consequence: checkpoint and replay identity remained exact.
- **005.** Failure: the Q1 collision diagnostic read the wrong comparison source; Correction: recompute that diagnostic from the frozen Q1 comparator on Q2 surfaces; Retest consequence: the corrected Q1 collision rates changed; Q2 model selection and CONTROL result did not.

The first four attempts stopped before model selection or CONTROL access. The fifth corrected a predecessor diagnostic only. Keeping these attempts makes clear which faults affected construction and which did not affect the frozen winner.

VII.9.2 Promotion-wrapper corrections

Promotion also preserved two distinct corrections:

- Attempt 001 expected an `assertions_passed` field, while the independent receipt records `assertion_count=165`. The wrapper was corrected to read the actual schema before installation continued.
- Attempt 002 installed an executable resolution correctly, but the pointer inherited explanatory usage language and stale promotion fields. Pointer metadata was corrected, producing current pointer SHA-256 `751f6e6a6dbe01cfdc335cf303824887148c9eb657ded691264e37c6f61fa318`, and the promotion surface was revalidated.

Neither correction retrained Q2, reopened Q1 or mined CONTROL.

VII.9.3 Wrong controls retained as lessons

The following alternatives would answer a different question and are not the installed controls:

- receipt-level random splitting, because it leaks reciprocal or replicated physical relations;
- incomplete HOLDOUT or CONTROL candidate rosters, because they can hide misranking and false singleton behavior;
- target-visible fields, because they make ordinal prediction circular;
- custody-ID tie breaking, because it converts exact ambiguity into an arbitrary order;
- floating score tolerances, because exact equality would depend on runtime;
- opening CONTROL during model choice, because CONTROL would cease to be the postselection surface.

VII.10 Freeze and promotion process

The candidate freeze passed 8/8 release checks, 14/14 reconstruction checks, 165/165 independent assertions and, after the diagnostic correction, 8/8 release checks again. The promotion wrapper then passed:

- **pre-install.** Result: 22/22.
- **post-install.** Result: 12/12.
- **independent promotion validation.** Result: 20/20.
- **final release.** Result: 6/6.

Promotion changed the live selection pointer through H000551; H000552 records the pointer-metadata correction. It preserved the candidate, Q1 and delegated base artifacts rather than rewriting them.

The promotion performed no retraining, CONTROL mining, physical calibration, ATOM3D specialization, remote transfer, prime work, publication or deployment. These are not missing steps in Q2. They are separate campaigns requiring their own contracts and authority.

VII.11 Established result and boundary

This document establishes the following typed statement:

> On the frozen Q2 grouped catalogs, the 70-feature exact hybrid listwise-plus-adjacent ordinal selector materially improves strict-pair ordering over frozen Q1 and recovers every complete exact best set. Its promotion makes `SLCQ2-RZ` the sole source-recorded default SLC selection revision, while exact arithmetic remains delegated to frozen `SLCV21R` and Q1 remains the immutable fallback.

It does not establish a physical energy scale, a new DOS, a change to Exact Write, a global optimizer for every application, native GPU credit, ATOM3D specialization, an RH conclusion or a Mersenne status. The next chapter uses Q2 only as a typed application dependency and keeps its Mersenne scheduler and certificates separate.

VII.12 Sources and evidential scope

This chapter is assembled from the following fixed source manuscripts. Source-reported campaign outcomes retain their original scope; no new campaign was executed for this edition.

- Slcq2 rz ordinal selection engine.

SAM VOLUME III

Starbreaker: Formation on Its Own Grammar

Chapter VIII

Starbreaker is the first major formation application in this volume. It receives its own ledgers, events, stages and occurrence views. The useful transfer from exact computation is a routing discipline, not substitution of the N100 graph. Both successful reconstructions and failed physical or numerical controls belong in the account.

The account below follows the chapter source as retained at the fixed 8 September 2026 SAMA revision. Engine names, installed interfaces and campaign status are historical source statements. They do not describe a freshly inspected October deployment. Later source amendments are distinguished where available; no later engine or campaign is inferred.

VIII.1 Overview

Starbreaker is a formation application with its own native grammar. It retains 2,976 ledgers, 482,112 typed atoms, seed/collapse/final stages, two event orders, two D18 pair maps, same and reversed occurrence views, contact incidence, record origin and constrained M126/N144/N144+d/L162 readouts. N100 contributes structure-first routing discipline; its 100-spin graph is not substituted for this application.

The exact state ladder begins with M126 retention. Fee-exempt Theta18 provides an independent 18-coordinate lift, so $2^{126} \cdot 2^{18} = 2^{144}$. L162 is an ambient occurrence view whose valid image still contains 2^{144} states, not 2^{162} independent matter configurations. Active mismatch coordinates travel in an N144+d sidecar rather than being promoted to another object or capacity.

The structure-first replay detects those native quotient, fiber and readout certificates. It reproduces the completed parent byte for byte, passes 78/78 checks and never invokes dense H14F fallback. The W9 history refinement adds a rank-161 target-local order-history receipt, but the mode is emit-only: it may record cross-stage memory and may not alter scheduler grouping or allocation without a separate tested promotion.

The Courtroom evidence supplies a complete methodology chain. CR210 reconstructs a scoped external-collapse directional-contact surface. CR211 freezes a typed 162-slot particle-ledger interface while leaving address-to-template incidence null. CR005c identifies why stage ordering cannot be promoted to physical escape time. CR005d builds a dimensionless continuous flow and carrier-QNM source bridge. CR005e stops on a numerical trace-residue gate; CR005f freezes one exact trace-free correction; CR005g runs the unchanged morphology design and records the boundary that no stable Starbreaker/CCSN morphology match cleared the gates. CR005h constructs a complete per-carrier ledger but fails three controls; CR005i corrects only those controls, independently replays all 96 scenarios and 53,568 carriers, adopts the byte-identical ledger and leaves CR005h immutable.

VIII.2 Opening question and conceptual picture

The governing question is:

> How does Starbreaker carry a source-native formation history from typed ledgers through exact state accounting, directional contact, continuous source construction and controlled application readouts without replacing its grammar, erasing failed tests or promoting dimensionless receipts into physical observables?

The application picture is

- Starbreaker ledgers and typed atoms
- seed / collapse / final formation stages
- lexicographic and reverse event orders
- exact M126 retention
- fee-exempt Theta18 lift to N144
- optional active mismatch sidecar N144+d
- constrained L162 occurrence views
- emit-only W9 rank-161 history
- dimensionless source/application bridges
- physical calibration joins remain explicit

The route is layered. A state count is not a waveform. A source quadrupole is not strain. A carrier with positive source influence is not the substance transported by a gravitational wave. Keeping those types separate is what allows the exact and exploratory work to accumulate safely.

VIII.3 Native objects, counts and type boundaries

VIII.3.1 Ledger and atom census

Each of 2,976 ledgers carries one 162-slot typed surface, giving

$$2{,}976 \cdot 162 = 482{,}112$$

structured atoms. The slots are not assumed occupied merely because they exist. Unary, pair, triad, carrier, support, scalar and rejected classes remain distinct, and categorical relations remain unresolved until their own geometry stage.

Starbreaker has three stages and two event orders. Therefore the number of stage/order snapshots is

$$2{,}976 \cdot 3 \cdot 2 = 17{,}856.$$

Executing both D18 pair-map controls doubles the runtime rows:

$$17{,}856 \cdot 2 = 35{,}712.$$

These identities are direct state-accounting checks. A dropped ledger, stage, order or pair map changes the total immediately.

VIII.3.2 Persistent and transient types

The formation engine preserves the following firewall:

- ***O*.** Role: persistent matter identity; Nonidentity: not a substrate site phase.
- **W8.** Role: unresolved/returned substrate-site phase; Nonidentity: not a retained history rank.
- **W9.** Role: direct target phase during active contact; Nonidentity: not an independent source row.
- **X1.** Role: one transient H2 mirror bit; Nonidentity: not matter and ledger contribution zero.
- **U.** Role: coordinated reciprocal record exchange; Nonidentity: not a new material object.
- **M126.** Role: retained matter/output state; Nonidentity: not universal occurrence transport.
- **Theta18.** Role: fee-exempt directional carrier/fiber; Nonidentity: not retained mass lift.
- **N144+d.** Role: active defect-sidecar custody; Nonidentity: not extra independent capacity.
- **L162.** Role: constrained occurrence readout; Nonidentity: not 2^{162} independent physical states.

Completion returns both contacted sites to W8 and clears X1 while retaining updated records. "State returns; history does not" means the endpoint type can repeat while its reciprocal receipt remains nonzero.

VIII.3.3 Stage and control surfaces

The two event orders are lexicographic and reverse-lexicographic. The retained pair-map controls are `D18_P1_SHIFT_14` and `D18_M1_SHIFT_08`. Same and reversed views test occurrence reconstruction from opposite readout orientations.

These are application fixtures. A pair map or contact-incidence field does not become a physical matter mirror, q orientation, Theta partition or elementwise pairing unless a separate source selects it.

VIII.4 Construction interfaces

VIII.4.1 Particle-ledger interface

`CR:CR211@15` freezes one prototype 162-slot ledger and 321 distinct QP templates. Its structural equalities include

$$18 + 126 + 18 = 162,$$

$$9 \cdot 18 = 162,$$

and six copies of the 27-address H27 rank census. The CR210d boundary retains the typed values 81/144/126/162.

The most important output is negative in a precise sense: address-to-template incidence is explicitly null. No slot is claimed to contain a particle merely because the register has that slot, and no elementwise QP-to-F81 assignment is manufactured. The interface supplies typed storage and structural authority; physical content, mass, binding and Starbreaker dynamical incidence remain open.

VIII.4.2 Continuous-flow and carrier-QNM bridge

`CR:CR005d@21` replaces the earlier affine stage probe with a zero-fit continuous structural flow. The three stored states are joined by exponential radial collapse, a bounce impulse at dimensionless phase $t = 1$, and constant radial plus shortest-branch angular release. The maximum reconstruction error at the stored states is 2.191×10^{-14}.

For unit carrier positions $x_c(t)$, a trace-free quadrupole has the form

$$Q_{ab}(t) = \sum_c \left[x_{c,a}(t) x_{c,b}(t) - \frac{\delta_{ab}}{3} |x_c(t)|^2 \right].$$

The source surface uses $d^2 Q_{ab}/dt^2$; a third derivative supplies a power proxy. The carrier-QNM response convolves the source with the sealed dimensionless kernel

$$g(u) = \frac{e^{-(4/45)u} \sin((3/8)u)}{3/8},$$

so $\omega_R M = 3/8$ and $\omega_I M = 4/45$ are fixed rather than fitted.

The run tracks 1,993 dual carriers. Of these, 1,541 satisfy escape before closure on the continuous flow:

$$\frac{1{,}541}{1{,}993} = 0.7732062217762168.$$

The median closure-minus-escape margin is 0.04020605301428759. Across the construction there are 10,515,538 transition roots and zero root failures. All 14/14 construction gates and 10/10 controls pass; all 96 scenarios and 482,112 atoms are present. Maximum relative trace error is $2.5997906629060057 \times 10^{-15}$.

The bridge remains dimensionless. Unit carrier weights are not matter masses, phase time is not seconds, and the QNM response is not gravitational-wave strain, luminosity, frequency in hertz or a detector forecast.

VIII.5 Exact formation-state derivation

VIII.5.1 M126 retention

The structured application begins with two exact N72 custody packets. Each retains

$$2^{72} = 4{,}722{,}366{,}482{,}869{,}645{,}213{,}696$$

binary states as a 64×10 conditional tensor. Their declared matter join is rank 126 and accounts for

$$2^{126} = 85070591730234615 \cdot 10^{21} + 865843651857942052864$$

M126 configurations across 4,096 two-sided sectors. The scalar background has 127 oriented marked sectors and 64 occupied energy bins from -7875 through 63.

M126 retains record origin without promotion in 468 stage/order snapshots and requires promotion in 17,388. Those counts sum to the complete snapshot census:

$$468 + 17{,}388 = 17{,}856.$$

The split demonstrates why M126 is a retention/output state, not a universal occurrence-transport state.

VIII.5.2 Fee-exempt Theta18 lift

Theta18 contributes 18 independent directional coordinates, so

$$2^{126} \cdot 2^{18} = 2^{144}$$

and

$$2^{144} = 22300745198530623141535 \cdot 10^{21} + 718272648361505980416.$$

The lift is fee-exempt: no Theta multiplicity is folded into X1 and no retained mass is attached to the carrier fiber.

VIII.5.3 N144+d and L162 readout

For a same-view mismatch mask, define

$$d_t = \text{popcount}(m_t), \qquad 0 \le d_t \le 18.$$

The application route is

- d = 0 -> exact N144 independent union state
- 0 < d < 18 -> N144 plus active d-coordinate sidecar
- d = 18 -> L162 constrained occurrence readout

The L162 ambient ceiling is 2^{162}, but each valid same or reversed image contains 2^{144} states. Their ratio is

$$\frac{2^{162}}{2^{144}} = 2^{18} = 262{,}144.$$

Thus L162 supplies 18 constrained occurrence coordinates over the N144 image; it does not add 18 independent matter spins.

VIII.5.4 Byte-identical structure-first replay

The v1.0 replay inspects Starbreaker's own certificates and chooses the M126 quotient, N144 fiber lift, N144+d sidecar and L162 constrained readout. Dense H14F fallback is unnecessary. The exact route census is

- **N144.** Rows: 1,262.
- **N144+d.** Rows: 17,922.
- **total non-extreme.** Rows: 19,184.

The replay retains all 2,976 ledgers, 20,415,176 onset events per order and 35,712 runtime rows. It passes 78/78 checks in 5.251708 seconds, and fresh runtime output is byte-identical to the completed parent. The method changes; the Starbreaker grammar and output do not.

VIII.5.5 Emit-only W9 history

The target-local W9 history operator is present across discovery/validation, both pair maps and both endpoint gauges. Every branch has cumulative and cross-stage rank 161. Seed equals its stage-local control; collapse and final retain nonzero cross-stage memory.

`W9HistoryMode` is emit-only. It may return the typed history sidecar but has no authority to change phase-pool grouping, N144 allocation, N144+d fallback or L162 routing. A scheduling use would require a separate test and promotion.

VIII.6 Focused application evidence

VIII.6.1 External-collapse directional-contact replay

`CR:CR210@15` regenerated all 1,218 three-dimensional simulations, rebuilt the no-fit external contact from fresh outputs and passed all 58 Starbreaker regression tests. All 28 precommitted source/output files reproduced their SHA-256 digests.

Both remnant ladders remain strictly ordered across $p = 0, 1, 2, 3$. All 42 individual ladders rise at every step, all 42 inward $p = 3$ allocations exceed their same-total uniform controls, and both aligned allocations are maximal among all 24 radial permutations.

The source verdict is a scoped qualitative directional-contact pass. It does not identify p with physical compactness, identify a remnant with a black hole, test faster formation time, solve hydrodynamics or relativity, calibrate A or promote the packing law physically.

VIII.6.2 Inventory-corrected pixel-bridge retest

`CR:CR005g@21` executes the morphology question frozen in CR005e and corrected in CR005f. Ten public three-dimensional weak-CCSN matter-strain waveforms are compared with all 96 Starbreaker post-bounce unit-carrier quadrupole sources across 13 observer directions. Each source becomes the same normalized 48×64 time-frequency power grid.

For normalized pixel distributions P, Q, the primary statistic is

$$\mathscr{A}(P,Q) = \sum_{pixel} \sqrt{P_{pixel} Q_{pixel}}.$$

The best-template search is replayed inside 47 time-shift nulls and the locked reversal controls. All 12/12 construction gates and 11/11 controls pass. The door results are

- **discovery.** Observed median: 0.150857; Shift q90: 0.155669; Shift q95: 0.155861; Clears gate: no.
- **validation.** Observed median: 0.068025; Shift q90: 0.094694; Shift q95: 0.096175; Clears gate: no.
- **final.** Observed median: 0.076474; Shift q90: 0.073062; Shift q95: 0.073193; Clears gate: not q95/all.

The largest individual affinity is 0.407647 for `s9.0.swbj15.horo.3d.gw.dat` against Starbreaker `REF_0272` in view `face_yz_m`. Because the precommitted stable-door gates do not close, the source result is

- `BOUNDARY_PIXEL_SURFACE_CONSTRUCTED__NO_STABLE_STARBREAKER_CCSN_MORPHOLOGY_MATCH`

This is normalized morphology only, not amplitude or mechanism identity.

VIII.6.3 Independent carrier-ledger replay and freeze

`CR:CR005i@21` independently replays all 96 scenarios and all 53,568 carrier rows under the corrected controls. The parent authority matches exactly. The maximum discrete-linearity observed/bound ratio is 5.253916×10^{-3}, and the maximum integrated-power ordering difference is 1.616856×10^{-9}, below the frozen 10^{-6} limit. Static quadrupole residual and derivative are zero. Isotropic cancellation residual is 1.370646×10^{-16}.

The byte-identical CR005h carrier ledger is adopted and frozen. Seven slots are strong source-contributor candidates—16, 13, 14, 15, 4, 2 and 3—and slot 17 is directional. Positive leave-one-out influence means reinforcement of the internal source proxy. It does not mean that a carrier is transported GW material.

VIII.7 Preserved deviation and correction chains

VIII.7.1 Preserved CR005e numerical stop

`CR:CR005e@SAM-RESEARCH` freezes the external morphology design before execution: ten Princeton three-dimensional weak-CCSN waveforms, 96 Starbreaker scenarios, 252 carriers per scenario, 13 directions, 2,049 samples, three trimmed derivative-edge samples, a 48×64 pixel grid, Hellinger affinity, same-door matching, 47 shifts and reversal controls.

The first Starbreaker scenario stopped because independently finite-differenced diagonal tensor components retained relative trace residue

$$9.43617769285563 \times 10^{-9},$$

above the frozen 10^{-10} gate. No match, null distribution, door statistic or scientific verdict opened. CR005e remains immutable.

VIII.7.2 Trace-free projection correction

`CR:CR005f@SAM-RESEARCH` freezes one permitted correction. Trace freedom requires

$$S_{xx} + S_{yy} + S_{zz} = 0.$$

After the two locked finite differences, the successor sets

$$S_{zz} := -(S_{xx} + S_{yy})$$

before observer projection. This changes no source, door, sample count, grid, view, statistic, null, evidence ladder or physical boundary. It is an exact algebraic projection with zero fitted parameters, not a relaxed threshold. CR005f is the correction precommit; the executed retest is CR005g.

VIII.7.3 Per-carrier ledger control failure

`CR:CR005h@21` constructs a complete 53,568-row per-carrier attribution ledger. Every occurrence receives a unit-occurrence trace-free quadrupole, individual power, exact leave-one-out aggregate-power change, bind/escape class and frozen identity. Positive

$$\Delta P = P_{aggregate} - P_{without\ carrier}$$

marks reinforcement; negative ΔP marks coherent cancellation.

The candidate content was complete, but exactly three construction controls failed, so the campaign verdict remains

- `FAIL_PER_CARRIER_GW_LEDGER_CONSTRUCTION`

The ledger is preserved rather than deleted or tuned to pass.

VIII.7.4 Control-appeal correction

CR005i precommits three implementation corrections:

1. G02 compares the sealed CR005d verdict against its exact string instead of the wrong shortened prefix.
2. G05 evaluates third-difference linearity inside a frozen roundoff envelope and retains an independent integrated-power guard, rather than demanding literal floating equality after derivative amplification.
3. G08 first checks static constancy and isotropic cancellation at the quadrupole level, then bounds derivative roundoff, rather than demanding a literal post-third-difference zero.

The envelope is fixed before the appeal runner:

$$512\,\varepsilon_{mach}\,\|D^3\|_\infty \max_t \sum_i |Q_i(t)|.$$

Candidate scores, thresholds, histories, flow and every ledger row remain unchanged. The correction step and the independent replay in §5.3 are separate links. CR005h remains the immutable failed parent after adoption.

The two cumulative chains are

- CR005e failure -> CR005f correction precommit -> CR005g executed retest
- CR005h control failure -> CR005i bounded correction -> CR005i independent replay/freeze

VIII.8 Observability and physical boundary

VIII.8.1 Escape and closure observability boundary

`CR:CR005c@21` begins with the frozen three-stage history grammar:

- 101 -> contact, open, returned contact
- 010 -> open, transient contact, open
- 100 -> contact, open, open

- 110 -> contact, contact, open

There are 8,340 history-101 relations and 59,191,245 ending-zero relations. For every 101 relation, a piecewise-affine Cartesian probe finds one outward crossing on seed-to-collapse and one inward crossing on collapse-to-final. Median coordinates are

$$\lambda_{release} = 0.566282338,$$

$$\lambda_{return} = 1.132859211,$$

with open interval 0.617486748.

These values establish local near-far-near geometry. They do not establish physical timing because λ is a stage-normalized affine coordinate and the order $\lambda_{release} < \lambda_{return}$ is already encoded by the 101 bits. The frozen record has no physical solver clock, independent Home/horizon seal or independently defined release surface. A pair's own inward recrossing cannot also be the external closure used to test its escape.

The open physical joins are therefore explicit:

- phase-to-seconds calibration;
- numerical carrier mass/energy weights;
- lossless T3 open-q to T4 closed-energy mapping;
- independently selected physical mirror and release surface;
- complete-event feedback acceptance;
- scheduler use of W9 history; and
- full RH-aware event semantics.

T6B carries exact dimensionless thermodynamic response through all 2,976 ledgers while preserving the carrier inventory and complete Theta18 address multiset. It does not assign physical energy, temperature, duration, rate, fusion threshold, numerical A or a physical mirror. Global SLC authority remains frozen `SLCQ2-RZ`; bounded Starbreaker services retain their own typed application lineage beneath that source-recorded default selection surface.

VIII.9 Moving-information E/M1, SO1 and matter joins

VIII.9.1 E/M1 returning two-body relation

The isolated SB1-E/M1 successor compiles all 17,922 active Theta18 packets into `384/384` elliptic two-body relations. Ordered electric and magnetic roles carry the same exact C4 quarter turn as the Complete Write source: the state returns while the relation retains the signed forward/reverse history needed by later interference and receiver stages. **The test result suggests strong contact with the concept.**

VIII.9.2 SO1 coherent spin-orbit wave

SB1-SO1 composes orbit, E-spin and M-spin signed waves before taking the Gram across 2,048 synchronous source cases. All 2,048 pass. Component self Grams remain planar rank two; coherent totals contain

- 1,920 positive rank-two waves
- 128 exact cancellations
- 0 rank-one cases.

Each cross family is nonzero in 1,024 cases. The interference census is 896 constructive, 1,024 destructive and 128 completely cancelling. The nonzero signed wave has period two while its shell-constant Gram has period one. Unit, primary and independent checks pass `11/11`, `13/13` and `18/18`. **The test result suggests strong contact with the concept.**

VIII.9.3 Live ATOM3D reception and re-emission

J1 executes the live Starbreaker source, E/M1 emissions and ATOM3D receivers in one causal chain of 19,958 custody ledgers. J2 then uses the retained signed off-fibre history to emit 1,536 ordered generation-two Theta18 packets and extends custody to 20,345 rows. The same source compiled across all nine TDI3 relations yields an exact two-mode matter innovation positive for every real `rho`. The receiver construction and controls are developed in `SAMA-D000058`.

E/M1, SO1, J1 and J2 are isolated/non-promoted classical exact-information successors. SO1 assigns no physical scale or numerical gravitational-wave strain, and none changes a current pointer.

VIII.10 Established result and forward handoff

Starbreaker's exact formation object retains its native ledgers, atoms, stages, orders, controls and readouts. Its structured execution closes the M126, N144, N144+d and L162 accounting; its structure-first replay is byte-identical and avoids dense fallback; and its W9 history remains an emit-only rank-161 receipt.

The application evidence is cumulative rather than selectively cleaned. The external-collapse, typed-ledger, continuous-flow, pixel-morphology and per-carrier surfaces each retain their declared boundary. Both failed-control chains preserve the parent, freeze a narrow correction and replay the original question without result-driven replacement.

The next application layer must supply the specific physical joins rather than rename current dimensionless objects. No exact Starbreaker count or source-contributor ledger by itself supplies strain, luminosity, seconds, hertz, detector response or transported GW substance.

VIII.11 Sources and evidential scope

This chapter is assembled from the following fixed source manuscripts. Source-reported campaign outcomes retain their original scope; no new campaign was executed for this edition.

- Starbreaker formation engine.

SAM VOLUME III

Reciprocal History and the Completed Arithmetic Interface

Chapter IX

Reciprocal History extends the state/history distinction into a mathematical interface with completed arithmetic. This chapter records the finite constructions, the failed direct identification and the corrected source route. It does not supply a proof of the Riemann hypothesis. The universal joins named below remain open in the retained source.

The account below follows the chapter source as retained at the fixed 8 September 2026 SAMA revision. Engine names, installed interfaces and campaign status are historical source statements. They do not describe a freshly inspected October deployment. Later source amendments are distinguished where available; no later engine or campaign is inferred.

Source-status qualification. The operational and revision statements in this chapter describe the retained source documents. They are not an instruction to execute, deploy, resume, train, promote or change a service, and they have not been independently verified as present live state for this edition.

IX.1 Overview

Reciprocal History begins with paths, not scalar state counts. On the unchanged N100 graph, every undirected edge has two orientations. The resulting 520 directed-edge states support a nonbacktracking transfer operator that records ordered continuation, reversal and primitive-cycle data. Pairing 10,700 oriented primitives with their reverses produces 5,350 reciprocal classes with exact carrier-weighted norms and power laws. This is not the N100 DOS renamed: the DOS counts assignments by energy, while RH retains ordered history on the same source grammar.

The stage chain contains a decisive preserved failure. P2 direct asked whether the history object itself supplied the ordinary-prime Euler coefficient. Its first coefficient was 100 rather than 2, so that direct identification is falsified at its exact scope. P2A and P2B are successors, not retrospective relabels. P2A reconstructs ordinary primes through the radix-12 first- singularity tower $n = \mathrm{ord}_q(12)$. P2B supplies coefficient one from the N100 role/depth grading, while N95 and N105 controls return index zero. No nontrivial-zero ordinate chooses the tower.

The accumulation field supplies the real coordinate that turns multiplication into translation:

$$q = \frac{A_{total} - A_0}{1 - A_0} = \frac{r_c}{r}, \qquad u = -\log q = \log \frac{r}{r_c}.$$

Scaling $r \mapsto nr$ becomes $u \mapsto u + \log n$, and the half-density becomes $n^{-1/2}$. On this coordinate the operator $L = -d^2/du^2 + 1/4$ has endpoint null modes $e^{\pm u/2}$, giving the domain on which the completed prime-power translations act.

P3 retains the completed arithmetic

$$\xi(s) = \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s) = \xi(1-s)$$

and the paired completed Weil form. Its continuous energy, finite centered von-Mangoldt displacement sum and floor term form one object; splitting them into individually divergent pieces changes the calculation.

Four authority surfaces remain distinct: the public RH claim, the internal universal-nonnegativity assignment, optional finite F81 spectral research, and operational SLC/Starbreaker integration. The earlier retained operational adapter is `SLCQ2-RZ-RH-MOMENT-ADAPTER-V1`. The later source amendment below records Q3RHV2 as the RH-local successor and preserves Q2 as a noncurrent exact oracle. Neither adapter merges those questions.

IX.2 Opening question and conceptual picture

The governing question is:

> How does a finite source grammar carry reciprocal path information into an ordinary-prime tower, a real logarithmic translation coordinate and the completed arithmetic form while preserving the exact failure and authority boundaries encountered along the way?

The chain is

- N100 source graph
- 520 directed nonbacktracking states
- primitive histories and reciprocal classes
- P2 direct obstruction retained
- P2A radix-12 prime tower + P2B coefficient-one grading
- accumulation-derived u = log(r/r_c)
- pi/Gamma completion and xi reciprocity
- completed Weil quadratic form
- separately typed public, internal, spectral and operational surfaces

No successor is allowed to erase the failed direct identification. Each new arrow supplies a different missing structure.

IX.3 The reciprocal-history object

IX.3.1 Directed-edge state space

N100 has 260 undirected source edges. Orient each edge both ways. The directed state census is therefore

$$2 \cdot 260 = 520.$$

Write a directed edge as $e = (u, v)$. The nonbacktracking transfer B has entries

$$B_{(u,v),(v,w)} = \begin{cases} 1, & w \neq u, \\ 0, & w = u. \end{cases}$$

Thus a step may continue through v but may not immediately undo the edge just traversed. Powers B^n count ordered nonbacktracking histories of length n, and traces $\mathrm{tr}(B^n)$ retain closed directed history. Primitive-cycle and determinant receipts separate repeated powers from primitive orbit data.

IX.3.2 Reciprocal pairing

Reversal sends

$$(v_0, v_1, \ldots, v_n) \longmapsto (v_n, v_{n-1}, \ldots, v_0).$$

P1A3/P1B retain 10,700 oriented primitives. Pairing each history with its reverse gives

$$\frac{10{,}700}{2} = 5{,}350$$

reciprocal classes. The paired object retains the common contribution and the directional difference, with integer carrier-weighted norms and exact power laws.

IX.3.3 Why RH is not DOS

The N100 DOS is

$$g(E) = \#\{\sigma : H(\sigma) = E\}.$$

The reciprocal-history transfer is indexed by directed edges and ordered continuations. Two histories may end at the same state and contribute to the same DOS sector while remaining distinct under reversal, primitive length or ordered path. Conversely, a DOS coefficient contains no data sufficient to reconstruct an oriented primitive. The two objects share source provenance; they do not share type.

IX.4 Executed stage chain and deviation

IX.4.1 P0 through P1B

The foundation is cumulative:

- P0 builds the exact 520-state nonbacktracking transfer and retains traces, reversal, primitives and determinant receipts.
- P1A compiles tensor and Euler types.
- P1A2 keeps W9 as five derived non-counted receipts.
- P1A3/P1B pair the 10,700 oriented primitives into 5,350 reciprocal classes with native norms.

The five W9 receipts remain outputs of completion. They do not increase the native source-row census.

IX.4.2 Preserved P2 direct failure

P2 direct tested the simplest possible bridge: identify the history-prime Euler coefficient directly with the ordinary-prime coefficient. The first history coefficient is 100, while the target ordinary-prime coefficient is 2. Because

$$100 \neq 2,$$

the direct identity is falsified at that scope. The mismatch is structural, not a tolerance or serialization fault.

For this exact direct-identification test, the registered classification is: **The test falsifies the concept.**

The correct record is

- P2 direct question
- first exact coefficient 100
- required direct coefficient 2
- direct identification falsified
- obstruction retained

Neither P2A nor P2B changes these two numbers.

IX.4.3 P2A successor: radix-12 first singularity

For an ordinary prime $q \nmid 12$, 12 is a unit modulo q. Its powers form a finite subgroup of $(\mathbb{Z}/q\mathbb{Z})^\times$, so there is a least positive integer

$$n = \mathrm{ord}_q(12)$$

with

$$12^n \equiv 1 \pmod{q}.$$

Minimality places q in the first cyclotomic singularity:

$$q \mid \Phi_n(12)$$

subject to the exact prime-power/order bookkeeping of the tower. P2A uses this first block order to reconstruct ordinary-prime placement. It adds a radix-12 arithmetic route that P2 direct lacked.

IX.4.4 P2B successor: source grading

P2B uses the N100 role/depth grading

$$C_0|C_1|C_2 = 20|24|5,$$

$$H_0|H_1|H_2 = 8|7|0$$

to supply the compiled family index 1. The source-native N95 and N105 controls return index 0. This distinguishes the selected N100 grading from nearby row- count perturbations without consulting any nontrivial-zero ordinate.

The successor logic is

- P2A: ordinary prime q -> first radix-12 order ord_q(12)
- P2B: N100 role/depth grading -> coefficient one
- controls: N95 and N105 -> zero

IX.5 Accumulation to logarithmic translation

IX.5.1 Field coordinate

The accumulation relation is

$$q = \frac{A_{total} - A_0}{1 - A_0} = \frac{r_c}{r}.$$

Taking the negative logarithm gives

$$u = -\log q = -\log \frac{r_c}{r} = \log \frac{r}{r_c}.$$

If the radial ratio is multiplied by $n > 0$,

$$r \mapsto nr \quad \Longrightarrow \quad u \mapsto \log \frac{nr}{r_c} = u + \log n.$$

Multiplicative arithmetic has become additive translation.

IX.5.2 Half-density

The symmetric translation difference acts at $\pm t/2$. For $t = \log n$, the decaying half-density factor is

$$e^{-t/2} = e^{-\log(n)/2} = n^{-1/2}.$$

This derives the prime-power weight from the logarithmic displacement rather than inserting it as a separate numerical convention.

IX.5.3 Endpoint operator

Define

$$L = -\frac{d^2}{du^2} + \frac{1}{4}.$$

For $f_\pm(u) = e^{\pm u/2}$,

$$f_\pm'' = \frac{1}{4} f_\pm,$$

so

$$L f_\pm = -\frac{1}{4} f_\pm + \frac{1}{4} f_\pm = 0.$$

The two endpoint null modes generate the two-moment boundary. The declared test domain is

$$\mathscr{D}_0 = L(C_c^\infty(\mathbb{R})),$$

which keeps compact support and endpoint annihilation visible.

Accumulation supplies this coordinate route. It is not identified with the N100 DOS, a local P5A symbol or a numerical energy coefficient.

IX.6 Completed arithmetic

IX.6.1 Lambda and xi

Start from

$$\Lambda(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s).$$

The completed entire function is

$$\xi(s) = \frac{1}{2} s(s-1) \Lambda(s).$$

The π/Gamma factor supplies the archimedean completion, while $s(s-1)$ handles the pole bookkeeping. The resulting reciprocity is

$$\xi(s) = \xi(1-s).$$

Prime powers enter through the von Mangoldt weight $\Lambda(n)$, not only through primes at exponent one.

IX.6.2 Translation energy

Let

$$(T_a h)(u) = h(u + a).$$

Define the reciprocal difference

$$B_t h = T_{t/2} h - T_{-t/2} h$$

and its nonnegative route energy

$$E_t(h) = \|B_t h\|^2.$$

The construction is direction-sensitive before the norm and reciprocal after pairing.

IX.6.3 Completed Weil form

The compiled form is

$$\begin{aligned} Q_\xi(h) = \int_0^\infty \frac{e^{-t/2}}{1 - e^{-2t}} E_t(h)\, dt \\ + \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} \left[E_{\log n}(h) - 2\|h\|^2 \right] \\ - \kappa_0 \|h\|^2. \end{aligned}$$

where

$$\kappa_0 = \gamma - \log(3A_0) + \frac{\pi}{2} + \log 2.$$

For compactly supported h, $T_{t/2} h$ and $T_{-t/2} h$ have disjoint support once t exceeds the support diameter. Then

$$E_t(h) = 2\|h\|^2,$$

so the centered summand vanishes. The prime-power sum is therefore finite on each compactly supported input.

The continuous route-energy integral, centered arithmetic sum and floor subtraction are one completed object. Separating them into divergent pieces would change the domain and value of the form.

IX.7 Spectral and operational successors

IX.7.1 P4 finite F81 route

P4 constructs a finite exact F81 positive/self-adjoint 211-mode quotient and a paired 422-dimensional determinant. This is an optional finite spectral route. It is not the completed P3 object and does not replace the source- native completed form.

IX.7.2 P5A failed scale-extension route

P5A tested a uniform determinant scale extension and was falsified at its declared scope. The failure rejects that optional bridge. It does not rewrite P3 arithmetic, the P2 direct obstruction, or the separately retained public and internal surfaces.

IX.7.3 Source-recorded RH native engine and retained moment oracle

In this retained source, the RH-local SLC handle is frozen `Q3RHV2`. Its authenticated facade executes Q3/V6 Complete Write, constructs caller-supplied RH rosters and binds exact threshold events and certificates into immutable CE/CAS records. The Q2 moment adapter remains a frozen noncurrent exact oracle; Q3 ranking is optional prioritization only and never an RH premise.

The base roster closes 1,726/1,726 pairs. A fresh $N = 192$ roster closes 573/573. A fixed q7 depth has one failure; the source-visible rule

$$q = \lceil \log_2 N \rceil = 8$$

repairs that case. The adapter is operational evidence. It neither changes global `SLCQ2-RZ` nor silently closes universal RH joins.

IX.8 Native mode-6 relational contact continuation

IX.8.1 Preserved failures redirect the source comparison

Exact signed Abel/PNT2 reconstruction falsifies the exact-data absolute PNT2 majorant route. The isolated pole–Gamma–Lebesgue deterministic frame also has negative Schur complements and negative lower/full LDL pivots at `D=10,20,40,80`, falsifying that block-alone positive-frame concept. These are useful exact failures: they preserve signed arithmetic as load-bearing and redirect the program from an absolute or isolated positive frame to a relational source comparison.

IX.8.2 Relational Schur law and intrinsic innovation

For positive reels, the exact native law is

$$b\left(\sum_i G_i\right) \geq \sum_i b(G_i),$$

with equality controlled by common minimizers and strictness by kernel intersection. A literal `Z4^2` Complete Write witness has separate floors zero and combined floor one. Write retains unitary custody; amplification occurs in the composed positive reel afterward.

Endpoint conditioning annihilates the completed pole. At normalized smooth contact, the debit-parallel connection

$$\Gamma_A = \frac{1}{2} A_\Gamma^{-1} A'_\Gamma$$

removes arbitrary transport and identifies the intrinsic current

$$I_A = q^*(C' - \Gamma_A^* C - C\Gamma_A)q = H\mu' = J.$$

Each admitted prime-power event separates into an exact linear debit and a triple-zero relational remainder, reducing the threshold cell to

$$z^*(K'_{bg} + R'_k)z > 4w.$$

The relational law and its validation are classified: **The test result suggests strong contact with the concept.**

IX.8.3 Particle spin, orbit wave and returning history

The N100 four-role reciprocal tensor aggregates to `2^99 I_2` across 177 occupied bins and is strict on 174. That tensor result suggests strong contact. Its unaugmented two-mode H000724 innovation has a negative least current at all four sampled supports, so positivity at all four for that unaugmented route is separately falsified.

The source-native orbit-wave successor carries the same exact C4 quarter turn:

$$P_{C4}T = TJ, \qquad T^T L_{C4} T = I_2, \qquad J_\Theta^* G_\Theta J_\Theta = L_{C4}.$$

Here `P_C4^4=I` while cumulative wave action has positive derivative. The state returns, but history does not; information does not sit still. Primary, precision, independent and release checks pass `84/84`, `12/12`, `141/141` and `151/151`, and the result suggests strong contact.

The active continuation is the exact weight and placement of accumulated dimensionless Theta/wave action in the completed current. Universal mode 6, product-cone, signed-tail, Schur/atlas and whole-domain joins remain open.

IX.9 Four authority surfaces

- **Public.** Object: documented SAM RH claim and public repository; Current boundary: external validation remains pending.
- **Internal.** Object: $Q_\xi(h) \geq 0$ for every $h \in \mathscr{D}_0$; Current boundary: universal derivation remains active; not an extra gate on the public surface.
- **Spectral.** Object: finite F81 determinant and related exact identifications; Current boundary: optional research; P5A uniform bridge falsified.
- **Operational.** Object: Q3RHV2 plus source-native Starbreaker/Theta continuations; Current boundary: exact event/custody engine installed; universal joins remain open.

The owner-designated internal status is

- `SAM_RH_CLOSURE_PENDING_EXTERNAL_VALIDATION`

The public repository, internal universal-sign program, finite spectral research and application engineering may inform one another, but none acquires authority merely by sharing vocabulary.

IX.10 Wrong controls and preserved boundaries

The route retains several decisive wrong controls:

1. Rename N100 DOS as RH: rejected because ordered directed history is absent.
2. Identify the P2 history coefficient directly with the ordinary-prime Euler coefficient: falsified by 100 versus 2.
3. Use N95 or N105 grading as the selected coefficient-one family: both return zero.
4. Let a nontrivial-zero ordinate select the prime tower: excluded; the tower is derived from $\mathrm{ord}_q(12)$.
5. Split continuous, arithmetic and floor terms before completion: rejected because the separate pieces are not the compiled object.
6. Promote the P5A uniform determinant bridge into a required closure: its fixed route is falsified and remains optional.
7. Use a fixed RH adapter depth when the source-visible rule changes with N: the q7 miss is preserved and q8 is the declared correction at $N = 192$.

These controls preserve rather than weaken the positive completed arithmetic already in custody.

IX.11 Established result and forward handoff

The established RH route begins with an exact 520-state nonbacktracking history operator, pairs 10,700 oriented primitives into 5,350 reciprocal classes, retains the P2 direct failure, supplies the P2A/P2B successor prime route, derives the logarithmic translation coordinate from accumulation and builds the completed ξ/Weil form.

The current operational adapter is separately typed, and the four authority surfaces remain explicit. `SAMA-D000056` uses prime coordinates for an exact Mersenne factor-search application; that application does not enter the RH completed-form chain retroactively.

IX.12 Sources and evidential scope

This chapter is assembled from the following fixed source manuscripts. Source-reported campaign outcomes retain their original scope; no new campaign was executed for this edition.

- Reciprocal history and the rh route.

SAM VOLUME III

Prime Coordinates and Exact Mersenne Factors

Chapter X

The prime application asks a narrower question than primality: can a declared structure expose exact factor opportunities? The geometry is an information-coordinate language. Modular certificates decide factors, and a finite failure to find one remains only that. Later bounded application amendments are recorded under their own provenance.

The account below follows the chapter source as retained at the fixed 8 September 2026 SAMA revision. Engine names, installed interfaces and campaign status are historical source statements. They do not describe a freshly inspected October deployment. Later source amendments are distinguished where available; no later engine or campaign is inferred.

Source-status qualification. The operational and revision statements in this chapter describe the retained source documents. They are not an instruction to execute, deploy, resume, train, promote or change a service, and they have not been independently verified as present live state for this edition.

X.1 Overview

The Mersenne application begins from an owner-defined change of coordinate: prime marks are informational radix coordinates on a closed two-dimensional sphere, not equally spaced marks on an ordinary integer ruler. Exact prime integers remain indispensable. They generate the marks, define operator composition and carry every modular certificate. The conceptual change is in the role assigned to a prime, not in its arithmetic identity.

The rational stereographic chart

$$x = \tan(\chi/2), \qquad S(x) = \left(\frac{2x}{1+x^2}, \frac{1-x^2}{1+x^2}\right)$$

turns multiplication into an exact sphere action $T_n(x) = nx$, inversion into $J(x) = 1/x$, and logarithmic displacement into $du = dx/x = d\chi/\sin\chi$. Prime-only marks $p^- \to p$ form reversible adjacent cells. On each cell the program places three reciprocal voices:

$$\alpha = 2+\sqrt{3}, \qquad \beta_p = \frac{p+\sqrt{3}}{p-\sqrt{3}}, \qquad \gamma_p = \beta_p \beta_{p^-}^{-1}.$$

Repeated squaring generates dyadic pulses. A contact at depths i, j has shallow depth $s = \min(i, j)$ and delay $d = |i-j|$. Transport into the shallow frame produces an exact echo E satisfying

$$E^{2^s} = 1 \pmod{q}, \qquad \operatorname{ord}_q(E) \mid 2^s.$$

The executed MP-S0–MP-S4B lineage compiles those objects into ECHO16 and exposes the genuine Mersenne factor coordinate

$$k = b2^e, \qquad q = 2kp+1.$$

A prime q is assigned as an exact factor of $M_p = 2^p - 1$ only through the order certificate $\operatorname{ord}_q(2) = p$. MP-S5's sealed structural k-order returns 103 factored objects against 100 for equal-budget ascending counting, an installed strong-contact result in the application-scheduler scope. MP-S6 assigns 1,282 exact factors among 5,390 exponents. MP-S7 and MP-S8 preserve a 1,858-row roster after official-factor removal and complete 9,595-coordinate shell misses. Those misses remain `PRIMALITY_UNASSIGNED`.

The clean-sheet route is a successor to two preserved failures. M4's 13,818 RH prime-sidecar ranking and M4A's 7,708 candidate-local Theta18/W9 ranking both fail their fixed direct-successor questions. Their classifications remain **The test falsifies the concept.** The clean-sheet compiler does not rename those outcomes.

The owner has resumed only the bounded CEV1/MP successor and its initial training. All other production, farm, frontier and publication work remains paused. This chapter records the installed CEV1 and MPV2 successors without launching arithmetic, remote work, promotion or distribution.

X.2 Opening question and conceptual picture

The governing question is:

> Can the prime-coordinate, reciprocal-history and dyadic structures be compiled into an exact ordering of genuine Mersenne factor coordinates, while keeping factor assignment, PRP status and Mersenne-primality certification strictly distinct?

The complete route is

- prime integer p and predecessor p^-
- prime-only reversible sphere cell
- alpha, beta_p and gamma_p reciprocal voices
- dyadic pulse depths and delayed echo
- structural k = b*2^e coordinate
- prime q = 2*k*p + 1
- exact modular order certificate
- factor assigned OR finite miss remains PRIMALITY_UNASSIGNED

This is an application compiler, not a universal replacement for the SLC substrate, Q2 selector, Reciprocal History, RH authority or Lucas–Lehmer certificate lane.

X.3 Informational radix with exact arithmetic custody

X.3.1 The owner-defined role

The owner interpretation is:

Prime numbers are informational radix coordinates used to express orbit on a closed two-dimensional sphere.

The formalization retains two layers:

- **arithmetic encoding/certificate.** Meaning: p and q are exact certified prime integers; modular powers and divisibility are exact.
- **SAM object role.** Meaning: each prime emits one indecomposable mark and multiplicative sphere action.

Uniform integer ordinal is not the ruler. Composite integers still appear inside arithmetic values such as n, k and (q-1); they simply do not emit prime-ruler marks of their own.

X.3.2 Orbit and harmony are typed terms

`Orbit` denotes the discrete generated family

$$z, z^2, z^4, \ldots, z^{2^n}$$

on the exact information-state surface. It is not a claim of physical motion around a sphere. `Harmony` denotes exact equality, reciprocal identity or modular closure between separately generated voices. It is not an acoustic analogy substituted for calculation.

X.4 Closed sphere and prime-only ruler

X.4.1 Rational sphere chart

Set $x = \tan(\chi/2)$. The point

$$S(x) = (X(x), Z(x)) = \left(\frac{2x}{1+x^2}, \frac{1-x^2}{1+x^2}\right)$$

lies on the unit circle section of the closed two-dimensional sphere because

$$X(x)^2 + Z(x)^2 = \frac{4x^2 + (1-x^2)^2}{(1+x^2)^2} = \frac{1+2x^2+x^4}{(1+x^2)^2} = 1.$$

The two limits $x = 0$ and $x = \infty$ close the chart at reciprocal poles.

X.4.2 Multiplicative actions and reversal

Define

$$T_n(x) = nx.$$

Then operator composition is exactly integer multiplication:

$$(T_m \circ T_n)(x) = m(nx) = (mn)x = T_{mn}(x).$$

With $J(x) = x^{-1}$,

$$(J \circ T_n \circ J)(x) = J\left(\frac{n}{x}\right) = \frac{x}{n} = T_{1/n}(x).$$

Thus reversal conjugates expansion by n to contraction by $1/n$.

Differentiating $x = \tan(\chi/2)$ gives

$$\frac{dx}{x} = \frac{\frac{1}{2}\sec^2(\chi/2)}{\tan(\chi/2)} d\chi = \frac{d\chi}{2\sin(\chi/2)\cos(\chi/2)} = \frac{d\chi}{\sin\chi}.$$

Therefore

$$du = \frac{dx}{x} = \frac{d\chi}{\sin\chi}.$$

The logarithmic coordinate $u = \log x$ converts T_n into translation by $\log n$, joining the sphere chart to the logarithmic translation language of Reciprocal History without merging their application authorities.

X.4.3 Marks and adjacent cells

MP-S0B emits one object for each prime p. A mark retains:

- the exact integer p and reciprocal point $1/p$;
- indecomposable action T_p;
- the two exact sphere points $S(p)$ and $S(1/p)$; and
- the adjacent arithmetic ports (p-1) and (p+1).

If p^- is the preceding prime, the directed adjacency $p^- \to p$ defines one reversible cell with scale

$$\rho_p = \frac{p}{p^-}, \qquad \rho_p^{-1} = \frac{p^-}{p}, \qquad \Delta u_p = \log\frac{p}{p^-}.$$

Its exact spherical chord is computed from the two rational sphere points. MP-S0A/S0B install 172 prime-only marks and 171 adjacent-prime cells. No composite between consecutive marks becomes an extra ruler object.

X.5 Three reciprocal voices

X.5.1 Native Lucas–Lehmer voice

Let

$$\alpha = 2 + \sqrt{3}, \qquad \alpha^{-1} = 2 - \sqrt{3}, \qquad \alpha\alpha^{-1} = 1.$$

Modulo $M_p = 2^p - 1$, define

$$L_i = \alpha^{2^i} = a_i + b_i\sqrt{3}.$$

The reciprocal trace is

$$s_i = L_i + L_i^{-1} = 2a_i.$$

Because $L_{i+1} = L_i^2$,

$$s_{i+1} = L_i^2 + L_i^{-2} = (L_i + L_i^{-1})^2 - 2 = s_i^2 - 2 \pmod{M_p}.$$

At $i = 0$,

$$s_0 = \alpha + \alpha^{-1} = 4,$$

so the common reciprocal mode is exactly the Lucas–Lehmer recurrence. MP-S0 reconstructs this mode for all 171 prime exponents through 1024.

X.5.2 Candidate prime-mark voice

The Cayley lift

$$\beta_p = \frac{p+\sqrt{3}}{p-\sqrt{3}}$$

has reciprocal conjugate

$$\beta_p^{-1} = \frac{p-\sqrt{3}}{p+\sqrt{3}}.$$

Its norm is one, so its dyadic family

$$R_j = \beta_p^{2^j}$$

lives on the same reciprocal quadratic-unit surface as the α family. The prime integer selects the voice; the modular and quadratic arithmetic retains exact custody.

X.5.3 Incoming transition beat

The transition from the previous prime cell is

$$\gamma_p = \beta_p \beta_{p^-}^{-1}.$$

This object records change between consecutive prime voices rather than a second copy of the current mark. Its inverse reverses the cell:

$$\gamma_p^{-1} = \beta_{p^-} \beta_p^{-1}.$$

The three families answer different questions: α carries the native Mersenne recurrence, β_p carries the current prime mark and γ_p carries the adjacent-prime transition.

X.6 Beats within beats

X.6.1 Dyadic depth and delay

Repeated squaring of any voice produces pulses at depths

$$z^{2^0}, z^{2^1}, \ldots, z^{2^n}.$$

For a contact between depth i in one voice and depth j in another, set

$$s = \min(i,j), \qquad d = |i-j|.$$

If, for example, $j = i+d$, then

$$z^{2^j} = \left(z^{2^i}\right)^{2^d}.$$

The exponent 2^d transports the deeper pulse into the shallower frame. The residual relation is the base echo E. Every retained MP-S2 and MP-S3 factor component satisfies

$$E^{2^s} = 1 \pmod{q}, \qquad \operatorname{ord}_q(E) \mid 2^s.$$

The echo is therefore an exact finite-order modular object with a dyadic closure depth, not a visual correlation.

X.6.2 Nested compiler

Factor the Mersenne multiplier coordinate as

$$k = b2^e,$$

where b is odd and $e = v_2(k)$. The structural base b and dyadic scale 2^e echo the prime-cell and pulse decomposition. The complete compiled route is

- adjacent-prime cell
- alpha/beta/gamma reciprocal relation
- shallow depth s and echo delay d
- base echo closure
- odd structural base b and dyadic scale 2^e
- k = b*2^e
- q = 2*k*p + 1

This is the exact content of `beats within beats`: nested coordinate relations are compiled into a candidate factor address that the modular lane can certify or reject.

X.7 Exact Mersenne factor certificate

X.7.1 Deriving the candidate form

Let p be an odd prime and suppose prime q divides

$$M_p = 2^p - 1.$$

Then

$$2^p \equiv 1 \pmod{q}.$$

The multiplicative order $r = \operatorname{ord}_q(2)$ divides p. For a nontrivial Mersenne factor, $r \neq 1$; since p is prime, $r = p$. Fermat's theorem also gives $r \mid q-1$, hence

$$p \mid q-1.$$

Because q is odd, (q-1) is even, so for some positive integer k,

$$q-1 = 2kp, \qquad q = 2kp+1.$$

Conversely, if the candidate $q = 2kp+1$ is prime and the exact order check returns

$$\operatorname{ord}_q(2) = p,$$

then $2^p \equiv 1 \pmod{q}$ and therefore $q \mid M_p$. The implementation may certify the modular closure with `pow(2,p,q)=1` together with the declared prime/order conditions. It never needs to materialize the enormous integer $2^p - 1$.

X.7.2 Result-type boundary

- **prime q, candidate form, exact order p.** Authorized type: exact factor of M_p.
- **finite k- or shell search finds no factor.** Authorized type: `PRIMALITY_UNASSIGNED`.
- **PRP result with its certificate.** Authorized type: production PRP/certificate status only.
- **exact Lucas–Lehmer terminal.** Authorized type: Mersenne-primality baseline/confirmation status.

A finite miss never crosses rows in this table. It is not a prime, probable prime or verified Mersenne-prime assignment.

X.8 Executed discovery-to-compiler chain

X.8.1 Object installation: MP-S0 through MP-S0B

MP-S0 installs the α reciprocal common mode on 171 prime exponents through 1024. MP-S0A installs the exact sphere actions. MP-S0B emits 172 prime-only marks and 171 reversible cells. These stages define the object language before evaluating factor-search advantage.

X.8.2 Harmony and transition discovery: MP-S1 through MP-S3

- **MP-S1.** Exact observation: self-harmony exposes factors for 52/158 composites and 0/13 exact positives; Classification: **The test result suggests the concept is possible.**.
- **MP-S2.** Exact observation: α/β_p harmony adds five factors beyond MP-S1; all 37 contacts are phase-shifted echoes; Classification: **The test result suggests the concept is possible.**.
- **MP-S3.** Exact observation: α/γ_p adds four held-out factors; all four are delayed echoes; Classification: **The test result suggests the concept is possible.**.

The zero counts on exact-positive controls remain part of the record. The factor contacts motivated compilation because they closed under an exact echo identity, not because every composite was detected.

X.8.3 Compilation and blind-range anatomy: MP-S4A/MP-S4B

MP-S4A compiles the three voices into candidate-local deterministic ECHO16 with checkpoints. MP-S4B applies it to a blind range and factors 50/136 composites and 0/1 exact prime. Inspecting the exact factor anatomy exposes $q = 2kp + 1$, turning the voice/echo structure into the genuine factor-coordinate interface. MP-S4B's classification remains **The test result suggests the concept is possible.** ECHO16 is a language and local compiler at this point, not yet an established global ordering.

X.8.4 Structural scheduler: MP-S5

MP-S5 seals a structural ordering of $k = b2^e$ before comparison. At equal budget it factors 103 objects versus 100 under ascending k counting and reaches shared contacts earlier. Its authorized classification is:

The test result suggests strong contact with the concept.

The result concerns this Mersenne factor-scheduling application. It does not promote the structural ruler to the global Q2, Starbreaker or RH scheduler.

X.8.5 Frontier deployment: MP-S6 through MP-S8

MP-S6 deploys the structural ruler on 5,390 prime exponents:

$$5{,}390 = 1{,}282 + 4{,}108.$$

It assigns 1,282 exact factors without constructing M_p; 4,108 rows remain campaign survivors. The classification is **The test result suggests the concept is possible.**

MP-S7 removes 2,250 survivors by independently verified official factors:

$$4{,}108 - 2{,}250 = 1{,}858.$$

MP-S8 executes every one of 9,595 first-singularity-shell coordinates for each of those 1,858 rows without assigning a factor. The 1,858-row roster is the immutable structural freeze, and every row remains `PRIMALITY_UNASSIGNED` at that boundary.

A later dated status triage yields a 1,204-row operational distribution derivative after removing 647 distinct verified PRP composites, routing two reliable-but-unverified PRP results to verification, removing two verified Lucas–Lehmer composites and holding three active assignments. That derivative does not rewrite the 1,858-row MP-S8 result.

X.9 Preserved direct-ranking failures and wrong controls

X.9.1 M4 direct prime-sidecar ranking

M4 joined 13,818 ordinary-prime spacing sidecars to the completed Mersenne curriculum. It compared pointwise, era-balanced pairwise, smooth-minimax and parent-blend routes across six held-out eras and 19 exact positives. Every non-parent route had a negative worst-era change, so M3E remained selected.

The exact direct-ranking concept is classified:

The test falsifies the concept.

X.9.2 M4A candidate-local Theta18/W9 ranking

An unsealed successor first treated Theta18 as an 18-neighbor prime window. That was the wrong object. The corrected source-visible bridge for each q30 survivor was

- static U63 + dynamic U63
- theta18_cross_hex
- reciprocal left/right F81 views
- W9HistoryMode
- nine forward + nine reverse Z4 maps

M4A then evaluated all 7,708 q30 survivors with exact reciprocal, directional and reconstruction checks. Mechanical validation passed 22/22. Its coupled channel used

$$q(p)=\frac{1}{p}, \qquad A_{lift}(p)=\frac{1-A_0}{p},$$
$$A_{total}(p)=A_0+A_{lift}(p), \qquad u(p)=\log p.$$

the weight $\log(p)/\sqrt{p}$ and the declared pi-bearing floor constant. The fixed candidate-local channel did not stably replace M3E. Its exact-scope classification is:

The test falsifies the concept.

Its receipt is typed `CANDIDATE_LOCAL_COMPLETED_CHANNEL_RECEIPT_NOT_FULL_Q_XI`. It neither evaluates the full completed RH form nor changes RH universal-nonnegativity authority.

X.9.3 Successor relationship

The clean-sheet route asks a different executable question. Instead of using an RH-derived scalar to rank Mersenne positives directly, it orders genuine trial-factor coordinates and retains exact modular factor decisions. This successor does not revise M4 or M4A. It preserves them as the deviation chain that redirected the application toward an operationally exact target.

Wrong controls for the clean-sheet route include:

- declaring a finite factor miss to be a primality result;
- comparing unequal search budgets in MP-S5;
- allowing official factor lookups to enter a blind factor-search surface;
- letting a receipt or exponent identity break a structural-score tie;
- treating the sphere as a physical trajectory rather than the declared exact information-state chart;
- calling the factor scheduler a global SLC/RH scheduler; and
- replacing Lucas–Lehmer certification with the structural filter.

X.10 Bounded CEV1/MP resumption and exact custody

H000725 resumes only the bounded CEV1/MP successor and its initial training. The remaining pause has the following literal operational consequences:

- no new prime benchmark, factor farm, production process or supplemental exploratory run is started;
- Farm V1.1, V9/V13 and the balanced factor pass remain inactive;
- the active remote M143064041 power-9 run is not stopped, altered, queried for new credit or incorporated into this documentation;
- H14F remains a fourteen-physical-worker contract; no 30-worker result exists;
- the 31/127 execution-arity observations remain unserialized diagnostics, while the formal binary-versus-adaptive-31/127-versus-fixed-L162 rotated comparison remains unexecuted; and
- no public prime output, pointer, assignment, promotion or distribution is changed here.

X.10.1 Native CEV1 MP domain

Frozen non-promoted `SLCQ3-RZ-CEV1` compiles `p`, `2^p-1`, `p[ln(2)]`, D18/D81/D162 and exact `q=2kp+1` coordinates into a source-visible 200-feature integer model distinct from the Q3 reciprocal weights. Its first checkpoint trains on 16 discovery runs and evaluates three disjoint held-out runs:

- 25,786 strict correct comparisons
- 190 ties
- 1,281 reversed comparisons
- 27,257 held-out comparisons total.

Base, independent and complete-source checks pass `8/8`, `22/22` and `12/12`; producer-independent reconstruction passes `19/19`. MP52 is barred from training. **The test result suggests strong contact with the concept.**

X.10.2 MPV2 certified frontier and provenance firewall

Isolated MPV2 separates discovery from certification and carries exact state through a chunked CAS certificate DAG. Component-local valuations, combined `F`, exact cyclotomic closure, every frontier edge, zero-yield work and H000723/H000724 wave learning from independently certified intermediate growth survive 252/252 integrated and 59/59 producer-independent checks. Exact closure alone supplies `ACTIVE -> EXHAUSTED_NO_MORE_CERTIFIED_MASS`; an ordinary completed bounded wave remains `ACTIVE`.

Conventional Prime95-compatible arithmetic, residues and checkpoints and PRP-derived scheduling examples remain useful under their exact provenance. They do not become native CEV1/Q3 execution or a primality premise by origin. Native custody begins only after independently certified factor growth.

Topology-derived endpoint observation is installed, but the authenticated remote-worker, CPU-replayed 780M and physical-T500 execution-custody join is not. Native deployment therefore remains fail-closed. No real-hardware benchmark or continuous MPV2 training process is running. **The test result suggests strong contact with the concept.**

For $p=143064041$, $k=320000$ would form

$$q=2(320000)(143064041)+1=91560986240001,$$

a 47-bit candidate coordinate. The active run did not use that gate. The calculation records a possible address only; it is not an executed factor result, and a miss at any finite gate would remain primality-unassigned.

The paused frontier-prime objective below ten minutes remains unfinished in the Lucas–Lehmer certificate lane. This statement preserves the live boundary without weakening any completed MP-S0–MP-S8 result.

X.11 Established result and forward boundary

This chapter establishes a continuous conceptual-to-technical chain:

1. exact prime integers emit informational sphere marks;
2. adjacent marks carry reciprocal α, β_p and γ_p voices;
3. dyadic voice contacts yield finite-order delayed echoes;
4. the echo anatomy factors $k = b2^e$ and forms $q = 2kp + 1$;
5. exact order p assigns a genuine factor of M_p;
6. the structural ruler supplies an application-specific search order; and
7. finite misses remain `PRIMALITY_UNASSIGNED`.

The installed application results and their exact classifications are retained stage by stage. No further computational step follows from this document while the owner pause is active. Any successor requires fresh owner authority and a new campaign that preserves frozen artifacts, exact status types and the untouched active-remote boundary.

X.12 Sources and evidential scope

This chapter is assembled from the following fixed source manuscripts. Source-reported campaign outcomes retain their original scope; no new campaign was executed for this edition.

- Prime ruler and mersenne application.

SAM VOLUME III

Jacobian Maps, Return and Retained History

Chapter XI

The Jacobian branch uses a polynomial map as a laboratory for the difference between returning endpoints and retained path history. The finite source results are preserved, together with corrections that prevent a chosen auxiliary grammar from becoming an unproved natural invariant. The proposed clean restart remains distinct from executed work.

The account below follows the chapter source as retained at the fixed 8 September 2026 SAMA revision. Engine names, installed interfaces and campaign status are historical source statements. They do not describe a freshly inspected October deployment. Later source amendments are distinguished where available; no later engine or campaign is inferred.

XI.1 Overview

The Jacobian program asks how an exact polynomial map can return its target state while retaining a nontrivial history of sheets, braids, escape and projective completion. That is a mathematical and computational question, so JC1 and HF1–HF6TH belong in Volume III. Their earlier matter-volume-labelled files remain immutable provenance, but relocation changes placement and current interpretation rather than any executed result.

Bounded status of the archived algebra. The following formulas and fiber claims are source-reported archived algebra, not a globally verified theorem certified by this edition. Their printed coefficients lie in the rational numbers. If intended as a global polynomial map over the characteristic-zero complex field, an everywhere constant nonzero Jacobian together with distinct preimages of one target would amount to a counterexample to the Jacobian conjecture. This edition does not certify that global claim. The intended domain, source interpretation and independent mathematical adjudication remain unresolved here. The finite Exact Write endpoint/history comparison below is logically separate and does not require accepting a global counterexample.

The JC1 record reports the polynomial formulas

$$P(x,y,z) = (1+xy)^3 z + y^2(1+xy)(4+3xy),$$
$$Q(x,y,z) = y + 3x(1+xy)^2 z + 3xy^2(4+3xy),$$
$$R(x,y,z) = 2x - 3x^2 y - x^3 z,$$

and reports that their Jacobian determinant simplifies to

$$\det D(P,Q,R) = -2.$$

An exact elimination chart produces a cubic fiber polynomial, its resultant and discriminant. On the frozen slice $V = -16/27, R = 1$, one winding around $U = 0$ transposes two escaping sheets, while double winding returns the sheet permutation to the identity. The same winding programs compile through Complete Exact Write on one Theta18 axis. Four positive writes, four negative writes and eight positive writes all have zero compiled translation across all 262,144 addresses, yet their ordered W9 receipts retain four, four and eight events with different orientations and semantic hashes. JC1 passes 18/18; independent validation passes 9/9. Its exact classification is **The test result suggests strong contact with the concept.**

The portable result is narrower and stronger than an endpoint slogan: endpoint equality does not imply program-history equality. It does not by itself install JC1 monodromy, a quotient, a Newton field or a tensor/path-jet field as a matter payload.

HF1–HF6TH then form a valuable but historically scaffolded discovery chain: local reducers and a chosen `Z4^2` cell grammar; nonproper meridians and exact polynomial cells; finite versus projective sheet custody; path-dependent escape corrections; blinded path-jet discriminators; fresh-map transfer; Newton-polygon signatures; and a complete mixed-jet compiler. Every executed finding and correction remains in custody. The current clean restart does not assume that the HF2 torus, its generators, `H_SAM`, W9 or any carrier/matter meaning arises naturally from the polynomial map.

The restart therefore reverses the order of dependence: reconstruct the source algebra independently, derive invariants from actual fibers and the regular-value set, freeze algebraically generated loops, examine monodromy, properness and infinity, and only then reveal legacy HF observations. An optional SAM comparison requires an explicit compiler. A later matter application requires carrier compilation, reciprocal preservation, an identified matter consumer and a load-bearing controlled output change. None has executed. This future bridge is not standalone SAMA Volume IV, which is the RH program.

XI.2 Placement, question and authority

XI.2.1 Governing question

The clean governing question is:

> What global algebraic or topological invariant arises naturally from the exact JC1 polynomial map before an SLC rewrite grammar, W9 receipt, A-field-carrier role or matter interpretation is imported?

The reading order is

- exact polynomial source
- affine fibers and eliminant
- regular-value loops and monodromy
- nonproper and projective behavior
- exact endpoint-return/history distinction
- preserved HF clue deck
- unexecuted source-derived restart
- optional explicit SAM compiler
- optional four-join future matter-applicability test

XI.2.2 Current placement

H000211 relocates JC1/HF1–HF6TH from a former matter-volume interpretation into Volume III computation. The current rule is:

- executable result artifacts control their exact outputs;
- immutable history preserves the route, including superseded literal bridges and corrections;
- `volume_III/SAM_VOLUME_III_JACOBIAN_PROGRAM.md` controls current program routing; and
- live SLC authority remains frozen `SLCQ2-RZ`, whose delegated base installs no Jacobian or physical-matter result.

No historical file is deleted or retroactively reclassified. In particular, H000195/H000196 remain the record of a literal quotient/carrier bridge route, while H000197 and H000211 remove that bridge interpretation from current authority.

XI.3 Exact JC1 source algebra

XI.3.1 Polynomial map and constant Jacobian

Let

$$F : \mathbb{A}^3 \to \mathbb{A}^3, \qquad F(x,y,z) = (P,Q,R)$$

with P,Q,R as stated above. The derivative matrix is formed by differentiating each component with respect to ((x,y,z)). The archived record reports that polynomial expansion and factorization of this determinant cancel the variable-dependent terms and leave

$$\det DF = -2.$$

Within the archived statement, this determinant is nonzero at every affine source point and is read as absence of a local Jacobian singularity. The record then investigates escaping branches and properness. These global interpretations remain subject to the bounded-status qualification above; they are not newly certified by the finite history experiment.

XI.3.2 Elimination chart

JC1 uses

$$x = \frac{1}{m}, \qquad y = m(s-1), \qquad z = (5-3s-m)m^2.$$

Substitution into P,Q,R, followed by elimination of m, gives the exact resultant

$$\mathrm{Res}_m(U-Q, V-P) = s^3\Phi(U,V;s),$$

where

$$\Phi(U,V;s) = K(U,V)s^3 + (U^2-12V)s - 4V$$

and

$$K(U,V) = U^3 - U^2 - 18UV + 27V^2 + 16V.$$

The cubic discriminant factors as

$$\mathrm{Disc}_s\,\Phi = -4\big(U^3 - 18UV + 54V^2\big)^2 K(U,V).$$

Thus the two load-bearing target components are

$$D(U,V) = U^3 - 18UV + 54V^2$$

and $K(U,V)$. Their repeated and simple factors distinguish finite-root braid behavior from coefficient loss and projective escape.

XI.3.3 Exact calibration and loss fibers

In the archived record, the target $(P,Q,R) = (-1,1,-1)$ is assigned the eliminated x-fiber

$$(x-1)(3x+1)(3x+2),$$

with three exact sources

$$(1,-2,9), \quad (-1/3,4,27), \quad (-2/3,-1/2,-9/8).$$

The record reports that each maps to the same target and has Jacobian -2. This is precisely the source-reported claim whose global scope is not certified here.

At the coefficient-loss target

$$(U,V,R) = (0,-16/27,1),$$

the unique finite source is

$$(x,y,z) = (1/2,-8/3,16).$$

The eliminated x-fiber drops to $(2x-1)/2$. Two other cubic branches have moved to projective infinity rather than collided at an affine Jacobian zero.

XI.4 Monodromy from the fiber

XI.4.1 Frozen slice

On $V = -16/27$,

$$K(U,-16/27) = \frac{U(3U^2-3U+32)}{3}.$$

At $U = 0$, the cubic coefficient vanishes while

$$\Phi(0,-16/27;s) = \frac{64}{9}s + \frac{64}{27}.$$

One branch therefore tends to $s = -1/3$. Balancing the cubic and linear terms for the other two branches gives

$$s^2U \longrightarrow -\frac{2}{3}.$$

Those branches escape with magnitude proportional to $|U|^{-1/2}$.

XI.4.2 Loop continuation

Take a small circle

$$U(\theta) = \rho e^{i\theta}, \qquad V = -16/27, \qquad R = 1.$$

At each step, numerically continue the three roots of Φ by minimum matching to the preceding roots. A positive winding and a negative winding both return the finite branch and exchange the two escaping branches:

$$[0,1,2] \longmapsto [0,2,1].$$

The opposite loop orientation changes path direction but not the transposition. A double winding applies the transposition twice:

$$(1\ 2)^2 = e,$$

so the sheet permutation returns to `[0,1,2]`. The offset zero-winding control also returns identity. Maximum recorded fiber residuals stay below 5×10^{-13} on the winding paths.

This chain separates target return, sheet permutation and path winding. Double winding can have identity permutation while still containing a nonempty ordered program.

XI.5 Exact Write compilation and retained history

XI.5.1 Theta18 event map

Theta18 addresses form $\mathbb{Z}_4^9$ in packed two-bit coordinates. On axis j, one positive or negative event applies the exact signed $\mathbb{Z}_4$ step. For axis zero and initial address zero, the positive program visits

- 0 -> 1 -> 512 -> 513 -> 0

while the negative program visits

- 0 -> 513 -> 512 -> 1 -> 0.

Four equal signed steps are zero modulo four. Hence the programs

$$w_+ = (0,+1)^4, \qquad w_- = (0,-1)^4, \qquad w_{++} = (0,+1)^8$$

all compile to translation zero. Because the compiled action is a group translation, zero translation returns every one of the 262,144 input addresses, not only address zero. Sequential and compiled native execution agree on the full address set.

XI.5.2 Ordered receipt construction

Before endpoint collapse, retain for event r

$$(j_r, \varepsilon_r, a_{r-1}, a_r).$$

The JC1-to-W9 adapter assigns event r to reciprocal channel r, with

$$L_r = \varepsilon_r, \qquad R_r = -\varepsilon_r.$$

The W9 common and directional components are

$$C_r = L_r + R_r = 0, \qquad A_r = L_r - R_r = 2\varepsilon_r.$$

Therefore the positive four-event receipt begins ((2,2,2,2)), the negative receipt begins ((-2,-2,-2,-2)), and the double receipt has eight positive entries. Their L^1 receipt norms are 8, 8 and 16, and their semantic hashes are distinct. Reciprocal reconstruction is

$$L_r = \frac{C_r + A_r}{2}, \qquad R_r = \frac{C_r - A_r}{2}.$$

The exact endpoint is the same; event count, sign, order and receipt remain different.

XI.5.3 JC1 result and boundary

The principal campaign passes 18/18 and independent validation passes 9/9. The result is classified:

The test result suggests strong contact with the concept.

The established portable statement is:

- endpoint compilation can erase which closed program occurred;
- an ordered event receipt can retain that distinction.

This directly joins the append-only reel of `SAMA-D000042`. It does not imply that a particular carrier, intersection or matter ledger consumes the full JC1 polynomial object.

XI.6 Preserved HF1–HF6TH lineage

XI.6.1 HF1: reducer foundation

HF1 separates endpoint action in $\mathbb{Z}_4^9$ from an unwrapped signed lift in $\mathbb{Z}^9$. For an endpoint-returning word w,

$$\omega(w) = \widetilde{z}(w)/4 \in \mathbb{Z}^9.$$

It reduces only adjacent exact reciprocal writes with matching route, payload, account anchor and completion type. Four same-orientation writes retain a completed turn. Generic freeze passes 42/42, JC1 calibration 41/41 and independent reconstruction 54/54. A first 41/42 receipt is preserved: its validator mistook boundary prose for embedded map coefficients. The corrected validator checks actual coefficient markers. HF1's classification is **The test result suggests strong contact with the concept.**

XI.6.2 HF2: chosen local `Z4^2` grammar

HF2 installs reciprocal bigons and signed commuting squares on a chosen `(x,y) Z4^2` surface. Termination yields normal form

$$x^m y^n,$$

while endpoint compilation retains only $(m \bmod 4, n \bmod 4)$. The resulting history quotient is

$$H_{local} \cong \mathbb{Z}^2.$$

Among 87,381 words through length eight, 8,909 return the endpoint, 5,341 are locally null and 3,568 retain nonzero history. This falsifies the fixed target that every endpoint-returning word in that installed grammar is locally exact-null. The result does not establish that the chosen torus or its two generators are natural invariants of JC1.

XI.6.3 HF3/HF4: nonproper obstruction and exact regular cell

HF3 follows the (K/x) meridian. Its radial filling reaches coefficient loss: one source stays finite while two escape with exponent $-1/2$. Positive, negative and double filling attempts remain `NONPROPER_OBSTRUCTION`; no exact cell is promoted. HF3 passes 28/28, separated promotion 24/24 and independent reconstruction 46/46, with strong contact at the obstruction-localization scope.

HF4 shows the obstruction is not a fixed-slice artifact by constructing the full nonproper hypersurface and its nonzero meridian linking. Separately, a (D/y) meridian has identity sheet permutation, pure-braid winding one and no escape. An exact Gaussian-rational source disk certifies boundary ((0,1)), so

$$\Lambda_{JC1}^{certified} = \langle (0,1) \rangle, \qquad \mathbb{Z}^2/\Lambda_{JC1}^{certified} \cong \mathbb{Z}.$$

Two implementation attempts are preserved: NumPy-boolean serialization and a closed-path final-sample tracking defect. Corrected execution passes 37/37; promotion and independent reconstructions pass their declared gates. The result suggests strong contact.

XI.6.4 HF5 and HF5A: selected-sheet fill versus global sheet custody

HF5 constructs an exact source disk with K-winding one and certifies a regular finite sheet through the coefficient-loss center. Its first interpretation promoted ((1,0)), combined it with ((0,1)) and concluded a full lattice. H000195/H000196 preserve that literal bridge route and its exact calculations.

HF5A corrects the current interpretation: the finite-sheet disk does not fill all three typed sheets. Homogenizing

$$\Phi^h = KS^3 + BST^2 - 4VT^3$$

gives at projective infinity

$$\Phi^h|_{T=0} = KS^3.$$

At the loss center the local infinity equation is

$$\frac{64}{27}t^2(t+3) = 0.$$

One finite root survives and a double root remains at infinity. The response $I_\infty(k,d) = k$ retains the (K/x) global-history class. Current preserved HF authority therefore returns to the rank-one certified lattice above. HF5A passes 26/26 with 22/22 independent reconstruction and suggests strong contact. Three pre-result field/dependency/alias faults remain preserved.

XI.6.5 HF6 and the path-specific correction

HF6 emits anonymous finite-plus-projective receipts before attaching sector labels. It reconstructs:

- **`D/y`.** finite/infinity sheets: 3/0; infinity index: 0; permutation: identity; custody: `GLOBALLY_DISCHARGED`.
- **`K/x`.** finite/infinity sheets: 1/2; infinity index: 1; permutation: `[0,2,1]`; custody: `MIXED`.

Seven nonslice deformations preserve K-winding one, the transposition and the surviving class. Execution passes 19/19 and independent validation 24/24; the result suggests strong contact.

H000199 corrects one overgeneralized coefficient. The escape exponent remains $-1/2$, but the leading coefficient depends on the exact path jet. The radial route alone has $s^2U \to -2/3$; six other slopes and the HF5 disk have their own exact values. The correction passes 15/15 and 14/14 and does not change HF6's classification or quotient.

XI.6.6 HF6T/HF6TF: blinded jet rule and transfer boundary

HF6T freezes the eliminant-conditioned rule

- a3(0)=0 and ord_path(a3)=1 and b1(0)!=0
- `PROJECTIVE_CUSTODY_CANDIDATE`
- otherwise
- `FINITE_OR_UNRESOLVED`

before reveal. It matches nine forward routes and nine reversals, reconstructs path-specific escape and preserves the failed pure-tensor partition. A first 18/19 run used the wrong determinant sign after swapping target coordinate order; correcting only that explicit sign gives 19/19. The result suggests strong contact for the JC1 path-jet discriminator, not a universal tensor law.

HF6TF freezes the rule unchanged on ten fresh maps. It returns four true positives, zero false positives, two false negatives and four true negatives. It is exact 7/7 on the admitted proper-cubic/simple-transverse domain, but it misses quadratic coefficient loss and simultaneous cubic/linear degeneration. Those misses remain evidence and motivate the next signature rather than a retroactive threshold change.

XI.6.7 HF6TG/HF6TH: Newton signature and mixed-jet compiler

For

$$\Phi_t(s) = \sum_j a_j(t) s^j, \qquad \nu_j = \operatorname{ord}_t a_j(t),$$

HF6TG takes the lower convex hull of (j, ν_j). Positive-slope edge width, slope and initial polynomial predict escaping-sheet count, exponent and regular cycle structure. On 15 fresh fibers, projective presence, sheet counts and coefficient winding close 15/15; 34/34 exponents and 10/10 regular cycle types reconstruct. A repeated edge remains `SINGULARITY_OBSTRUCTION`. The result suggests strong contact without installing a universal Newton law.

HF6TH then works in the triangular affine-path chart and freezes the complete mixed coefficient

$$\mathscr{T}_{rj} = \frac{1}{r!\,j!} D^{r+j} F^2(v^r, w^j).$$

For

$$G(x,y) = \sum_{p,j} c_{pj} x^p y^j,$$

exact Taylor expansion gives

$$\mathscr{T}_{rj} = \sum_{p \geq r} c_{pj} \binom{p}{r} u_0^{p-r}, \qquad \nu_j = \min\{r : \mathscr{T}_{rj} \neq 0\}.$$

The complete mixed jet therefore reconstructs the Newton signature in that chart. Known calibration closes 15/15; twelve fresh maps reconstruct all valuation tables, hulls, edge signatures and projective counts, with 40/40 escape exponents. This is a chart-specific exact compiler identity. Low-order compression, nontriangular maps, nonaffine paths and coordinate-invariant transfer remain open.

XI.7 What the legacy lineage does and does not install

The HF lineage contains exact reducers, cells, obstructions, projective receipts, blind-transfer results and compiler identities. Its stages remain valid in their declared charts. The current program makes three distinctions:

1. the HF2 grammar is a constructed SLC object, not yet a source-derived JC1 invariant;
2. projective and mixed-jet findings do not constitute a universal tensor law;
3. historically matter-volume-labelled carrier supplements are preserved route artifacts, not current matter interfaces.

No current JC1-specific history, sheet, quotient, Newton or tensor field is consumed by AFC2–AFC4. No `q_A`, mass, SI energy or matter-generation relation is installed.

XI.8 Clean restart: frozen method, no execution

XI.8.1 Premise firewall

JCR1 begins without the former matter-volume interpretation, A-field carrier, q_A, $A(r)$, W8/X1/W9, HF1 reducer roles, the HF2 torus, its generators, `H_SAM`, projective custody as information custody or tensor-to-matter interpretation. These remain available only after clean definitions are frozen.

XI.8.2 Stage sequence

- **JCR1-0.** Work: normalize exact source map, fields, coordinates, determinant, fibers, discriminant and assumptions; Required output or boundary: source normal form; no SLC encoding.
- **JCR1-1.** Work: independently reconstruct determinant, resultant, fibers, degree-loss set and homogenization; Required output or boundary: algebraic facts only; no history labels.
- **JCR1-2.** Work: derive candidates from regular-value complement, covering, monodromy, braid, homology, infinity divisors and cycles; Required output or boundary: typed invariant roster with domains and equivalences.
- **JCR1-3.** Work: freeze meridians, reversals, concatenations, null loops and map controls from algebraic components; Required output or boundary: clean loop campaign; failures retained.
- **JCR1-4.** Work: account for finite/infinity multiplicity, branch continuation, divisors and asymptotics; Required output or boundary: projective receipts with exact/numerical separation.
- **JCR1-5.** Work: ask the two-dimensional invariant question with all dimensions separated; Required output or boundary: no `Z4^2` substitution for polynomial geometry.
- **JCR1-6.** Work: reveal JC1/HF3–HF6TH observations one at a time; Required output or boundary: rediscovered/compatible/scaffold-dependent/conflicting/unresolved classifications.
- **JCR1-7.** Work: optionally compare stable invariant with SAM/SLC history; Required output or boundary: explicit compiler both ways or precise information loss.
- **JCR1-8.** Work: optionally test future matter applicability; Required output or boundary: four native bridge joins required.

XI.8.3 Candidate-invariant contract

Every candidate invariant must state its definition, domain, equivalence relation, composition law, base-point and coordinate dependence, loop-reversal and concatenation behavior, ordinary invertible-map value and JC1 value or unresolved status. A familiar shape is not enough to rename it `H_SAM`.

The legacy reveal occurs only after the clean source, definitions and first predictions are sealed. Agreement, disagreement, partial rediscovery and irrelevance are all retained. This prevents legacy success from becoming a hidden premise.

XI.8.4 Current status

JCR1 is a draft for owner review and has not executed. It installs no new Volume III authority, invariant, tensor law, SLC training change, carrier payload or matter mapping. This chapter does not authorize its execution.

XI.9 Future matter-bridge boundary

A future Jacobian object reaches a matter application only if one bridge-native campaign establishes all four joins:

1. an exact compiler from the source-derived invariant into a declared carrier payload;
2. exact preservation of that payload through reciprocal delivery;
3. an identified intersection or matter-ledger consumer that reads it; and
4. a load-bearing output change when the payload is altered under a controlled comparison.

The first two establish transport; the third establishes consumption; the fourth establishes consequence. A history hash that survives delivery without a consumer is not the completed bridge. A consumer that changes for an undeclared correlated field is not the controlled payload result.

No current JC1/HF field has completed this chain. Depth training, Q2 selection, W9 availability or A-field-carrier existence alone does not validate or promote a Jacobian-to-matter interpretation.

The source program called this a prospective `Volume IV` boundary. That exact wording remains in the immutable source and `SAMA-C000256-R001`. In this standalone atlas, Volume IV is the Riemann Hypothesis program. The matter bridge therefore remains unnumbered, unexecuted and outside the Volume IV dependency graph unless a later source-bound RH relation is separately established.

XI.10 Established result and forward handoff

The established current result is the isolated JC1 algebra/monodromy and endpoint-history distinction, together with the preserved HF findings in their exact charts. The relocated program establishes where those results belong and which former interpretations are historical rather than current.

The next mathematical step is the unexecuted source-derived restart. Its first output is independent affine/projective reconstruction, not an SLC or matter bridge. Any later comparison must make the compiler and information loss explicit.

XI.11 Sources and evidential scope

This chapter is assembled from the following fixed source manuscripts. Source-reported campaign outcomes retain their original scope; no new campaign was executed for this edition.

- Jacobian program and clean restart.

SAM VOLUME III

Relational Composition and the Boundary of the Volume

Chapter XII

The final chapter changes evidential level. It preserves two source-originated proposals about composition and interaction, rather than presenting a newly validated universal law. Their role is to state the intended conceptual relation among the finite objects already developed. The volume can end with these questions explicit; they do not create an obligation to run another campaign.

The account below follows the chapter source as retained at the fixed 8 September 2026 SAMA revision. Engine names, installed interfaces and campaign status are historical source statements. They do not describe a freshly inspected October deployment. Later source amendments are distinguished where available; no later engine or campaign is inferred.

XII.1 Orientation

SAM already contains three ingredients that are usually discussed at different scales: a nonzero accumulation field sourced by matter, an exact W8/X1/W9 write lifecycle with Theta carrier structure, and the accumulated effect of many stellar sources in galaxy halos. The Nonzero Amplification Principle identifies these as one recursive mechanism.

The central statement is:

> **Matter writes. The substrate carries. Nonzero writes accumulate.**

Matter does not have to occupy the route between two sources for information to cross that route. A source changes the substrate accumulation; the lifted substrate carries the source-conditioned write through matter-free space. The same composition that turns particle-scale writes into a body-scale presence can turn stellar writes into a galactic accumulation halo.

This document records an owner-originated conceptual bridge for continued mathematical development. It does not assign a test result classification.

XII.2 Established inputs and new principle

XII.2.1 Established inputs

The new bridge begins from established, separately typed SAM structures:

- the universal accumulation floor `A_0`;
- the spherical source accumulation lift `A(r)`;
- the closed-matter return to accumulation through source support and carrier compression;
- the W8/X1/W9 substrate-atom relation;
- Theta18 as a tensor carrier; and
- the Complete Exact Write computation surface.

These inputs retain their established definitions. The new concept does not replace `A_0`, redefine W9 as a separate object, turn Theta18 into matter, or replace the Exact Write with a continuum metaphor.

XII.2.2 Owner-originated Nonzero Amplification Principle

> **Every nonzero constituent substrate write survives coherent composition, allowing particle-scale information to accumulate recursively into macroscopic and galactic accumulation structure.**

The principle adds a cross-scale interpretation to the established inputs:

1. a particle contributes a nonzero source-conditioned write;
2. many particle writes compose into the coherent write of a body;
3. the body-scale write is carried by matter-free substrate;
4. overlapping body writes establish local informational contact;
5. body writes compose again at larger scales; and
6. billions of stellar writes form the extended galactic accumulation halo.

XII.3 The information-bearing substrate

Introduce provisional notation for the source-conditioned lift:

$$\Delta A_S(x) = A_{\text{total},S}(x) - A_0.$$

This notation distinguishes three states that must not be collapsed:

- **Matter present.** Meaning: A local material source participates in creating or changing the write..
- **Matter absent and `Delta A_S(x)>0`.** Meaning: The substrate remains source-conditioned and carries information without an intervening material messenger..
- **`Delta A_S(x)=0`, equivalently the source route has returned to `A=A_0`.** Meaning: The active source-conditioned connection is absent at that point..

The substrate is therefore not a passive geometric backdrop. Changes in `A` change the local write environment. In the owner-originated mechanism, those changes continually affect the W8/X1/W9 lifecycle and Theta carrier activity throughout the connected matter-free region.

Define the source-conditioned contact domain provisionally by

$$\Omega_S = \{x : \Delta A_S(x) > 0\}.$$

A route between two systems is in active source contact when it remains in the appropriate connected component of this lifted domain. A point at which the source-conditioned lift returns to zero supplies the proposed disconnection condition.

XII.4 Exact composition rather than point-particle messaging

For a composite source `S`, write

$$\Delta A_S(x) = \mathscr{C}_S(\{\delta A_i(x) : i \in S\}),$$

where each constituent contribution is nonzero on its active route and `mathcal C_S` denotes the exact composition law.

The symbol `mathcal C_S` is deliberately not replaced by ordinary linear addition. SAM has an Exact Write, retained histories, typed carriers and directional computation. The correct composition law should be recovered or derived from that algebra so that cross terms, history and carrier structure remain intact.

The conceptual content is nevertheless exact enough to distinguish the mechanisms:

- an individual Earth particle does not send an isolated command to an individual Moon particle;
- all particle writes in Earth compose into an Earth-scale accumulated write;
- all particle writes in the Moon compose into a Moon-scale accumulated write; and
- their source-conditioned domains overlap in the matter-free substrate, where continuing changes in `A`, W8/X1/W9 and Theta provide local contact.

The Moon responds inside the Earth's accumulated substrate write while also contributing its own write to the same substrate.

XII.5 One recursive bridge across three scales

XII.5.1 Quantum to macroscopic

The quantum bridge is the passage from exact local substrate transactions to a coherent composite source:

$$\{\text{particle Exact Writes}\} \xrightarrow{\mathscr{C}} \text{body-scale accumulated write.}$$

Object-level informational presence need not be introduced as an unrelated primitive if it is the coherent closure of the constituent writes.

XII.5.2 Earth to Moon

Earth and Moon participate in one shared lifted substrate. Their continuing internal particle activity changes their composite writes; those changes are carried through matter-free space by the substrate itself. The orbital response is therefore downstream of an information-bearing write process, not merely a description of the resulting path.

XII.5.3 Stars to galaxy

The galactic bridge applies the same composition recursively:

$$\{\text{stellar body writes}\} \xrightarrow{\mathscr{C}} \text{galaxy-scale accumulated halo.}$$

Each star contributes a nonzero effect at large distance. Billions of such contributions produce the extended galactic accumulation structure. The particle-to-body and star-to-galaxy steps are thus two applications of the same nonzero-preserving composition law.

XII.6 Technical continuation

The next investigation begins with the exact algebra already present in SAM:

1. **Composition operator.** Identify `mathcal C` inside Complete Exact Write and SLC execution, including retained-history and directional terms.
2. **Nonzero preservation.** Determine the exact condition under which a constituent nonzero write survives composite closure.
3. **A-dependent lifecycle.** State the operator by which a local change in `A` changes W8/X1/W9 execution and Theta carrier activity.
4. **Contact domain.** Derive the connected `A>A_0` route and the exact `A=A_0` disconnection condition.
5. **Recursive closure.** Determine when a composite write is admitted as a source input to the same operation at the next scale.
6. **Existing SLC orbit.** Inspect whether the current particle-orbit emulator already contains part of this composition and transport law.
7. **Cross-scale comparison.** Express particle/body, Earth–Moon and star/galaxy examples through one shared receipt format.

This route is intended to discover the exact operator already implied by the combined SAM structures, rather than introduce separate quantum, orbital and galactic mechanisms.

XII.7 Relational interaction and completed composition

XII.8 Orientation

The first Nonzero Amplification document proposes that constituent substrate writes can compose recursively from particles to bodies and from stellar bodies to galaxies. The relational proposal adds the following intended correction: composition depends on how the constituents interact.

Two planets light-years apart and the same two planets closely orbiting have the same constituent inventory. They do not have the same reciprocal history, overlap, directional relation or completed substrate write. The nearby orbiting pair forms a coherent composite source; the distant pair approaches two separately carried source writes.

The central statement is:

> **Nonzero amplification is relational. Interacting writes create completed information that is absent from an inventory-only sum.**

XII.9 Inventory and interaction are differently typed

Let `S` be a composite system. Its source data includes at least:

- constituent writes `{w_i}`;

- a typed active relation network `R_S`;
- retained reciprocal or directional history `H_S`;
- the local and route-dependent accumulation state; and
- the completed W8/X1/W9 and Theta carrier state.

The proposed completed composition is therefore written schematically as

$$\Delta A_S = \mathscr{C}_S(\{w_i\}, R_S, H_S).$$

An inventory-only control would instead evaluate

$$\Delta A_{\mathrm{diag}} = \sum_i \delta A_i.$$

The difference

$$\mathscr{I}_S = \Delta A_S - \Delta A_{\mathrm{diag}}$$

is a useful diagnostic representation of the interaction innovation. It does not define `mathcal C_S` as ordinary addition. The native composition must retain Exact Write types, histories, cross terms and completion.

XII.10 Separation changes the completed write

For the same two constituents, consider two relational states.

XII.10.1 Distant state

The two source-conditioned writes are carried separately. Their active overlap and continuing reciprocal history are reduced, and a route returning to the universal floor supplies the proposed disconnection condition.

XII.10.2 Nearby orbiting state

Each source continually changes the accumulation environment of the other. Those changes alter the matter-free substrate between them, the local W8/X1/W9 lifecycle and Theta carrier/intersection activity. The pair therefore generates a continuing relational history and becomes a new composite source.

The distinction is not "two sources versus more sources." It is:

$$(w_1, w_2, R_{\mathrm{distant}}, H_{\mathrm{distant}}) \neq (w_1, w_2, R_{\mathrm{orbit}}, H_{\mathrm{orbit}}).$$

XII.11 Many-body amplification

For many particles, the number of possible relations grows faster than the constituent count. Pair, cycle, face and higher closure information can therefore become load-bearing even when every individual particle write is small.

A schematic cluster display is

$$\Delta A_S \sim \sum_i A_i + \sum_{i<j} \mathscr{I}_{ij} + \sum_{i<j<k} \mathscr{I}_{ijk} + \cdots,$$

but the Exact Write composition owns the actual algebra. The display only shows what a spreadsheet-style constituent sum omits: interaction terms whose values depend on the completed relational state.

This gives Nonzero Amplification two inseparable aspects:

1. each constituent contributes a nonzero write; and
2. coherent interaction among those writes generates additional completed information.

XII.12 ATOM3D consequence

ATOM3D has frequently represented candidate assemblies through counts, independent relation rows and separately scored receipt features. The current Li-6 action-frame frontier exposes the limit of that representation: complete W9P custody retains each relation but supplies no relation-to-frame identity, leaving 13,824 reciprocal frame assignments.

Relational Nonzero Amplification changes the unit evaluated by a successor. The unit should be the whole active relation network, with each relation read in the context of all co-active writes:

$$(e, f_e, R_S, H_S, \Delta A_S, \Theta_S) \longmapsto \mathscr{I}_e(S).$$

Here `e` is a relation, `f_e` its action frame and `mathcal I_e(S)` its context-dependent innovation inside the completed assembly. The same isolated edge placed in a different network need not produce the same innovation.

The direct successor is therefore an **interaction-aware composite-write receipt**, not another isolated edge selector. Candidate Li-6 assemblies can then be compared by exact completed closure while preserving the current bars against geometry fitting and opaque selection.

XII.13 RH consequence

The corresponding RH distinction is between eventwise inventory and completed coupling. If `W` is assembled from event writes `W_i`, then

$$W^*W = \sum_i W_i^* W_i + \sum_{i \neq j} W_i^* W_j.$$

The off-diagonal products carry relational information. Current RH work already retains completed Gamma–prime–Robin coupling where eventwise positivity is insufficient, and its support extensions are expressed through transported carriers and entrant Schur innovations.

Relational Nonzero Amplification supplies a shared candidate origin for those cross terms: they are the mathematical receipt of coexisting source writes interacting inside one completed substrate history.

XII.14 Technical continuation

1. Recover `mathcal C_S` from Complete Exact Write and current SLC receipts.
2. Compile whole-network interaction innovations without reducing relations to independent rows.
3. Compare identical inventories under distant, nearby, orbiting and relation-deleted configurations.

4. Determine which interaction data selects or reduces ATOM3D action frames.
5. Express the resulting operator as a completed Gram or Schur innovation.
6. Cross-walk that operator to the RH source-append architecture.

XII.15 Sources and evidential scope

This chapter is assembled from the following fixed source manuscripts. Source-reported campaign outcomes retain their original scope; no new campaign was executed for this edition.

- Nonzero amplification quantum galactic bridge.
- Relational nonzero amplification interaction closure.